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Our ref: E-39210-47753

02-September-2016

Kind attention: S B Marine

Company : S B Marine

Dear Sir/Madam,

Your document (detailed below) has been reviewed.

Project details	
Project number	NC001969
Project name	Chisti Marine Engineering Works – CHISTI/002
Project manager	Varun Yalvatkar

Document Details	
Plan No.	SBM-16-1011-701
Plan Title	Preliminary Intact Stability Booklet
Rev.No.	0
Upload.No.	1
Submitted Date	22-July-2016
Review Status	Approved
Parent Plan Title	

The plan has been examined for compliance with:

. IS code 2008

Notes

1. The review status Approved indicates:
The Preliminary Intact Stability booklet has been examined for compliance with IS Code 2008 and found in order.
Final Stability Booklet is to be submitted for our approval incorporating approved lightweight parameters resulting from Inclining Test.
2. The list of remarks is part of this letter. Please address these comments by reply to us

Thanking you,

Yours Sincerely,

For INDIAN REGISTER OF SHIPPING

Parvathi Rajalakshmi

Principal Surveyor

This is an electronically published document and does not require signature.

Our Ref: E-39210-47753

Review Comments			
Sr.No	Comment Date	Description	Clearing Responsibility
1	02-September-2016	Draft marks plan is to be submitted for approval and the same is to be included in the final stability booklet.	Head Office
2	02-September-2016	Only unprotected openings are to be considered as downflooding points.	For Info
3	02-September-2016	The engine room supply vent opening to be modified as per IRS rules 2015 Part3 ch13 sec 2.3.2, "the ventilators which are necessary to continuously supply the machinery space and on demand ..need to be taken up to a height as specified in 2.3.1 with sufficient heights specified in 2.3.1, without having to fit weather tight closing appliances.	Head Office
4	02-September-2016	Details of ventilators are to be submitted for approval.	Head Office



21.30M TUG BOAT

PRELIMINARY INTACT STABILITY BOOKLET



SEE LETTER
E - 39210 - 47753

0	7/21/2016		SIVA	SB	SC
Rev	Date	Description	Prepared	Checked	Reviewed
Client:			Title:		
K.B. SHIPPING & CO.			PRELIMINARY INTACT STABILITY BOOKLET		
Builder:			21.30M TUG BOAT		
CHISTI MARINE ENGINEERING WORKS, SIKKA					
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			21-Jul-16	Page 1 of 74	16-1011
			Document: 16-1011-701		Rev: 0
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1 Introduction:

This Booklet consists of preliminary intact stability analysis of the vessel "21.30M Tug Boat" with a bollard pull of 10.00 Tons complying the necessary stability requirements.

2 Main Particulars of the vessel:

2.1 General particulars

Name of the Vessel	:	21.30m Tug
Client	:	KB SHIPPING
Type	:	Tug Boat
Flag	:	Indian
Class	:	Indian Register of Shipping
Yard Number	:	002
Engine	:	2 x 400 BHP
Complement	:	12 Persons
Class Notation	:	[SUL TUG, [IY, INDIAN RIVER SEA VESSEL TYPE 4

2.2 Principal particulars

Length O.A.	:	21.30 M
Length B.P	:	19.76 m
Breadth (MLD)	:	06.50 M.
Depth (MLD)	:	02.80 m
Design Draft	:	1.90 m
Scantling Draft	:	2.00 m
Full Load Displacement	:	164.717 T
Deadweight	:	70.717 T
Design speed	:	10 knots
Bollard Pull	:	10 T

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3 Notes on the Use of Stability Booklet

3.1 Conversion Tables

The Government of India strongly recommends use of S.I (Systems International Units). The following table shows conversion factors for converting British Units to Metric Units and vice versa.

Multiply by	To Convert from	To Obtain	
0.03937	Millimeters	Inches	25.4
0.3937	Centimeters	Inches	2.54
3.2808	Meters	Feet	0.3048
2.2046	Kilograms	Pound	0.45359
0.0009842	Kilograms	Tons (2240 lbs)	1016.047
0.9842	Metric Tonnes	Tones (2240 lbs)	1.016
2.4998	Metric Tonne per centimeter of immersion	Tones per inch of immersion	0.04
8.2014	Moment to change trim by one centimeter (tonne-metre)	Moment to change trim by one inch (Ton-foot)	0.122
0.0174532	Degree	Radians	57.2957795
	TO OBTAIN	TO CONVERT FROM	MULTIPLY BY

Relation between Weight and Volume:

10 ³ mm	=	1 cm ³
1 cm ³ of fresh water	=	1 Gram
1000 cm ³ of fresh water	=	1 Kilogram (1000 Grams)
1 m ³ of fresh water	=	1 Metric tonne (1000 Kilograms)
1 m ³ of salt water (Specific Gravity = 1.025 t/m ³)	=	1.025 Metric Tonnes
1 Metric Tonne of salt water	=	0.975 m ³
1 m ³	=	35.316 ft ³
1 ft ³	=	0.0283 m ³



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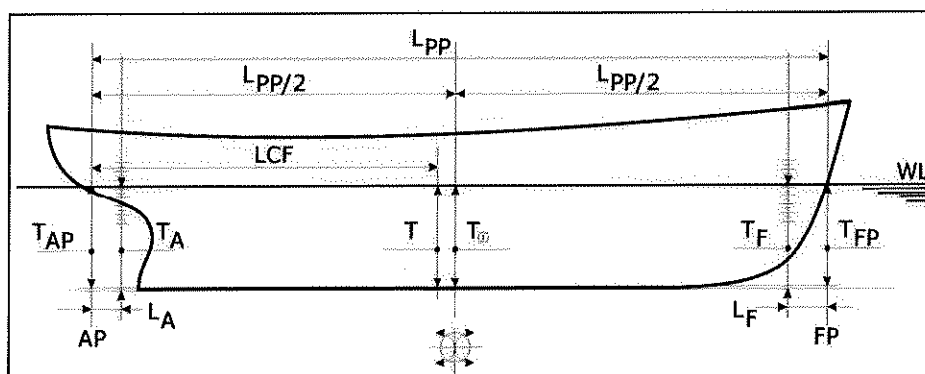
3.2 Symbols and Sign Convention

SYMBOL	DESCRIPTION	UNIT
Δ	Displacement	t
$\bar{v}$	Volume of Displacement	m ³
AP	Aft perpendicular	m
A _{WP}	Area of Water Plane	m ²
B	Breadth Moulded	m
D	Depth Moulded (Upto Main Deck at Side)	m
Dwt	Deadweight	tonnes
FP	Forward perpendicular	m
L _A	Distance from AP to aft draft mark (positive value if draft mark is positioned forward of AP)	m
LBP	Length between perpendiculars	m
LCB	Longitudinal Centre of Buoyancy (from A.P. i.e., Fr-2)	m
LCF	Longitudinal Centre of Flotation (from A.P. i.e., Fr-2)	m
LCG	Longitudinal centre of gravity (from A.P. i.e., Fr-2)	m
L _F	Distance from FP to forward draft mark (positive value if draft mark is positioned forward of FP)	m
TCG	Transverse centre of gravity from centre line	m
GG _o	Free surface correction	m
GM _T	Transverse metacentric height	m
GoM _T	Corrected transverse metacentric height	m
GM _L	Longitudinal metacentric height	m
GZ	Righting arm	m
GT	Gross Tonnage	-
G _o Z	Corrected righting arm	m
G _o Z _{Max}	Corrected maximum righting arm	m
KB	Vertical Centre of Buoyancy (from Base Line)	m
KG/VCG	Vertical centre of gravity from base line	m

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KG_o	Corrected vertical centre of gravity from base line	m
KM_{τ}	Transverse Metacentric Height (from Base Line)	m
KN	Distance from base line to metacentric axis	m
MCT_{1cm}	Moment to change trim by one centimeter	tones x m/cm
NT	Net Tonnage	-
TPC	Tones per centimeter immersion	tonnes/cm
t	Total Trim ($T_{FP} - T_{AP}$)	m
T	Draft at LCF	m
T_{AP}	Draft Aft at AP	m
T_A	Draft aft draft marks from underside of keel	m
T_{FP}	Draft Fwd at FP	m
T_F	Draft forward draft marks from underside of keel	m
T_M	Draft Mean at Midship	m
W_{llool}	Wind heeling arm	m
θ	Angle of inclination in degrees	deg
ρ	Density of liquid	tonnes/m ³
θ_{Max}	Angle at which maximum G_oZ occurs	deg

3.3 Sign Convention:



LCB, LCG, LCF	Fwd of AP is positive Aft of AP is Negative
Trim by Aft	Negative
Trim by Forward	Positive

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3.4 General Instructions to Master against Capsizing

The fact that the vessel's stability has been calculated for several conditions of loading in which the vessel normally operates, and that such calculations have proven that the stability is good and in excess of the IMO requirements, and the fact that the calculations are checked for approval does not mean that the vessel is now immune against capsizing. A stamped and approved copy of this booklet must be on board at all times. The following measures should be considered as preliminary guidance on matters influencing safety and stability.

1. Compliance with the stability criteria does not ensure immunity against capsizing, regardless of the circumstances, or absolve the master from his responsibilities. Masters should therefore exercise prudence and good seamanship having regard to the season of the year weather forecasts and the navigational zone and should take appropriate action as to speed and course warranted by the prevailing circumstances.
2. Before a voyage commences, care should be taken to ensure that the cargo and sizeable pieces of equipment have been properly stowed or lashed so as to minimize the possibility of both longitudinal and shifting, while at sea, under the effect of acceleration caused by rolling and pitching.
3. The stability criteria recommended as per IS Code 2008 and set minimum values, but no maximum values are recommended. It is advisable to avoid excessive values of metacentric height, since these might lead to acceleration forces which could be prejudicial to the vessel, its complement, its equipment and to safe carriage of the cargo.
4. All doorways and other openings through which can enter into the hull or deckhouses, forecastle, etc., should be suitably closed in adverse weather conditions and accordingly all appliances for this purpose should be maintained on board and in good condition.
5. Weather tight and watertight hatches, doors, etc., should be kept closed during navigation, except when necessarily opened for the working of the vessel and should always be ready for immediate closure and be clearly marked to indicate that these fittings are to be kept closed except for access. All portable deadlights should be maintained in good condition and securely closed in bad weather.
6. Any closing devices provided for vent pipes to fuel tanks should be secured in bad weather.
7. Reliance on automatic steering may be dangerous as this prevents ready changes to course which may be needed in bad weather.
8. In all conditions of loading necessary care should be taken to maintain a seaworthy freeboard.
9. In severe weather, the speed of the vessel should be reduced if excessive rolling, propeller emergence, shipping of water on deck or heavy slamming occurs. Six heavy slamming or 25 propeller emergences during 100 pitching motions should be considered dangerous.

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10. Special attention should be paid when a vessel is sailing in following or quartering seas because dangerous phenomena such as parametric resonance, broaching to reduction of stability on the wave crest, and excessive rolling may occur singularly in sequence or simultaneously in a multiple combination, creating a threat of capsizing. Particularly dangerous is the situation when the wave length is of the order of 1.0-1.5 vessel's length. A vessel's speed and / or course should be altered appropriately to avoid the above mentioned phenomena.
11. Water trapping in deck wells should be avoided. If freeing ports are not sufficient for draining of the well, either speed of the vessel should be reduced, or course should be changed, or both. Freeing ports provided with closing appliances should always be capable of functioning and are not to be locked.
12. Masters should be aware that steep or breaking waves may occur in certain areas, or in certain wind and current combinations.

3.5 Notes on Stability

Stability

The stability of a vessel in general is its ability to maintain an upright position or to re-establish this after disturbing by an external force or moment. For an undamaged vessel at small angle of heel up to 5°, it is sufficient to evaluate the stability in transverse direction. This depends on position of two points relative to each other, centre of gravity (G) and transverse metacenter (M), see figure. 2.4.1. The metacentric height GM, the distance between the points G and M, means the stability at small angles and is given by the following equation:

$$GM_T = KM_T - KG.$$

The centre of gravity of the vessel (KG) depends on distribution of cargo and liquids in tanks in the vessel. Vessel KG for a particular loading condition is obtained by taking vertical moments (sum of weight of an item x centre of gravity of that item) and by division of total moments with the total weights.

The position of the transverse metacentre varies with the draft. For a certain draft corresponding to a certain displacement, the transverse metacentric position (M) above the keel point K is fixed and is denoted as KM_T . The location of the metacentre has neither to do with the nature nor the distribution of cargo & other weights onboard. The transverse metacentre above keel, only dependent on the lines of the vessel, shall be obtained from hydrostatics curves or tables.

Cross curves of stability

Position of transverse metacentre (M) is not fixed as the angle of heel exceeds beyond 5° and GM is no longer a stability criterion for measuring the stability of the vessel. Therefore righting lever or arm (GZ) shall be used for this purpose.

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These curves show the variation of the righting arms (GZ) with displacement and are drawn for various angles of inclination, each curve being drawn for a particular angle of inclination. When these curves are drawn for a fixed value of the vertical center of gravity position, they are called "GZ Curves". When the vertical center of gravity is assumed to be zero, these curves are called "KN Curves" and the same are used to derive the statical stability curve.

When the angle of inclination exceeds small angles, 4° or 5° the intersection of the vertical through the center of buoyancy in the inclined condition with the center line of vessel, point "M" can no longer be regarded as a fixed point relative to the vessel. The quantum "GM" no longer a suitable criterion for measuring the stability of the vessel and it is usual to use the value of the righting arm "GZ" for this purpose. The Cross Curves of stability provide a means of presenting the righting arm "GZ" for any probable value of displacement and for several angles of heel. The cross curves of stability depends on the geometry of the vessel and is independent of the loading of the vessel.

Curve of Statical Stability:

A curve of righting arm GZ to a base of angle of inclination for a fixed displacement is known as a curve of statical stability. This curve is provided for every intended loading condition in the booklet. To illustrate the righting levers at various angles of heel in a condition, the effective righting lever GZ is derived from:

$$GZ = KN - KG \times \sin \theta$$

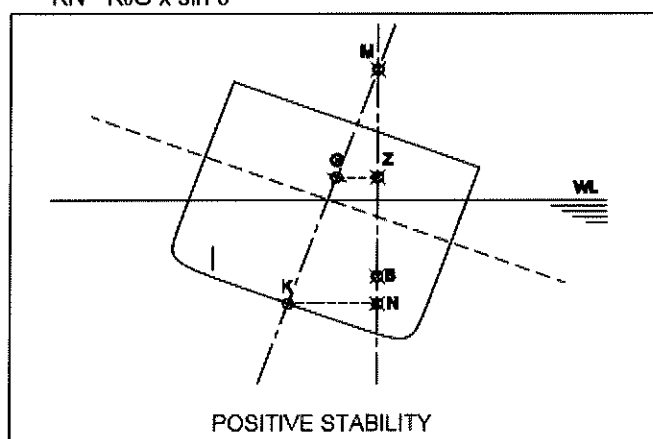
KN = Horizontal distance from keel to the vertical line passing through centre of Buoyancy.

KG = Centre of gravity from base line.

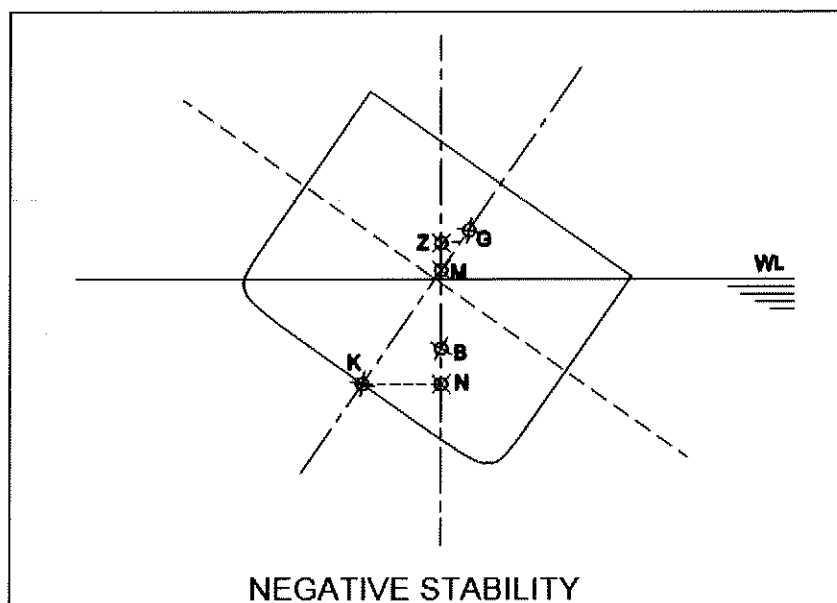
θ = Angle of inclination (heel).

Righting arm after correction of free surface effects is given by G_0Z .

$$G_0Z = KN - K_0G \times \sin \theta$$



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Use of Tables:

The tables for hydrostatics, cross curves and maximum permissible KG should be entered with the ship's moulded draught amidships, T_M , measured from the upper side of the keel plate.

$$T_M = \frac{T_{AP} + T_{FP}}{2}$$

Draft and Trim Calculation:

For calculation of trim and draft based on a loading condition calculation, the following data are required for ascertaining trim and draft.

- Longitudinal centre of gravity of the ship calculated as follows.

$$LCG = \frac{\text{SUMMATION OF LONGITUDINAL WEIGHT MOMENTS}}{\text{DISPLACEMENT}}$$

- Longitudinal centre of buoyancy, LCB
- Displacement, Δ
- Moment to change trim, MCT
- Longitudinal centre of Floatation, LCF.

LCB, LCF and MCT are taken from Hydrostatic tables (trim = 0 metres) corresponding to the actual displacement.

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Trim is calculated according to the following formula:

$$\text{Trim (t)} = \frac{LBP \times (LCG - LCB)}{GM_L}$$

The Moulded Drafts at AP (T_{AP}) and FP (T_{FP}) are calculated to following formulas:

$$\text{Draft at LCF (m)} T_{LCF} = T_m - \left(\frac{\frac{LBP}{2} - LCF}{LBP} \right) \times TRIM$$

$$T_{AP} = T_{LCF} - \frac{LCF \times TRIM}{LBP}$$

$$T_{FP} = T_{AP} + TRIM$$

Where T is the draft taken from Hydrostatic Tables ($t = 0$ metres) corresponding to the actual displacement.

The draft at draft marks is as calculated as follows:

$$T_A = T_{AP} + \frac{L_A \times TRIM}{LBP} + \text{keel plate thickness}$$

$$T_F = T_{FP} + \frac{L_F \times TRIM}{LBP} + \text{keel plate thickness}$$

If the righting lever curve (GZ Curve) has to be established, the KM_r and KN values to be used for this purpose are calculated for the calculated trim value. This is done by interpolation for the KMT and KN values, respectively, between the two tabulated trim values nearest to calculated trim.

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4 Stability Criteria

4.1 IMO Intact Stability Criteria

4.1.1 General Criteria

The vessel is required to comply with the general intact stability criteria laid down in MSC.267 (85) (IS Code 2008) Part A, Chapter 2.

Recommended General Criteria:

2.2.1 The area under the righting lever curve (GZ curve) should not be less than 0.05 meter Radians up to an angle of 30° and not be less than 0.09 meter radians up to $\theta = 40^\circ$ or the angle of flooding θ_f . If this angle is less than 40° . Additionally, the area under the righting lever curve (GZ curve) between the angles of heel of 30° and 40° or between 30° and θ_f , if this angle is less than 40° , should not be less than 0.030 meter-radians.

2.2.2 The right lever GZ should be at least 0.20 m at an angle of heel equal to or greater than 30° .

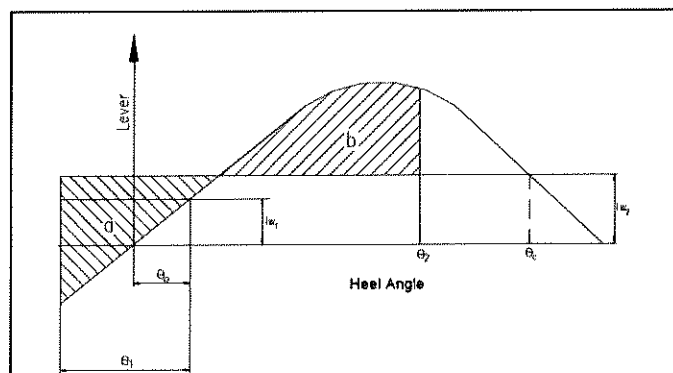
2.2.3 The maximum righting lever shall occur at an angle of heel not less than 25° . If this is not practicable, alternative criteria, based on an equivalent level of safety, may be applied subject to the approval of the Administration.

2.2.4 The initial metacentric height GM_0 should not be less than 0.15 m.

4.1.2 Severe Wind and Rolling Criteria (Weather Criteria)

The following criteria is extracted from IS Code 2008, Part A Chapter 2.3

The ability of a ship to withstand the combined effects of beam wind & rolling should be demonstrated for each standard condition of loading, with reference to the figure as following:



Area- B should be greater than Area- A

Maximum allowable static heel angle should be less than 16° or an angle corresponding to 80% of the deck immersion angle whichever is less.



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Where,

Area-B is the area under the righting curve up to an angle of 50° or second intercept of lw2 or Down Flooding angle whichever is less.

Area- A is the area under the curve up to the angle of roll back measured backward from equilibrium angle due to steady wind as shown in figure.

Intermediate value in tables 1-4 should be obtained by liner interpolation

Rolling period $T = \frac{2XCXB}{\sqrt{GM}}$ seconds

Where, $C = 0.373 + 0.023X\left(\frac{B}{d}\right) - 0.043X\left(\frac{L}{100}\right)$

The symbols in the above tables and formula for the rolling period are defined as follows:

L	=	waterline length of the ship (m)
B	=	moulded breadth of the ship (m)
D	=	moulded draft of the ship (m)
C _b	=	block coefficient
A _k	=	total overall area of bilge keels, or area of the lateral projection of the bar keel, or sum of these areas. (m ²)
GM	=	metacentric height corrected for free surface effect (m)

The wind heeling levers lw1 and lw2 referred to are constant values at all angles of inclination and are calculated as follows:

$$lw1 = \frac{P \times A \times Z}{1000 \times g \times Disp} \text{ and}$$

$$lw2 = 1.5 \times lw1$$

Where,

P	=	Pressure, 504 N/m ²
A	=	Projected lateral area of the portion of the ship and deck cargo if any above the waterline (m ²)
Z	=	vertical distance from the center of A to the center of the underwater lateral area or approximately to a point at one half the draft (m).
Disp	=	Displacement (t)

$$\text{Roll back angle} = 109 \times X_1 \times X_2 \times \sqrt{r \times s} \text{ (degrees)}$$

Where:

X1	=	factor as shown in table 1
X2	=	factor as shown in table 2



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- K = factor as follows
- K = 1.0 for round-bilged ship having no bilge or bar keels;
- K = 0.7 for ship having sharp bilge
- K = as shown in table 3 for a ship having bilge keels, a bar keel or both
- r = 0.73 ± 0.6 OG/d

With:

- OG = distance between the center of gravity and the waterline (-ve if it is below).
- D = mean moulded draft of the ship (m)
- S = factor as shown in table 4.

TABLE-1	
B/d	X1
≤ 2.4	1
2.5	0.98
2.6	0.96
2.7	0.95
2.8	0.93
2.9	0.91
3	0.9
3.1	0.88
3.2	0.86
3.3	0.84
3.4	0.82
≥ 3.5	0.8

TABLE-2	
Cb	X2
≤ 0.45	0.75
0.5	0.82
0.55	0.89
0.6	0.95
0.65	0.97
≥ 0.70	1
3.5	0.72
≥ 4.0	0.7

TABLE-3	
(Akx100)/(LxB)	k
0	1
1	0.98
1.5	0.95
2	0.88
2.5	0.79
3	0.74
18	0.038
≥ 20	0.035

TABLE-4	
T	s
≤ 6	0.1
7	0.098
8	0.093
12	0.065
14	0.053
16	0.044



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4.2 Towing criteria:

The vessel is required to comply with the general intact stability criteria laid down in *Rules and Regulations for the Construction and Classification of Steel Ships – 2016, Part 5 section 4.1*.

The stability of tugs is to be assessed considering the effect of transverse heeling force caused when the tow rope is not in line with the tug's longitudinal centerline.

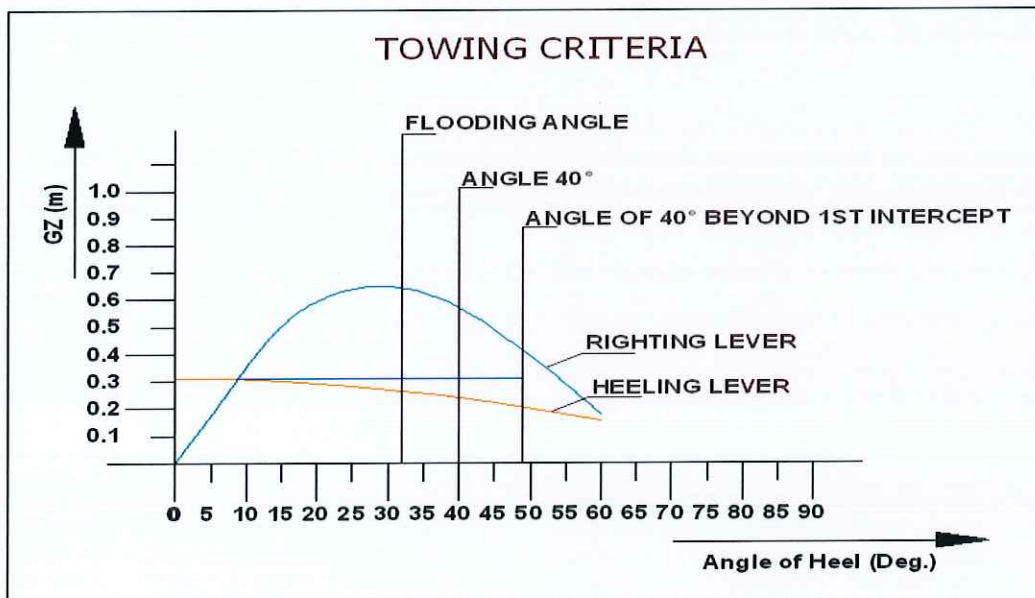
Compliance with the following criteria is recommended, as a minimum:

- (i) The residual area between a righting lever curve and a heeling lever curve developed from 70% of the maximum bollard pull force acting in 90° to the ship length direction is not to be less than 0.09 [m-rad]; The residual areas under the curves are to be calculated from the 1st intercept up to an angle of 40° beyond the first intercept or up to the down-flooding angle, whichever is less.

OR

- (ii) Alternatively, the area under a righting lever curve should not be less than 1.4 times the area under a heeling lever curve developed from 70% of the maximum bollard pull force acting in 90° to ship-length direction.

In this case the areas under the curves are to be calculated from upright condition up to an angle of 40° or the down-flooding angle, whichever is less.





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The heeling lever curve is to be derived by using the following formula:

$$b_h = 0.7 TH \cos\theta / 9.81\Delta$$

Where,

b_h = heeling arm [m];

T = maximum bollard pull [kN];

H = vertical distance [m] between the towing hook and the center of the propeller;

Δ = loading condition displacement [t];

θ = angle of heel.

5 Closures & Down flooding Points

5.1 Down flooding Points:

1. The angle of flooding is the angle of immersion of the sill or coaming of the opening on or above the freeboard deck through which progressive flooding can occur as the vessel heels.
2. If this opening has a weather tight closing device which cannot be kept closed and secured at all times while the vessel is underway or has no weather tight closing device, progressive flooding will occur when the sill of the opening immerses. This angle has to be marked on the GZ curve if it comes within the apparent positive range of the GZ curve terminates at this angle i.e. the actual angle of flooding becomes the range of the GZ curve.
3. If the lower sill of the opening through which Down flooding can occur immerses at 40° or smaller angles of heel, its closing appliances must be kept closed and secured at all times when the vessel is underway or at sea.

Description	Down flooding points			Symmetric
	X coordinate (m)	Y coordinate (m)	Z coordinate (m)	
Engine Room ventilation sup/Exhaust	6.000	2.119	3.612	Yes
Cowl Vent 1	16.455	1.597	3.822	Yes
Engine Room Funnel	7.052	2.100	7.700	Yes
Sky light	6.100	0.775	3.470	Yes

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6 Hydrostatics:

Trim = 1.00 m by aft

Trim: -1.000 (m)										
Draft (m)	Vol. mould (m ³)	Displ. (tonnes)	LCB (m)	VCB (m)	LCF (m)	S (m ²)	KMt (m)	KMI (m)	MCT (t ^m /cm)	TpCm (t/cm)
0.500	25.952	26.601	7.875	0.358	8.645	72.50	7.292	30.38	0.404	0.698
0.550	29.440	30.176	7.973	0.384	8.757	76.79	6.830	31.02	0.468	0.734
0.600	33.097	33.924	8.065	0.412	8.849	80.95	6.422	31.51	0.534	0.767
0.650	36.912	37.835	8.150	0.439	8.913	84.96	6.064	31.82	0.601	0.799
0.700	40.876	41.898	8.225	0.466	8.946	88.83	5.749	31.90	0.667	0.828
0.750	44.979	46.103	8.292	0.494	8.950	92.65	5.474	31.94	0.734	0.856
0.800	49.220	50.450	8.348	0.523	8.927	96.55	5.233	32.09	0.806	0.885
0.850	53.604	54.944	8.393	0.551	8.867	100.68	5.025	32.53	0.889	0.915
0.900	58.151	59.605	8.425	0.581	8.705	105.76	4.849	34.15	1.013	0.955
0.950	62.948	64.521	8.433	0.612	8.425	112.06	4.716	37.14	1.193	1.006
1.000	67.941	69.640	8.427	0.645	8.284	116.86	4.614	37.90	1.313	1.041
1.050	73.066	74.893	8.415	0.677	8.271	120.15	4.474	36.94	1.374	1.060
1.100	78.264	80.220	8.406	0.709	8.282	123.06	4.334	35.68	1.420	1.074
1.150	83.530	85.618	8.399	0.740	8.291	125.93	4.210	34.50	1.463	1.087
1.200	88.854	91.076	8.393	0.771	8.298	128.75	4.099	33.39	1.504	1.099
1.250	94.237	96.593	8.388	0.802	8.304	131.52	3.999	32.33	1.541	1.111
1.300	99.674	102.165	8.383	0.832	8.313	134.22	3.905	31.29	1.575	1.121
1.350	105.155	107.784	8.380	0.862	8.337	136.79	3.809	30.20	1.600	1.128
1.400	110.673	113.439	8.378	0.891	8.358	139.34	3.724	29.19	1.625	1.136
1.450	116.221	119.127	8.378	0.920	8.378	141.89	3.648	28.26	1.648	1.143
1.500	121.807	124.852	8.378	0.949	8.396	144.43	3.581	27.41	1.672	1.150
1.550	127.429	130.615	8.379	0.978	8.412	146.97	3.521	26.61	1.694	1.157
1.600	133.083	136.410	8.381	1.006	8.427	149.50	3.467	25.87	1.717	1.164
1.650	138.768	142.237	8.383	1.035	8.441	152.04	3.419	25.18	1.738	1.170
1.700	144.485	148.097	8.386	1.063	8.454	154.57	3.377	24.54	1.760	1.177
1.750	150.237	153.993	8.389	1.091	8.466	157.10	3.339	23.94	1.780	1.183
1.800	156.020	159.921	8.392	1.119	8.477	159.63	3.306	23.37	1.801	1.190
1.850	161.829	165.875	8.395	1.147	8.487	162.17	3.277	22.84	1.821	1.196
1.900	167.669	171.861	8.398	1.175	8.496	164.71	3.251	22.35	1.841	1.202
1.950	173.539	177.877	8.402	1.202	8.506	167.25	3.229	21.88	1.861	1.208
2.000	179.444	183.930	8.405	1.230	8.514	169.79	3.210	21.44	1.881	1.214
2.050	185.372	190.006	8.409	1.258	8.523	172.35	3.193	21.03	1.902	1.221
2.100	191.335	196.118	8.413	1.285	8.532	174.91	3.180	20.65	1.922	1.227
2.150	197.329	202.262	8.416	1.313	8.541	177.48	3.168	20.29	1.942	1.233
2.200	203.353	208.437	8.420	1.340	8.551	180.05	3.159	19.95	1.963	1.239

Note: Density of water = 1.025 tonnes/m³

Draft, VCB, KMt, KMI is from moulded base line.

LCB and LCF values are given relative to A.P. (Fr. No-2)

Trim is taken according to LBP (=19.76m).



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Trim = 0.75 m by aft

Trim: -0.750 (m)										
Draft	Vol. mould	Displ.	LCB	VCB	LCF	S	KMt	KMI	MCT	TpCm
(m)	(m ³)	(tonnes)	(m)	(m)	(m)	(m ²)	(m)	(m)	(t ^m /cm)	(t/cm)
0.500	25.059	25.685	8.247	0.333	9.058	73.34	7.536	33.78	0.435	0.710
0.550	28.605	29.321	8.354	0.360	9.165	77.45	7.008	33.86	0.497	0.744
0.600	32.306	33.114	8.452	0.388	9.235	81.26	6.550	33.50	0.555	0.774
0.650	36.145	37.048	8.537	0.416	9.279	84.87	6.153	32.96	0.610	0.801
0.700	40.111	41.114	8.612	0.443	9.305	88.38	5.807	32.43	0.666	0.826
0.750	44.198	45.303	8.677	0.471	9.314	91.84	5.506	31.94	0.721	0.851
0.800	48.402	49.612	8.732	0.500	9.307	95.26	5.244	31.51	0.779	0.874
0.850	52.720	54.038	8.778	0.528	9.282	98.71	5.015	31.18	0.838	0.897
0.900	57.153	58.582	8.815	0.557	9.238	102.24	4.816	31.00	0.902	0.921
0.950	61.704	63.247	8.844	0.586	9.168	105.95	4.643	31.04	0.975	0.946
1.000	66.386	68.045	8.863	0.615	9.053	110.07	4.495	31.54	1.065	0.976
1.050	71.241	73.022	8.868	0.646	8.779	115.83	4.375	33.94	1.230	1.021
1.100	76.310	78.218	8.855	0.677	8.586	120.79	4.287	35.06	1.361	1.057
1.150	81.528	83.566	8.834	0.709	8.483	124.73	4.203	35.01	1.450	1.082
1.200	86.833	89.004	8.812	0.741	8.474	127.58	4.090	33.87	1.492	1.094
1.250	92.196	94.501	8.793	0.772	8.469	130.37	3.988	32.76	1.530	1.106
1.300	97.615	100.055	8.775	0.803	8.465	133.12	3.897	31.74	1.567	1.117
1.350	103.087	105.664	8.758	0.834	8.462	135.85	3.815	30.78	1.601	1.127
1.400	108.606	111.321	8.743	0.864	8.462	138.54	3.740	29.87	1.634	1.137
1.450	114.166	117.020	8.730	0.894	8.471	141.14	3.665	28.94	1.661	1.145
1.500	119.768	122.762	8.718	0.924	8.487	143.66	3.593	28.01	1.683	1.152
1.550	125.396	128.531	8.708	0.953	8.501	146.18	3.530	27.15	1.704	1.158
1.600	131.060	134.337	8.699	0.982	8.514	148.71	3.473	26.36	1.725	1.165
1.650	136.752	140.171	8.692	1.011	8.526	151.23	3.423	25.62	1.746	1.171
1.700	142.477	146.039	8.685	1.040	8.537	153.75	3.378	24.94	1.766	1.178
1.750	148.234	151.940	8.680	1.068	8.548	156.27	3.339	24.30	1.786	1.184
1.800	154.018	157.869	8.675	1.097	8.557	158.80	3.304	23.70	1.806	1.190
1.850	159.833	163.829	8.671	1.125	8.567	161.33	3.274	23.15	1.826	1.196
1.900	165.678	169.819	8.667	1.153	8.576	163.86	3.247	22.63	1.846	1.202
1.950	171.553	175.842	8.664	1.181	8.585	166.40	3.224	22.14	1.865	1.208
2.000	177.462	181.898	8.662	1.209	8.594	168.95	3.204	21.69	1.885	1.214
2.050	183.396	187.980	8.660	1.237	8.603	171.51	3.188	21.27	1.906	1.221
2.100	189.362	194.096	8.658	1.265	8.613	174.07	3.173	20.88	1.927	1.227
2.150	195.356	200.240	8.657	1.293	8.624	176.65	3.162	20.52	1.948	1.233
2.200	201.381	206.416	8.656	1.321	8.635	179.24	3.152	20.18	1.970	1.240

Note: Density of water = 1.025 tonnes/m³

Draft, VCB, KMt, KMI is from moulded base line.

LCB and LCF values are given relative to A.P. (Fr. No-2)

Trim is taken according to LBP (=19.76m).

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Trim = 0.50 m by aft

Trim: -0.500 (m)										
Draft (m)	Vol. mould (m ³)	Displ. (tonnes)	LCB (m)	VCB (m)	LCF (m)	S (m ²)	KMt (m)	KMI (m)	MCT (t ^m /cm)	TpCm (t/cm)
0.500	24.512	25.125	8.674	0.314	9.455	74.35	7.723	37.10	0.468	0.724
0.550	28.113	28.816	8.779	0.343	9.520	78.06	7.140	35.99	0.520	0.753
0.600	31.850	32.647	8.869	0.371	9.568	81.58	6.641	34.89	0.570	0.779
0.650	35.711	36.604	8.946	0.399	9.599	84.95	6.212	33.85	0.620	0.804
0.700	39.687	40.679	9.012	0.428	9.616	88.22	5.842	32.88	0.668	0.826
0.750	43.770	44.864	9.069	0.456	9.620	91.41	5.523	32.01	0.716	0.848
0.800	47.954	49.153	9.117	0.484	9.612	94.57	5.246	31.23	0.765	0.868
0.850	52.238	53.544	9.157	0.513	9.594	97.71	5.005	30.56	0.814	0.888
0.900	56.620	58.035	9.189	0.542	9.564	100.86	4.795	30.00	0.865	0.908
0.950	61.096	62.623	9.215	0.570	9.521	104.06	4.611	29.55	0.918	0.928
1.000	65.670	67.312	9.235	0.599	9.463	107.33	4.451	29.22	0.975	0.948
1.050	70.346	72.105	9.247	0.628	9.385	110.75	4.312	29.07	1.038	0.969
1.100	75.130	77.009	9.253	0.658	9.280	114.45	4.191	29.22	1.113	0.993
1.150	80.044	82.045	9.250	0.687	9.115	118.79	4.090	30.00	1.217	1.023
1.200	85.138	87.266	9.234	0.718	8.850	124.25	4.012	31.88	1.376	1.065
1.250	90.404	92.664	9.206	0.749	8.694	128.60	3.951	32.41	1.485	1.093
1.300	95.783	98.177	9.175	0.780	8.630	131.93	3.880	31.96	1.549	1.111
1.350	101.224	103.755	9.146	0.811	8.619	134.68	3.798	30.99	1.585	1.122
1.400	106.720	109.388	9.118	0.842	8.608	137.40	3.724	30.09	1.619	1.132
1.450	112.266	115.072	9.093	0.872	8.599	140.10	3.657	29.24	1.652	1.142
1.500	117.855	120.801	9.069	0.903	8.592	142.77	3.597	28.44	1.683	1.151
1.550	123.486	126.573	9.047	0.932	8.589	145.41	3.540	27.65	1.712	1.159
1.600	129.152	132.381	9.028	0.962	8.597	147.94	3.483	26.83	1.733	1.166
1.650	134.856	138.228	9.009	0.991	8.608	150.45	3.430	26.05	1.753	1.172
1.700	140.587	144.102	8.993	1.021	8.618	152.96	3.384	25.33	1.772	1.178
1.750	146.346	150.005	8.979	1.050	8.627	155.48	3.343	24.65	1.792	1.184
1.800	152.132	155.935	8.966	1.078	8.637	158.00	3.307	24.03	1.811	1.190
1.850	157.952	161.901	8.954	1.107	8.646	160.53	3.275	23.45	1.831	1.196
1.900	163.805	167.900	8.943	1.136	8.655	163.07	3.248	22.92	1.851	1.203
1.950	169.682	173.924	8.933	1.164	8.665	165.62	3.224	22.42	1.871	1.209
2.000	175.591	179.981	8.924	1.192	8.675	168.18	3.203	21.96	1.892	1.215
2.050	181.530	186.068	8.916	1.221	8.686	170.74	3.186	21.54	1.913	1.221
2.100	187.505	192.193	8.909	1.249	8.698	173.32	3.171	21.14	1.935	1.228
2.150	193.510	198.347	8.902	1.277	8.709	175.90	3.159	20.77	1.957	1.235
2.200	199.544	204.533	8.897	1.305	8.721	178.50	3.149	20.43	1.979	1.241

Note: Density of water = 1.025 tonnes/m³

Draft, VCB, KMt, KMI is from moulded base line.

LCB and LCF values are given relative to A.P. (Fr. No-2)

Trim is taken according to LBP (=19.76m).

K.B. SHIPPING & CO.	21.30M TUG BOAT	16-1011-701
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Trim = 0.25 m by aft

Trim: -0.250 (m)										
Draft (m)	Vol. mould (m ³)	Displ. (tonnes)	LCB (m)	VCB (m)	LCF (m)	S (m ²)	KMt (m)	KMI (m)	MCT (t ^m /cm)	TpCm (t/cm)
0.500	24.288	24.895	9.146	0.304	9.800	75.10	7.836	39.13	0.489	0.733
0.550	27.930	28.628	9.234	0.333	9.841	78.61	7.213	37.39	0.537	0.760
0.600	31.698	32.490	9.308	0.361	9.869	81.92	6.684	35.79	0.583	0.784
0.650	35.577	36.466	9.370	0.390	9.885	85.09	6.234	34.35	0.627	0.806
0.700	39.560	40.549	9.422	0.419	9.891	88.16	5.849	33.07	0.670	0.827
0.750	43.640	44.731	9.466	0.448	9.887	91.16	5.518	31.94	0.713	0.846
0.800	47.812	49.008	9.502	0.476	9.875	94.13	5.232	30.93	0.755	0.864
0.850	52.073	53.375	9.532	0.505	9.855	97.06	4.984	30.04	0.798	0.882
0.900	56.416	57.827	9.556	0.533	9.826	99.99	4.769	29.25	0.840	0.899
0.950	60.845	62.367	9.574	0.562	9.788	102.92	4.580	28.56	0.884	0.916
1.000	65.354	66.988	9.588	0.590	9.743	105.90	4.415	27.99	0.929	0.933
1.050	69.949	71.697	9.596	0.619	9.690	108.94	4.271	27.55	0.977	0.951
1.100	74.627	76.493	9.600	0.648	9.627	112.07	4.144	27.24	1.029	0.968
1.150	79.398	81.383	9.599	0.677	9.550	115.33	4.034	27.07	1.087	0.988
1.200	84.265	86.372	9.594	0.706	9.455	118.79	3.938	27.11	1.154	1.008
1.250	89.240	91.471	9.583	0.735	9.326	122.61	3.856	27.47	1.237	1.033
1.300	94.353	96.712	9.564	0.765	9.120	127.29	3.789	28.59	1.362	1.066
1.350	99.640	102.131	9.534	0.795	8.905	132.12	3.739	29.75	1.497	1.099
1.400	105.061	107.688	9.498	0.826	8.786	135.95	3.695	29.90	1.584	1.122
1.450	110.568	113.333	9.461	0.856	8.747	138.92	3.638	29.29	1.631	1.135
1.500	116.124	119.027	9.427	0.887	8.731	141.62	3.579	28.50	1.663	1.145
1.550	121.732	124.775	9.394	0.917	8.718	144.31	3.526	27.76	1.695	1.154
1.600	127.385	130.570	9.364	0.947	8.706	146.98	3.479	27.05	1.725	1.163
1.650	133.078	136.405	9.335	0.976	8.698	149.62	3.435	26.38	1.753	1.171
1.700	138.809	142.279	9.309	1.006	8.697	152.22	3.393	25.71	1.779	1.179
1.750	144.570	148.185	9.285	1.035	8.706	154.74	3.351	25.01	1.798	1.185
1.800	150.367	154.126	9.263	1.064	8.716	157.26	3.313	24.37	1.818	1.191
1.850	156.191	160.096	9.243	1.093	8.726	159.80	3.281	23.78	1.838	1.197
1.900	162.047	166.099	9.224	1.122	8.737	162.35	3.252	23.23	1.859	1.203
1.950	167.927	172.125	9.208	1.150	8.749	164.91	3.227	22.73	1.880	1.210
2.000	173.845	178.191	9.192	1.179	8.760	167.47	3.206	22.26	1.901	1.216
2.050	179.794	184.289	9.178	1.207	8.772	170.05	3.188	21.83	1.923	1.223
2.100	185.777	190.422	9.165	1.236	8.785	172.63	3.172	21.43	1.946	1.230
2.150	191.793	196.588	9.153	1.264	8.797	175.23	3.160	21.05	1.968	1.236
2.200	197.834	202.780	9.143	1.292	8.810	177.83	3.149	20.70	1.991	1.243

Note: Density of water = 1.025 tonnes/m³

Draft, VCB, KMt, KMI is from moulded base line.

LCB and LCF values are given relative to A.P. (Fr. No-2)

Trim is taken according to LBP (=19.76m)

K.B. SHIPPING & CO.	21.30M TUG BOAT	16-1011-701
7/21/2016	PRELIMINARY INTACT STABILITY BOOKLET	Rev No: 0

Trim = 0.00 m

Trim: 0.000 (m)										
Draft (m)	Vol. mould (m ³)	Displ. (tonnes)	LCB (m)	VCB (m)	LCF (m)	S (m ²)	KMt (m)	KMI (m)	MCT (t*m/cm)	TpCm (t/cm)
0.500	24.361	24.970	9.647	0.301	10.117	75.73	7.867	40.21	0.504	0.740
0.550	28.034	28.735	9.710	0.330	10.134	79.05	7.221	38.01	0.548	0.765
0.600	31.821	32.616	9.761	0.360	10.142	82.19	6.677	36.10	0.590	0.787
0.650	35.712	36.605	9.802	0.388	10.143	85.21	6.216	34.43	0.631	0.808
0.700	39.699	40.691	9.836	0.417	10.137	88.14	5.825	32.95	0.670	0.827
0.750	43.773	44.868	9.864	0.446	10.124	91.00	5.489	31.64	0.708	0.844
0.800	47.932	49.131	9.886	0.474	10.104	93.80	5.201	30.46	0.745	0.861
0.850	52.170	53.474	9.902	0.503	10.078	96.59	4.951	29.41	0.782	0.876
0.900	56.481	57.893	9.915	0.531	10.048	99.37	4.735	28.51	0.820	0.892
0.950	60.871	62.393	9.923	0.560	10.014	102.17	4.545	27.74	0.858	0.907
1.000	65.334	66.967	9.928	0.588	9.976	104.99	4.379	27.08	0.898	0.923
1.050	69.872	71.619	9.930	0.617	9.932	107.83	4.233	26.52	0.939	0.938
1.100	74.485	76.347	9.928	0.645	9.882	110.72	4.104	26.05	0.982	0.953
1.150	79.174	81.153	9.924	0.673	9.826	113.67	3.991	25.68	1.027	0.969
1.200	83.941	86.040	9.917	0.702	9.761	116.70	3.892	25.41	1.076	0.986
1.250	88.789	91.008	9.906	0.730	9.686	119.83	3.805	25.26	1.130	1.003
1.300	93.729	96.072	9.892	0.759	9.597	123.12	3.730	25.24	1.190	1.022
1.350	98.761	101.230	9.875	0.788	9.486	126.66	3.665	25.42	1.262	1.043
1.400	103.907	106.504	9.852	0.817	9.337	130.64	3.611	25.95	1.355	1.068
1.450	109.192	111.922	9.822	0.846	9.110	135.48	3.569	27.19	1.492	1.102
1.500	114.636	117.501	9.784	0.876	8.956	139.62	3.537	27.71	1.596	1.128
1.550	120.184	123.189	9.743	0.906	8.868	143.07	3.505	27.63	1.666	1.146
1.600	125.800	128.945	9.704	0.936	8.843	145.85	3.462	27.03	1.703	1.156
1.650	131.457	134.743	9.667	0.966	8.828	148.54	3.422	26.40	1.734	1.166
1.700	137.171	140.600	9.631	0.995	8.816	151.23	3.385	25.80	1.765	1.175
1.750	142.923	146.496	9.598	1.025	8.808	153.90	3.352	25.24	1.795	1.183
1.800	148.718	152.436	9.567	1.054	8.805	156.55	3.322	24.69	1.824	1.192
1.850	154.545	158.408	9.539	1.083	8.810	159.14	3.291	24.14	1.848	1.199
1.900	160.407	164.417	9.512	1.112	8.822	161.70	3.261	23.58	1.870	1.205
1.950	166.303	170.461	9.488	1.141	8.835	164.27	3.235	23.07	1.891	1.212
2.000	172.232	176.538	9.466	1.169	8.847	166.84	3.213	22.59	1.914	1.218
2.050	178.188	182.642	9.445	1.198	8.860	169.43	3.194	22.14	1.936	1.225
2.100	184.178	188.782	9.426	1.227	8.874	172.02	3.178	21.73	1.959	1.232
2.150	190.202	194.957	9.409	1.255	8.888	174.63	3.165	21.35	1.983	1.239
2.200	196.265	201.171	9.393	1.283	8.902	177.24	3.154	20.99	2.006	1.245

Note: Density of water = 1.025 tonnes/m³

Draft, VCB, KMt, KMI is from moulded base line.

LCB and LCF values are given relative to A.P. (Fr. No-2)

Trim is taken according to LBP (=19.76m)

K.B. SHIPPING & CO.	21.30M TUG BOAT	16-1011-701
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Trim = 0.25 m by forward:

Trim: 0.250 (m)										
Draft (m)	Vol. mould (m ³)	Displ. (tonnes)	LCB (m)	VCB (m)	LCF (m)	S (m ²)	KMt (m)	KMI (m)	MCT (t ^m /cm)	TpCm (t/cm)
0.500	24.713	25.331	10.158	0.307	10.406	76.20	7.815	40.33	0.513	0.745
0.550	28.404	29.114	10.191	0.336	10.404	79.36	7.164	37.91	0.554	0.768
0.600	32.200	33.005	10.215	0.365	10.394	82.38	6.618	35.84	0.593	0.789
0.650	36.094	36.996	10.234	0.394	10.379	85.28	6.159	34.04	0.630	0.808
0.700	40.076	41.078	10.247	0.423	10.358	88.09	5.770	32.44	0.666	0.825
0.750	44.139	45.242	10.256	0.451	10.334	90.84	5.439	31.03	0.700	0.841
0.800	48.279	49.486	10.262	0.479	10.308	93.56	5.154	29.81	0.734	0.857
0.850	52.494	53.806	10.264	0.507	10.281	96.27	4.907	28.74	0.769	0.871
0.900	56.779	58.198	10.265	0.536	10.250	98.96	4.692	27.81	0.803	0.886
0.950	61.136	62.664	10.262	0.564	10.217	101.66	4.504	27.00	0.838	0.900
1.000	65.561	67.200	10.258	0.592	10.180	104.36	4.339	26.29	0.874	0.914
1.050	70.054	71.806	10.252	0.620	10.140	107.08	4.194	25.67	0.910	0.928
1.100	74.615	76.481	10.244	0.648	10.096	109.83	4.066	25.13	0.948	0.942
1.150	79.244	81.225	10.234	0.676	10.048	112.61	3.953	24.68	0.987	0.956
1.200	83.943	86.042	10.222	0.704	9.996	115.43	3.854	24.29	1.027	0.970
1.250	88.713	90.931	10.208	0.732	9.938	118.30	3.766	23.99	1.070	0.985
1.300	93.556	95.895	10.192	0.760	9.873	121.25	3.689	23.76	1.116	1.000
1.350	98.473	100.935	10.175	0.788	9.800	124.30	3.622	23.62	1.166	1.016
1.400	103.471	106.058	10.155	0.816	9.716	127.47	3.564	23.59	1.222	1.034
1.450	108.558	111.272	10.132	0.845	9.618	130.83	3.514	23.69	1.287	1.052
1.500	113.745	116.588	10.106	0.873	9.496	134.48	3.472	24.01	1.365	1.074
1.550	119.042	122.018	10.075	0.902	9.334	138.60	3.439	24.66	1.467	1.101
1.600	124.491	127.603	10.038	0.931	9.136	143.18	3.415	25.64	1.596	1.131
1.650	130.066	133.317	9.997	0.961	9.017	146.99	3.395	25.94	1.686	1.154
1.700	135.733	139.126	9.954	0.990	8.959	150.17	3.372	25.76	1.744	1.169
1.750	141.455	144.991	9.914	1.019	8.944	152.91	3.342	25.26	1.779	1.178
1.800	147.225	150.906	9.876	1.048	8.934	155.62	3.315	24.77	1.812	1.188
1.850	153.045	156.871	9.840	1.077	8.927	158.33	3.290	24.30	1.844	1.197
1.900	158.907	162.880	9.806	1.106	8.924	161.01	3.268	23.85	1.875	1.206
1.950	164.806	168.926	9.774	1.135	8.926	163.68	3.247	23.41	1.904	1.214
2.000	170.741	175.009	9.745	1.164	8.937	166.28	3.225	22.94	1.929	1.221
2.050	176.715	181.133	9.718	1.193	8.951	168.88	3.205	22.49	1.952	1.228
2.100	182.719	187.287	9.693	1.221	8.966	171.49	3.188	22.06	1.975	1.235
2.150	188.760	193.479	9.670	1.250	8.981	174.10	3.175	21.67	2.000	1.242
2.200	194.830	199.701	9.649	1.278	8.994	176.72	3.163	21.29	2.023	1.248

Note: Density of water = 1.025 tonnes/m³

Draft, VCB, KMt, KMI is from moulded base line.

LCB and LCF values are given relative to A.P. (Fr. No-2)

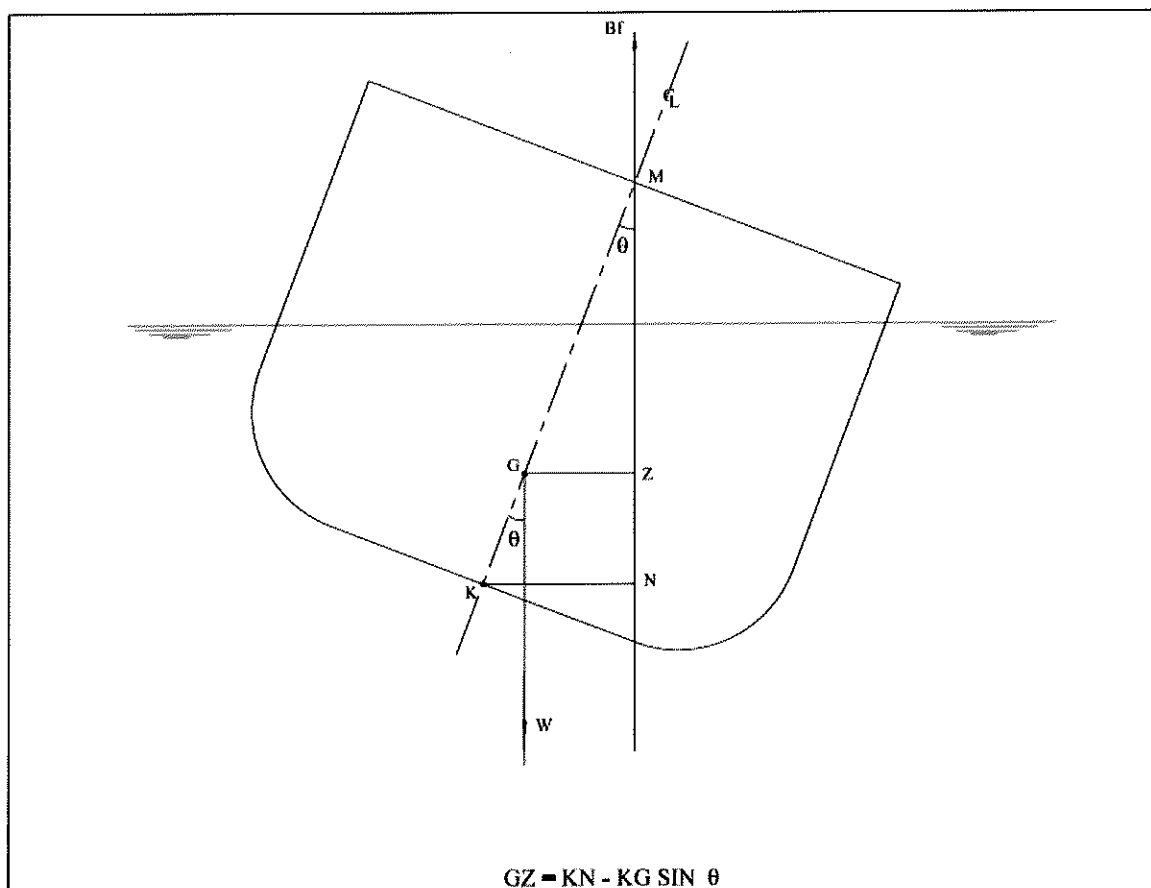
Trim is taken according to LBP (=19.76m)

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7 Cross curves of stability:

7.1 KN Concept diagram:

Volume considered for calculation of KN is depicted below:



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7.2 KN Values:

KN Values for Initial trim: 1.00 m by aft:

Displ. Deg.	0.0° Deg.	5.0° Deg.	10.0° Deg.	15.0° Deg.	20.0° Deg.	25.0° Deg.	30.0° Deg.	40.0° Deg.	50.0° Deg.	60.0° Deg.	70.0° Deg.
50.000	0.000	0.457	0.902	1.297	1.612	1.849	2.033	2.297	2.423	2.401	2.263
55.000	0.000	0.438	0.868	1.256	1.575	1.820	2.013	2.291	2.415	2.390	2.252
60.000	0.000	0.424	0.838	1.219	1.541	1.793	1.994	2.283	2.404	2.378	2.241
65.000	0.000	0.411	0.811	1.185	1.509	1.769	1.977	2.273	2.389	2.363	2.229
70.000	0.000	0.399	0.788	1.155	1.479	1.746	1.961	2.260	2.373	2.347	2.216
75.000	0.000	0.388	0.766	1.127	1.451	1.725	1.946	2.243	2.354	2.329	2.202
80.000	0.000	0.378	0.747	1.101	1.425	1.704	1.931	2.225	2.333	2.310	2.187
85.000	0.000	0.368	0.729	1.077	1.401	1.684	1.915	2.204	2.311	2.291	2.172
90.000	0.000	0.359	0.712	1.055	1.378	1.664	1.897	2.181	2.288	2.270	2.156
95.000	0.000	0.350	0.697	1.035	1.357	1.645	1.878	2.157	2.263	2.249	2.140
100.000	0.000	0.343	0.682	1.016	1.336	1.626	1.857	2.132	2.238	2.228	2.124
105.000	0.000	0.336	0.669	0.998	1.317	1.607	1.835	2.105	2.213	2.206	2.108
110.000	0.000	0.329	0.657	0.982	1.299	1.587	1.811	2.078	2.186	2.183	2.091
115.000	0.000	0.323	0.646	0.967	1.281	1.566	1.786	2.050	2.160	2.161	2.075
120.000	0.000	0.318	0.636	0.953	1.265	1.545	1.761	2.021	2.132	2.138	2.058
125.000	0.000	0.313	0.626	0.940	1.250	1.523	1.734	1.992	2.105	2.115	2.041
130.000	0.000	0.308	0.618	0.928	1.234	1.500	1.706	1.962	2.077	2.092	2.024
135.000	0.000	0.304	0.610	0.917	1.218	1.477	1.678	1.932	2.049	2.068	2.008
140.000	0.000	0.300	0.603	0.907	1.202	1.453	1.649	1.901	2.021	2.045	1.991
145.000	0.000	0.297	0.596	0.897	1.186	1.428	1.619	1.870	1.992	2.022	1.974
150.000	0.000	0.294	0.591	0.889	1.169	1.404	1.590	1.839	1.964	1.998	1.957
155.000	0.000	0.291	0.585	0.880	1.152	1.379	1.560	1.807	1.935	1.975	1.940
160.000	0.000	0.289	0.580	0.872	1.134	1.353	1.530	1.774	1.906	1.952	1.924
165.000	0.000	0.287	0.576	0.863	1.116	1.327	1.500	1.742	1.877	1.928	1.907
170.000	0.000	0.285	0.572	0.853	1.098	1.301	1.469	1.710	1.848	1.905	1.890
175.000	0.000	0.283	0.568	0.843	1.079	1.275	1.439	1.678	1.819	1.881	1.873
180.000	0.000	0.282	0.564	0.831	1.059	1.249	1.408	1.646	1.790	1.857	1.857
185.000	0.000	0.280	0.561	0.820	1.039	1.223	1.378	1.613	1.761	1.834	1.840
190.000	0.000	0.279	0.557	0.807	1.018	1.197	1.348	1.581	1.732	1.811	1.824
195.000	0.000	0.278	0.552	0.793	0.997	1.170	1.318	1.549	1.703	1.787	1.807
200.000	0.000	0.277	0.546	0.779	0.976	1.143	1.288	1.518	1.674	1.764	1.791



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KN Values for Initial trim: 0.75 m by aft:

Displ. Dag.	0.0° Deg.	5.0° Deg.	10.0° Deg.	15.0° Deg.	20.0° Deg.	25.0° Deg.	30.0° Deg.	40.0° Deg.	50.0° Deg.	60.0° Deg.	70.0° Deg.
50.000	0.000	0.454	0.893	1.285	1.599	1.834	2.018	2.282	2.413	2.393	2.257
55.000	0.000	0.433	0.857	1.244	1.562	1.805	1.997	2.277	2.405	2.384	2.247
60.000	0.000	0.415	0.825	1.206	1.527	1.778	1.977	2.270	2.395	2.371	2.236
65.000	0.000	0.401	0.798	1.172	1.495	1.753	1.960	2.260	2.381	2.357	2.224
70.000	0.000	0.389	0.775	1.141	1.465	1.729	1.943	2.247	2.365	2.341	2.211
75.000	0.000	0.379	0.754	1.114	1.437	1.708	1.928	2.232	2.347	2.325	2.198
80.000	0.000	0.371	0.735	1.088	1.411	1.688	1.914	2.216	2.328	2.307	2.184
85.000	0.000	0.362	0.719	1.065	1.388	1.669	1.900	2.197	2.308	2.288	2.170
90.000	0.000	0.355	0.703	1.044	1.365	1.650	1.885	2.177	2.286	2.269	2.155
95.000	0.000	0.347	0.689	1.024	1.344	1.632	1.869	2.155	2.263	2.249	2.140
100.000	0.000	0.340	0.676	1.006	1.325	1.615	1.851	2.131	2.239	2.229	2.124
105.000	0.000	0.333	0.664	0.989	1.306	1.597	1.832	2.107	2.215	2.208	2.109
110.000	0.000	0.327	0.652	0.974	1.289	1.580	1.811	2.082	2.190	2.186	2.093
115.000	0.000	0.321	0.642	0.960	1.272	1.562	1.788	2.055	2.164	2.164	2.077
120.000	0.000	0.316	0.632	0.946	1.257	1.543	1.765	2.028	2.138	2.142	2.061
125.000	0.000	0.311	0.623	0.934	1.242	1.524	1.740	2.000	2.111	2.120	2.044
130.000	0.000	0.307	0.615	0.923	1.229	1.503	1.714	1.971	2.085	2.098	2.028
135.000	0.000	0.303	0.607	0.913	1.215	1.482	1.687	1.942	2.058	2.075	2.012
140.000	0.000	0.299	0.600	0.903	1.201	1.460	1.659	1.913	2.031	2.052	1.995
145.000	0.000	0.296	0.594	0.894	1.187	1.437	1.631	1.883	2.003	2.030	1.979
150.000	0.000	0.293	0.588	0.885	1.173	1.413	1.603	1.853	1.976	2.007	1.962
155.000	0.000	0.290	0.583	0.878	1.158	1.390	1.574	1.822	1.948	1.984	1.946
160.000	0.000	0.288	0.578	0.870	1.142	1.366	1.545	1.791	1.920	1.962	1.929
165.000	0.000	0.286	0.574	0.863	1.126	1.341	1.516	1.760	1.892	1.939	1.913
170.000	0.000	0.284	0.570	0.856	1.109	1.317	1.486	1.728	1.864	1.916	1.897
175.000	0.000	0.282	0.566	0.848	1.091	1.292	1.457	1.697	1.836	1.893	1.881
180.000	0.000	0.281	0.563	0.839	1.073	1.267	1.427	1.665	1.808	1.870	1.864
185.000	0.000	0.279	0.560	0.829	1.054	1.241	1.398	1.634	1.779	1.848	1.848
190.000	0.000	0.278	0.557	0.818	1.035	1.216	1.368	1.602	1.751	1.825	1.832
195.000	0.000	0.277	0.555	0.807	1.015	1.190	1.339	1.571	1.722	1.802	1.817
200.000	0.000	0.276	0.551	0.794	0.995	1.164	1.310	1.539	1.694	1.779	1.801

K.B. SHIPPING & CO.	21.30M TUG BOAT	16-1011-701
7/21/2016	PRELIMINARY INTACT STABILITY BOOKLET	Rev No: 0

KN Values for Initial trim: 0.50 m by aft:

Displ. Deg.	0.0° Deg.	5.0° Deg.	10.0° Deg.	15.0° Deg.	20.0° Deg.	25.0° Deg.	30.0° Deg.	40.0° Deg.	50.0° Deg.	60.0° Deg.	70.0° Deg.
50.000	0.000	0.452	0.888	1.275	1.584	1.817	2.001	2.267	2.400	2.383	2.249
55.000	0.000	0.429	0.849	1.232	1.547	1.788	1.980	2.263	2.393	2.374	2.239
60.000	0.000	0.411	0.816	1.193	1.513	1.762	1.961	2.255	2.383	2.362	2.229
65.000	0.000	0.395	0.787	1.159	1.481	1.737	1.943	2.245	2.369	2.348	2.217
70.000	0.000	0.382	0.762	1.128	1.451	1.714	1.927	2.232	2.354	2.333	2.205
75.000	0.000	0.370	0.741	1.100	1.423	1.693	1.912	2.218	2.336	2.316	2.191
80.000	0.000	0.361	0.722	1.074	1.397	1.672	1.897	2.201	2.317	2.299	2.178
85.000	0.000	0.354	0.706	1.051	1.373	1.654	1.884	2.184	2.298	2.281	2.164
90.000	0.000	0.347	0.692	1.031	1.351	1.635	1.871	2.165	2.277	2.262	2.150
95.000	0.000	0.341	0.679	1.012	1.331	1.618	1.856	2.145	2.256	2.243	2.135
100.000	0.000	0.335	0.667	0.995	1.312	1.602	1.841	2.124	2.233	2.224	2.120
105.000	0.000	0.330	0.656	0.979	1.294	1.585	1.823	2.102	2.210	2.204	2.105
110.000	0.000	0.324	0.646	0.964	1.278	1.569	1.804	2.078	2.186	2.183	2.090
115.000	0.000	0.319	0.636	0.951	1.262	1.553	1.784	2.053	2.162	2.162	2.075
120.000	0.000	0.314	0.627	0.938	1.247	1.536	1.762	2.028	2.137	2.141	2.059
125.000	0.000	0.310	0.619	0.927	1.234	1.518	1.739	2.001	2.112	2.120	2.043
130.000	0.000	0.306	0.611	0.917	1.221	1.500	1.715	1.974	2.087	2.098	2.028
135.000	0.000	0.302	0.604	0.907	1.209	1.481	1.689	1.947	2.061	2.077	2.012
140.000	0.000	0.298	0.597	0.898	1.197	1.461	1.663	1.919	2.035	2.055	1.996
145.000	0.000	0.295	0.591	0.889	1.185	1.440	1.637	1.890	2.009	2.033	1.980
150.000	0.000	0.292	0.586	0.882	1.172	1.418	1.609	1.861	1.983	2.012	1.964
155.000	0.000	0.289	0.581	0.874	1.159	1.396	1.582	1.831	1.956	1.990	1.948
160.000	0.000	0.287	0.576	0.867	1.145	1.373	1.554	1.801	1.929	1.968	1.933
165.000	0.000	0.285	0.572	0.861	1.131	1.350	1.526	1.771	1.902	1.946	1.917
170.000	0.000	0.283	0.568	0.855	1.115	1.326	1.498	1.740	1.874	1.924	1.901
175.000	0.000	0.281	0.565	0.849	1.099	1.303	1.469	1.710	1.847	1.902	1.885
180.000	0.000	0.280	0.562	0.842	1.082	1.278	1.440	1.679	1.820	1.879	1.870
185.000	0.000	0.279	0.559	0.834	1.065	1.254	1.412	1.648	1.792	1.857	1.854
190.000	0.000	0.277	0.556	0.825	1.047	1.229	1.383	1.617	1.764	1.835	1.839
195.000	0.000	0.277	0.554	0.815	1.028	1.205	1.354	1.586	1.736	1.813	1.823
200.000	0.000	0.276	0.552	0.804	1.009	1.180	1.326	1.555	1.709	1.791	1.808

K.B. SHIPPING & CO.	21.30M TUG BOAT	16-1011-701
7/21/2016	PRELIMINARY INTACT STABILITY BOOKLET	Rev No: 0

KN Values for Initial trim: 0.25 m by aft:

Displ. Deg.	0.0° Deg.	5.0° Deg.	10.0° Deg.	15.0° Deg.	20.0° Deg.	25.0° Deg.	30.0° Deg.	40.0° Deg.	50.0° Deg.	60.0° Deg.	70.0° Deg.
50.000	0.000	0.450	0.884	1.267	1.570	1.800	1.984	2.253	2.387	2.371	2.239
55.000	0.000	0.427	0.844	1.222	1.533	1.772	1.963	2.248	2.379	2.362	2.230
60.000	0.000	0.407	0.809	1.181	1.499	1.746	1.944	2.240	2.368	2.350	2.220
65.000	0.000	0.391	0.779	1.146	1.467	1.722	1.927	2.229	2.355	2.337	2.208
70.000	0.000	0.377	0.753	1.114	1.437	1.699	1.912	2.217	2.339	2.321	2.196
75.000	0.000	0.365	0.730	1.086	1.409	1.678	1.897	2.202	2.322	2.305	2.183
80.000	0.000	0.355	0.711	1.061	1.383	1.658	1.883	2.186	2.304	2.288	2.169
85.000	0.000	0.346	0.694	1.038	1.359	1.640	1.870	2.168	2.284	2.270	2.155
90.000	0.000	0.339	0.680	1.018	1.337	1.622	1.856	2.150	2.264	2.251	2.141
95.000	0.000	0.333	0.667	0.999	1.317	1.604	1.842	2.131	2.243	2.233	2.127
100.000	0.000	0.328	0.656	0.982	1.298	1.588	1.826	2.111	2.222	2.214	2.112
105.000	0.000	0.324	0.646	0.967	1.281	1.572	1.810	2.090	2.200	2.195	2.098
110.000	0.000	0.320	0.637	0.954	1.266	1.557	1.792	2.068	2.178	2.175	2.083
115.000	0.000	0.315	0.629	0.941	1.251	1.542	1.774	2.045	2.155	2.155	2.069
120.000	0.000	0.311	0.621	0.929	1.237	1.526	1.754	2.021	2.132	2.136	2.054
125.000	0.000	0.308	0.613	0.919	1.224	1.509	1.732	1.997	2.108	2.115	2.039
130.000	0.000	0.304	0.607	0.909	1.212	1.492	1.709	1.971	2.084	2.095	2.024
135.000	0.000	0.300	0.600	0.900	1.201	1.475	1.686	1.945	2.060	2.075	2.009
140.000	0.000	0.297	0.594	0.892	1.190	1.456	1.661	1.918	2.035	2.054	1.994
145.000	0.000	0.294	0.589	0.884	1.179	1.437	1.636	1.891	2.010	2.033	1.978
150.000	0.000	0.291	0.584	0.877	1.168	1.417	1.610	1.863	1.984	2.012	1.963
155.000	0.000	0.289	0.579	0.871	1.156	1.396	1.584	1.834	1.958	1.991	1.948
160.000	0.000	0.287	0.575	0.864	1.143	1.375	1.557	1.805	1.932	1.970	1.933
165.000	0.000	0.285	0.571	0.859	1.130	1.353	1.530	1.776	1.906	1.949	1.918
170.000	0.000	0.283	0.567	0.853	1.116	1.330	1.503	1.746	1.880	1.927	1.903
175.000	0.000	0.281	0.564	0.848	1.102	1.308	1.476	1.717	1.853	1.906	1.888
180.000	0.000	0.280	0.561	0.842	1.086	1.284	1.448	1.686	1.826	1.885	1.873
185.000	0.000	0.278	0.559	0.835	1.070	1.261	1.420	1.656	1.799	1.863	1.858
190.000	0.000	0.277	0.556	0.828	1.053	1.237	1.392	1.626	1.773	1.841	1.843
195.000	0.000	0.276	0.554	0.819	1.035	1.213	1.364	1.596	1.745	1.820	1.827
200.000	0.000	0.276	0.552	0.810	1.017	1.189	1.336	1.566	1.718	1.798	1.812

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KN Values for Initial trim: 0.00 m

Displ. Deg.	0.0° Deg.	5.0° Deg.	10.0° Deg.	15.0° Deg.	20.0° Deg.	25.0° Deg.	30.0° Deg.	40.0° Deg.	50.0° Deg.	60.0° Deg.	70.0° Deg.
50.000	0.000	0.448	0.879	1.259	1.557	1.785	1.967	2.239	2.374	2.358	2.228
55.000	0.000	0.424	0.839	1.213	1.520	1.756	1.947	2.233	2.365	2.348	2.220
60.000	0.000	0.405	0.803	1.172	1.485	1.730	1.929	2.224	2.353	2.336	2.209
65.000	0.000	0.388	0.772	1.135	1.453	1.707	1.912	2.213	2.339	2.323	2.198
70.000	0.000	0.374	0.746	1.103	1.423	1.685	1.897	2.200	2.323	2.308	2.186
75.000	0.000	0.361	0.723	1.075	1.395	1.664	1.883	2.185	2.305	2.291	2.173
80.000	0.000	0.351	0.702	1.049	1.369	1.645	1.869	2.169	2.287	2.274	2.159
85.000	0.000	0.342	0.685	1.026	1.346	1.626	1.856	2.152	2.268	2.257	2.145
90.000	0.000	0.334	0.670	1.006	1.324	1.609	1.842	2.134	2.249	2.238	2.131
95.000	0.000	0.327	0.657	0.987	1.304	1.592	1.828	2.115	2.228	2.220	2.117
100.000	0.000	0.322	0.646	0.971	1.286	1.575	1.812	2.095	2.207	2.201	2.102
105.000	0.000	0.317	0.636	0.956	1.269	1.559	1.796	2.075	2.186	2.182	2.088
110.000	0.000	0.313	0.627	0.942	1.254	1.544	1.778	2.054	2.165	2.164	2.074
115.000	0.000	0.310	0.620	0.931	1.240	1.529	1.759	2.032	2.143	2.145	2.060
120.000	0.000	0.307	0.613	0.920	1.227	1.513	1.740	2.009	2.121	2.126	2.046
125.000	0.000	0.304	0.607	0.910	1.215	1.498	1.720	1.986	2.099	2.107	2.032
130.000	0.000	0.301	0.601	0.901	1.203	1.482	1.698	1.962	2.076	2.088	2.018
135.000	0.000	0.298	0.595	0.893	1.192	1.465	1.676	1.937	2.053	2.068	2.003
140.000	0.000	0.296	0.590	0.886	1.182	1.447	1.653	1.912	2.029	2.049	1.989
145.000	0.000	0.293	0.586	0.879	1.172	1.429	1.629	1.885	2.005	2.029	1.975
150.000	0.000	0.291	0.581	0.872	1.161	1.410	1.604	1.859	1.980	2.009	1.960
155.000	0.000	0.288	0.577	0.866	1.150	1.390	1.579	1.831	1.956	1.989	1.946
160.000	0.000	0.286	0.573	0.861	1.138	1.370	1.554	1.803	1.931	1.968	1.931
165.000	0.000	0.284	0.570	0.856	1.126	1.350	1.529	1.775	1.905	1.948	1.917
170.000	0.000	0.283	0.566	0.851	1.113	1.328	1.502	1.746	1.880	1.927	1.902
175.000	0.000	0.281	0.564	0.846	1.099	1.307	1.476	1.717	1.854	1.906	1.887
180.000	0.000	0.280	0.561	0.840	1.085	1.285	1.449	1.688	1.828	1.885	1.873
185.000	0.000	0.279	0.558	0.834	1.070	1.262	1.422	1.658	1.801	1.864	1.858
190.000	0.000	0.277	0.556	0.827	1.054	1.239	1.395	1.629	1.775	1.843	1.843
195.000	0.000	0.277	0.554	0.819	1.037	1.216	1.367	1.599	1.748	1.822	1.828
200.000	0.000	0.276	0.552	0.810	1.020	1.193	1.340	1.570	1.722	1.801	1.814

K.B. SHIPPING & CO.	21.30M TUG BOAT	16-1011-701
7/21/2016	PRELIMINARY INTACT STABILITY BOOKLET	Rev No: 0

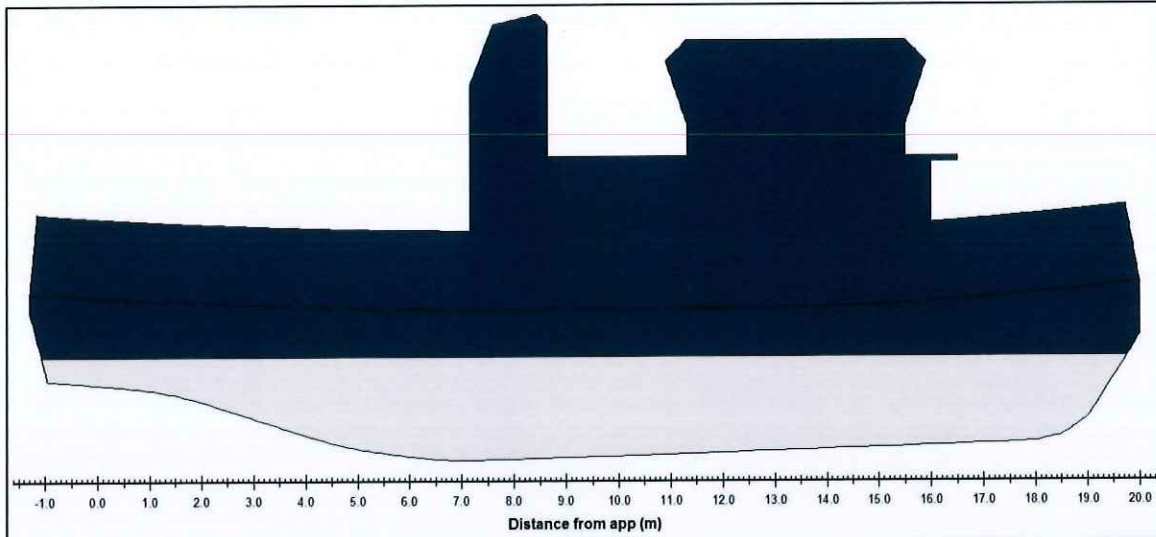
KN Values for Initial trim: 0.25 m by forward:

Displ. Deg.	0.0° Deg.	5.0° Deg.	10.0° Deg.	15.0° Deg.	20.0° Deg.	25.0° Deg.	30.0° Deg.	40.0° Deg.	50.0° Deg.	60.0° Deg.	70.0° Deg.
50.000	0.000	0.445	0.873	1.251	1.545	1.770	1.952	2.226	2.360	2.345	2.217
55.000	0.000	0.422	0.833	1.204	1.508	1.742	1.932	2.218	2.350	2.334	2.208
60.000	0.000	0.402	0.798	1.163	1.473	1.716	1.914	2.209	2.337	2.322	2.197
65.000	0.000	0.385	0.767	1.126	1.441	1.693	1.898	2.197	2.322	2.307	2.186
70.000	0.000	0.371	0.740	1.094	1.411	1.671	1.883	2.184	2.305	2.292	2.174
75.000	0.000	0.358	0.716	1.065	1.383	1.651	1.869	2.169	2.288	2.276	2.161
80.000	0.000	0.348	0.696	1.040	1.357	1.632	1.856	2.153	2.270	2.259	2.148
85.000	0.000	0.338	0.678	1.016	1.333	1.614	1.843	2.135	2.251	2.241	2.134
90.000	0.000	0.330	0.662	0.996	1.312	1.596	1.829	2.117	2.232	2.224	2.120
95.000	0.000	0.323	0.649	0.977	1.293	1.580	1.814	2.098	2.212	2.206	2.106
100.000	0.000	0.317	0.637	0.960	1.275	1.563	1.798	2.078	2.192	2.188	2.091
105.000	0.000	0.312	0.627	0.946	1.259	1.547	1.781	2.058	2.171	2.169	2.077
110.000	0.000	0.308	0.619	0.933	1.244	1.532	1.763	2.037	2.150	2.151	2.064
115.000	0.000	0.304	0.611	0.921	1.230	1.516	1.744	2.015	2.129	2.132	2.050
120.000	0.000	0.301	0.605	0.911	1.217	1.500	1.724	1.993	2.107	2.114	2.036
125.000	0.000	0.299	0.599	0.901	1.205	1.485	1.704	1.971	2.085	2.095	2.022
130.000	0.000	0.297	0.594	0.893	1.194	1.468	1.683	1.947	2.063	2.077	2.009
135.000	0.000	0.295	0.590	0.886	1.183	1.452	1.661	1.924	2.041	2.058	1.995
140.000	0.000	0.293	0.586	0.879	1.173	1.435	1.639	1.899	2.018	2.039	1.981
145.000	0.000	0.291	0.582	0.873	1.163	1.417	1.616	1.874	1.995	2.020	1.968
150.000	0.000	0.289	0.578	0.867	1.152	1.399	1.593	1.848	1.971	2.001	1.954
155.000	0.000	0.288	0.575	0.862	1.141	1.380	1.569	1.822	1.947	1.982	1.940
160.000	0.000	0.286	0.572	0.857	1.130	1.361	1.545	1.795	1.923	1.962	1.926
165.000	0.000	0.284	0.569	0.852	1.118	1.341	1.521	1.768	1.899	1.942	1.912
170.000	0.000	0.283	0.566	0.848	1.106	1.320	1.496	1.740	1.874	1.922	1.898
175.000	0.000	0.281	0.563	0.843	1.093	1.299	1.470	1.712	1.849	1.902	1.883
180.000	0.000	0.280	0.561	0.838	1.079	1.278	1.444	1.683	1.823	1.881	1.869
185.000	0.000	0.279	0.558	0.831	1.064	1.257	1.418	1.654	1.798	1.861	1.855
190.000	0.000	0.278	0.556	0.824	1.049	1.235	1.391	1.626	1.772	1.840	1.840
195.000	0.000	0.277	0.554	0.816	1.034	1.213	1.364	1.597	1.746	1.819	1.826
200.000	0.000	0.276	0.552	0.807	1.017	1.191	1.338	1.568	1.719	1.798	1.811



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8 Wind Moments:

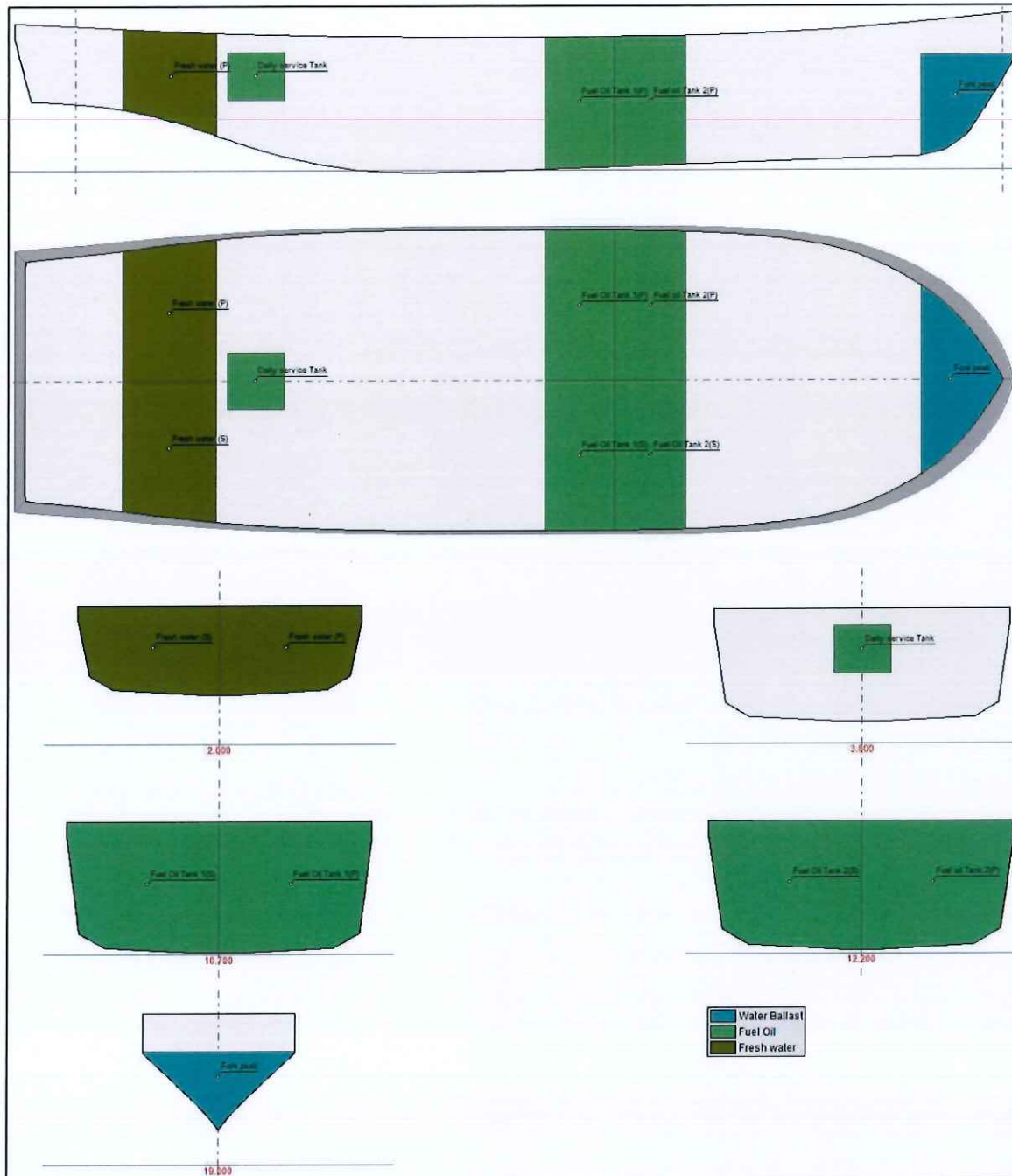


Wind pressure: 51.40 (kg/m ²) Bilge type: Sharp bilges								
Draft (m)	Lateral area (m ²)	LCE (m)	VCE (m)	Wind area (m ²)	LCE (m)	VCE (m)	Wind lever (m)	Wind moment (Tonnes*m)
0.700	8.766	10.747	0.397	101.17	10.136	3.469	3.071	15.970
0.750	9.302	10.874	0.416	100.37	10.129	3.490	3.074	15.860
0.800	10.339	10.795	0.451	99.561	10.122	3.512	3.061	15.667
0.850	11.060	10.837	0.475	98.744	10.116	3.535	3.060	15.529
0.900	11.780	10.875	0.499	97.919	10.110	3.557	3.058	15.392
0.950	12.849	10.793	0.534	97.084	10.104	3.580	3.046	15.200
1.000	13.694	10.788	0.561	96.238	10.099	3.603	3.042	15.046
1.050	14.440	10.813	0.585	95.383	10.094	3.626	3.041	14.909
1.100	15.336	10.791	0.613	94.519	10.089	3.649	3.036	14.750
1.150	16.289	10.754	0.643	93.643	10.086	3.673	3.030	14.583
1.200	17.176	10.736	0.670	92.756	10.082	3.697	3.026	14.428
1.250	17.990	10.740	0.695	91.855	10.080	3.721	3.025	14.284
1.300	18.996	10.689	0.726	90.937	10.079	3.745	3.020	14.114
1.350	19.903	10.665	0.753	89.995	10.080	3.771	3.018	13.959
1.400	20.907	10.612	0.783	89.025	10.084	3.797	3.014	13.791
1.450	21.910	10.558	0.812	87.603	10.120	3.835	3.023	13.611
1.500	22.933	10.500	0.842	87.000	10.101	3.852	3.010	13.458
1.550	23.957	10.447	0.871	85.975	10.111	3.879	3.008	13.293
1.600	24.984	10.400	0.900	84.948	10.121	3.907	3.007	13.130
1.650	26.013	10.356	0.929	83.919	10.131	3.935	3.006	12.968
1.700	27.045	10.316	0.957	82.888	10.142	3.963	3.006	12.807
1.750	28.078	10.279	0.985	81.854	10.152	3.991	3.006	12.647
1.800	29.112	10.245	1.014	80.817	10.163	4.020	3.006	12.489
1.850	30.153	10.214	1.042	79.780	10.174	4.048	3.007	12.330
1.900	31.193	10.185	1.069	78.740	10.185	4.077	3.008	12.173
1.950	32.235	10.157	1.097	77.697	10.196	4.106	3.009	12.017
2.000	33.278	10.132	1.125	76.654	10.207	4.135	3.011	11.862



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9 Tank Capacities



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Fuel Oil							
Tank description	Relative density	Volume (m ³)	Weight (tonnes)	VCG (m)	LCG (m)	TCG (m)	Max FSM (Tonnes*m)
Fuel Oil Tank 1(P)	0.850	12.299	10.454	1.481	10.747	1.536 (PS)	3.573
Fuel Oil Tank 1(S)	0.850	12.299	10.454	1.481	10.747	-1.536 (SB)	3.573
Fuel oil Tank 2(P)	0.850	12.011	10.210	1.518	12.247	1.525 (PS)	3.569
Fuel Oil Tank 2(S)	0.850	12.012	10.210	1.518	12.247	-1.525 (SB)	3.569
Daily service Tank	0.850	1.411	1.200	1.987	3.850	0.000 (CL)	0.144
Total		50.032	42.527	1.513	11.273	0.000 (CL)	14.428

Fresh water							
Tank description	Relative density	Volume (m ³)	Weight (tonnes)	VCG (m)	LCG (m)	TCG (m)	Max FSM (Tonnes*m)
Fresh water (P)	1.000	10.304	10.304	2.028	2.051	1.421 (PS)	4.517
Fresh water (S)	1.000	10.304	10.304	2.028	2.051	-1.421 (SB)	4.517
Total		20.608	20.608	2.028	2.051	0.000 (CL)	9.035

Water Ballast							
Tank description	Relative density	Volume (m ³)	Weight (tonnes)	VCG (m)	LCG (m)	TCG (m)	Max FSM (Tonnes*m)
Fore peak	1.025	5.300	5.432	1.768	18.604	0.000 (CL)	6.582
Total		5.300	5.432	1.768	18.604	0.000 (CL)	6.582

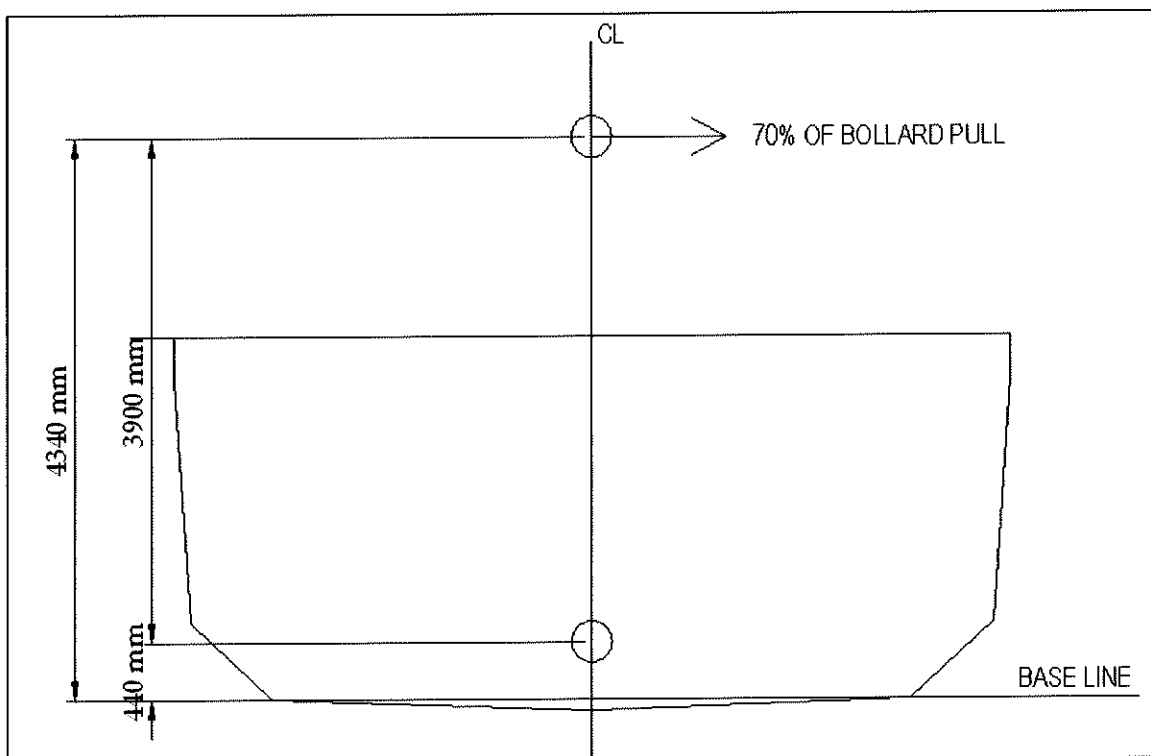
K.B. SHIPPING & CO.	21.30M TUG BOAT	16-1011-701
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10 Bollard Pull:

Maximum Bollard pull = 10.00 Ton

Lever above propeller axis = 03.90 m

Heeling Moment ($0.7 \cdot T \cdot H$) = 27.30 T-m



K.B. SHIPPING & CO.	21.30M TUG BOAT	16-1011-701
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11 Estimation of Lightship:

Lightship weight is estimated as 94.00 tonnes approximately. The LCG is estimated at 9.20 m (0.45L) ahead of A.P. The VCG is estimated at 2.40m above baseline as conservative.

12 Intact stability:

For the purpose of Intact Stability Calculations, the following Loading Conditions are considered.

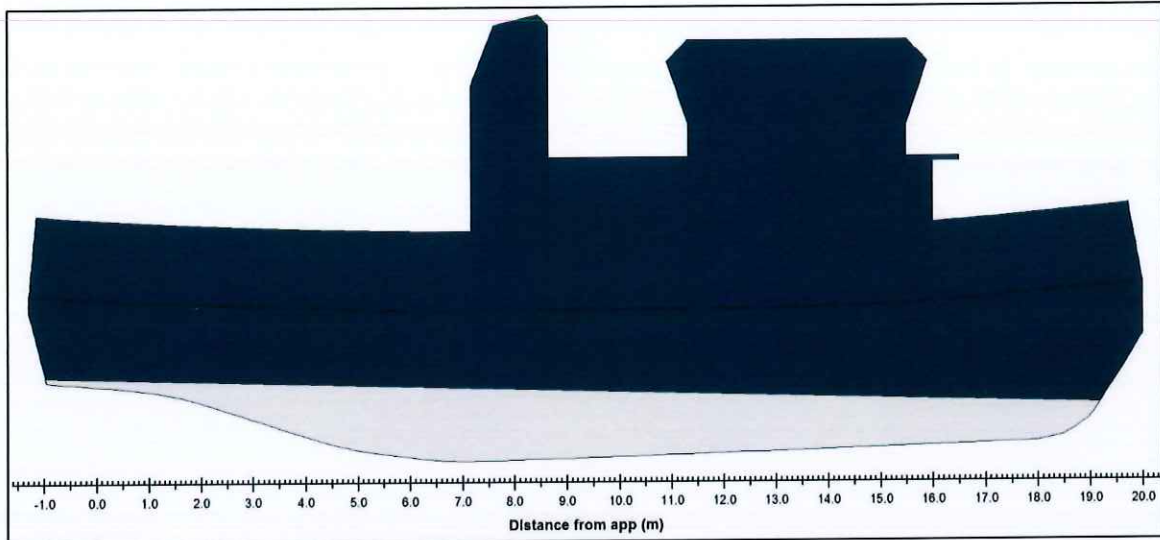
1. Lightship condition
2. Fully Loaded Departure condition
3. Fully Loaded Arrival condition
4. Ballast Departure condition
5. Ballast Arrival condition



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12.1 Lightship condition

The below load case for the Intact Stability Analysis has been done by considering the Lightship weight of the vessel.

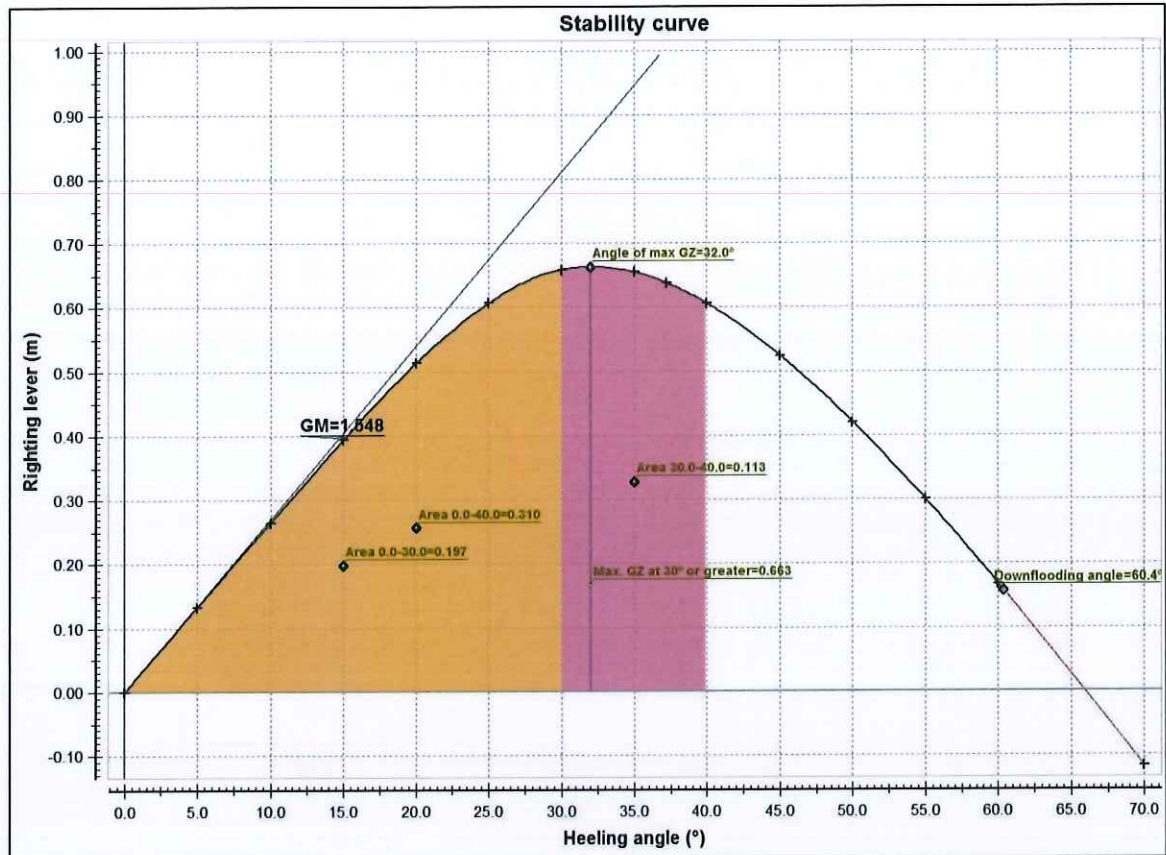


Hydrostatic particulars			
List	0.00 (CL)	(Degr.)	KM 3.948 (m)
Draft aft pp	1.524	(m)	GM solid 1.548 (m)
Mean draft	1.261	(m)	GG' 0.000 (m)
Draft forward pp	0.997	(m)	GM liquid 1.548 (m)
Trim	-0.526	(m)	MCT 1.515 (t*m/cm)
Volume	91.705	(m ³)	Relative water density 1.0250
LCF	8.635	(m)	Immersion rate 1.101 (t/cm)

GZ values	
Heeling angle (Degr.)	GZ (m)
0.00 (CL)	0.000
5.00 (PS)	0.133
10.00 (PS)	0.265
15.00 (PS)	0.394
20.00 (PS)	0.514
25.00 (PS)	0.607
30.00 (PS)	0.659
40.00 (PS)	0.606
50.00 (PS)	0.421
60.00 (PS)	0.168
70.00 (PS)	-0.118



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Description	Weight (tonnes)	VCG (m)	LCG (m)	TCG (m)	FSM (Tonnes*m)
Lightship	94.000	2.400	9.200	0.000 (CL)	
Deadweight	0.000	0.000	0.000	0.000 (CL)	0.000
Displacement	94.000	2.400	9.200	0.000 (CL)	0.000

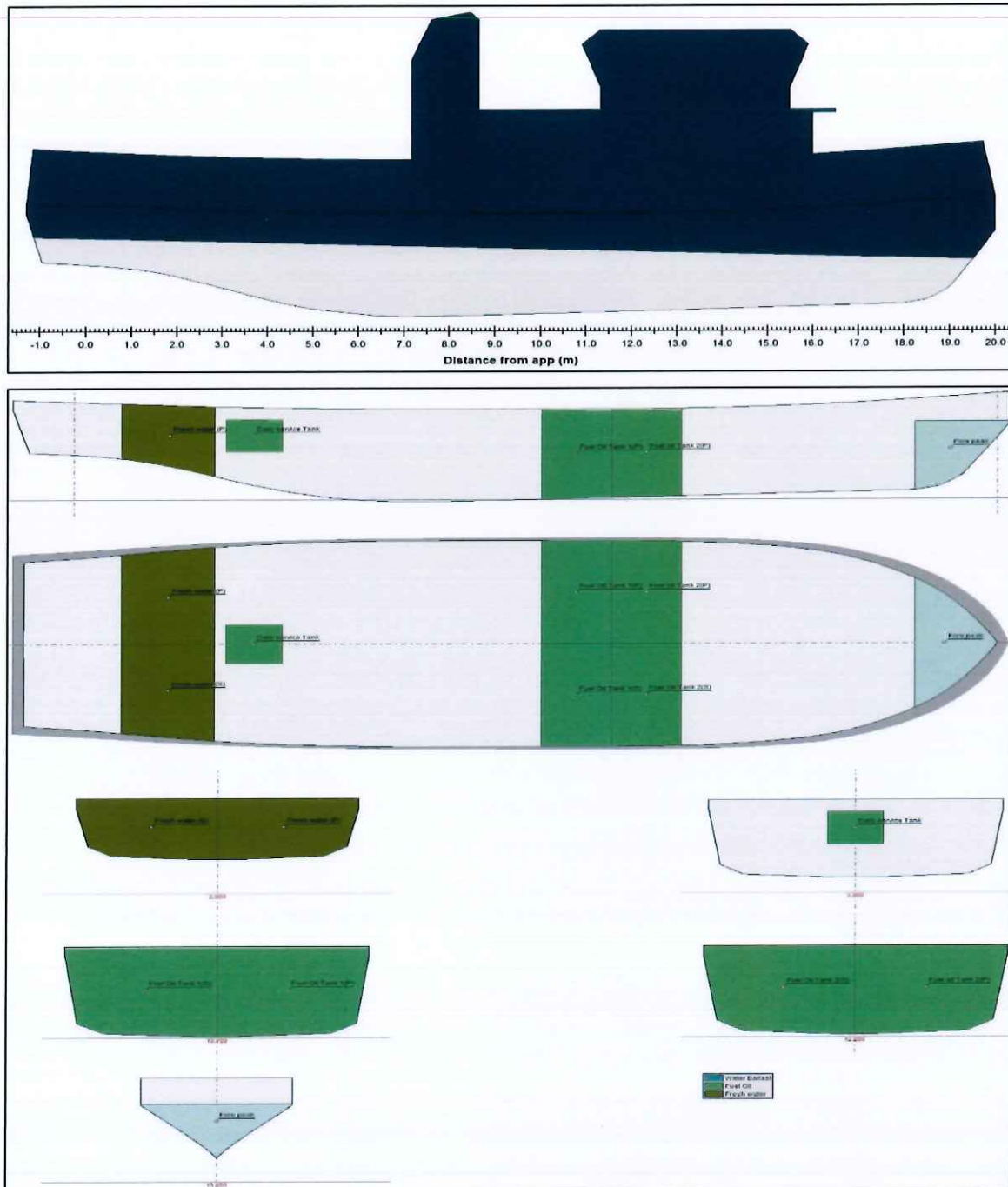
Critical points			
Description	Type	Dist. to wl (m)	Submersion angle (Degr.)
Engine Room ventilation sup/Exhaust	Downflooding	2.247	60.37

Evaluation of criteria						
Minimum design criteria applicable to Tugs						
Description	Attained value		Criterion	Required value		Complies
Area 0° - 30°	0.1966	(mrad)	>=	0.0550	(mrad)	YES
Area 0° - 40°	0.3095	(mrad)	>=	0.0900	(mrad)	YES
Area 30° - 40°	0.1130	(mrad)	>=	0.0300	(mrad)	YES
Max. GZ at 30° or greater	0.663	(m)	>=	0.200	(m)	YES
Angle of max GZ	32.0	(Degr.)	>=	25.0	(Degr.)	YES
Initial metacentric height	1.548	(m)	>	0.150	(m)	YES
The vessel complies with the stability criteria						

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12.2 Fully Loaded Departure condition

The below load case gives the Intact Stability Analysis of the vessel loaded with consumables at their maximum capacity and other miscellaneous items.



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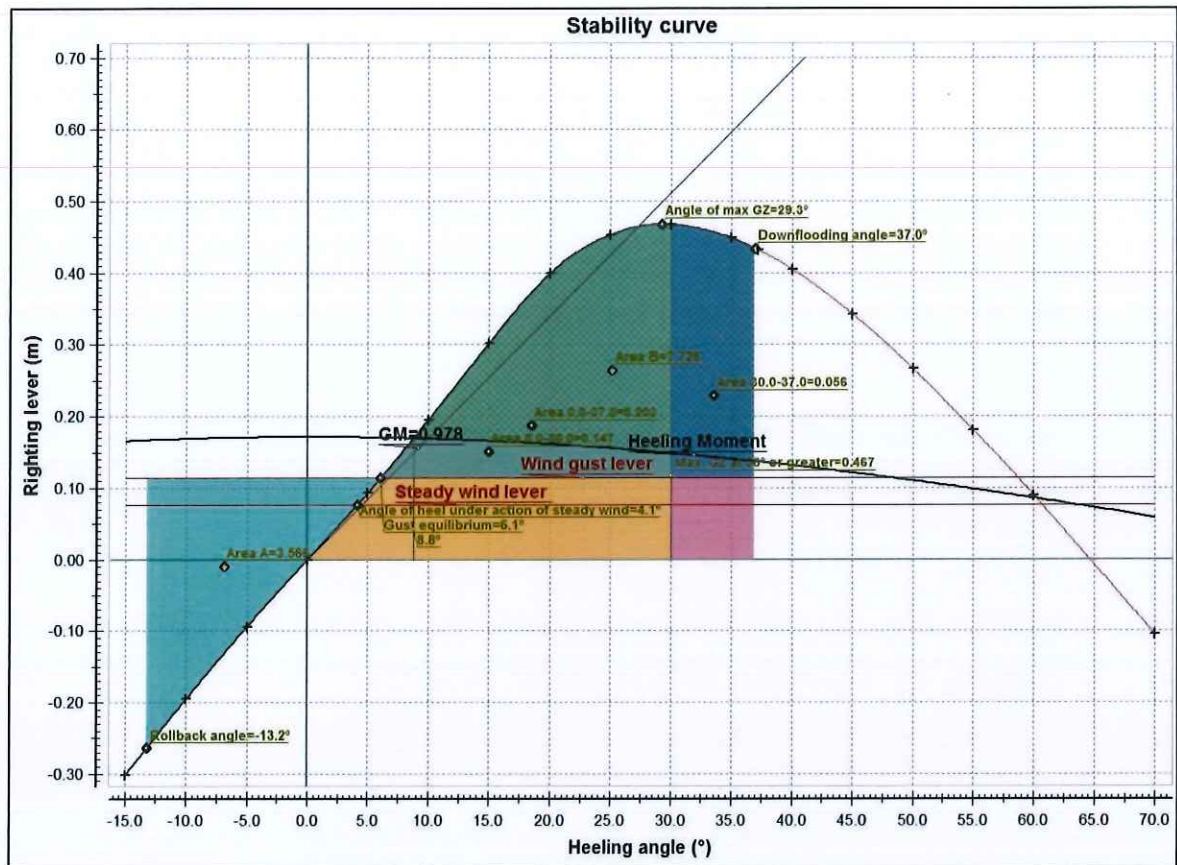
Hydrostatic particulars					
List	0.00 (CL)	(Degr.)	KM	3.291	(m)
Draft aft pp	2.122	(m)	GM solid	1.126	(m)
Mean draft	1.821	(m)	GG'	0.147	(m)
Draft forward pp	1.521	(m)	GM liquid	0.978	(m)
Trim	-0.601	(m)	MCT	1.818	(t*m/cm)
Volume	155.38	(m^3)	Relative water density	1.0250	
LCF	8.609	(m)	Immersion rate	1.193	(t/cm)

GZ values	
Heeling angle (Degr.)	GZ (m)
0.00 (CL)	0.000
5.00 (PS)	0.094
10.00 (PS)	0.195
15.00 (PS)	0.302
20.00 (PS)	0.400
25.00 (PS)	0.453
30.00 (PS)	0.467
35.00 (PS)	0.449
37.20 (PS)	0.432
40.00 (PS)	0.405
45.00 (PS)	0.342
50.00 (PS)	0.266
55.00 (PS)	0.181
60.00 (PS)	0.089
70.00 (PS)	-0.105

Description	Density (t/m^3)	Fill%	Weight (tonnes)	VCG (m)	LCG (m)	TCG (m)	FSM (Tonnes*m)
Water Ballast							
Fore peak	1.0250	0.0	0.000	0.000	0.000	0.000 (CL)	0.000
Miscellaneous							
Crew and Misc.			3.000	5.200	12.300	0.000 (CL)	
Fuel Oil							
Fuel Oil Tank 1(P)	0.8500	98.0	10.245	1.455	10.747	1.534 (PS)	3.573
Fuel Oil Tank 1(S)	0.8500	98.0	10.245	1.455	10.747	-1.534 (SB)	3.573
Fuel oil Tank 2(P)	0.8500	98.0	10.006	1.492	12.246	1.523 (PS)	3.569
Fuel Oil Tank 2(S)	0.8500	98.0	10.006	1.492	12.246	-1.523 (SB)	3.569
Daily service Tank	0.8500	98.0	1.176	1.977	3.850	0.000 (CL)	0.144
Totals for Fuel Oil			41.677	1.487	11.272	0.000 (CL)	14.428
Fresh water							
Fresh water (P)	1.0000	100.0	10.304	2.028	2.051	1.421 (PS)	4.517
Fresh water (S)	1.0000	100.0	10.304	2.028	2.051	-1.421 (SB)	4.518
Totals for Fresh water			20.608	2.028	2.051	0.000 (CL)	9.035
Lightship			94.000	2.400	9.200	0.000 (CL)	
Deadweight			65.284	1.829	8.409	0.000 (CL)	23.463
Displacement			159.284	2.166	8.876	0.000 (CL)	23.463



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Description	Critical points		
	Type	Dist. to wl (m)	Submersion angle (Degr.)
Engine Room ventilation sup/Exhaust	Downflooding	1.672	37.00
Cowl Vent 1	Downflooding	2.199	55.45
Engine Room Funnel	Downflooding	5.790	67.52
Sky light	Downflooding	1.533	55.90



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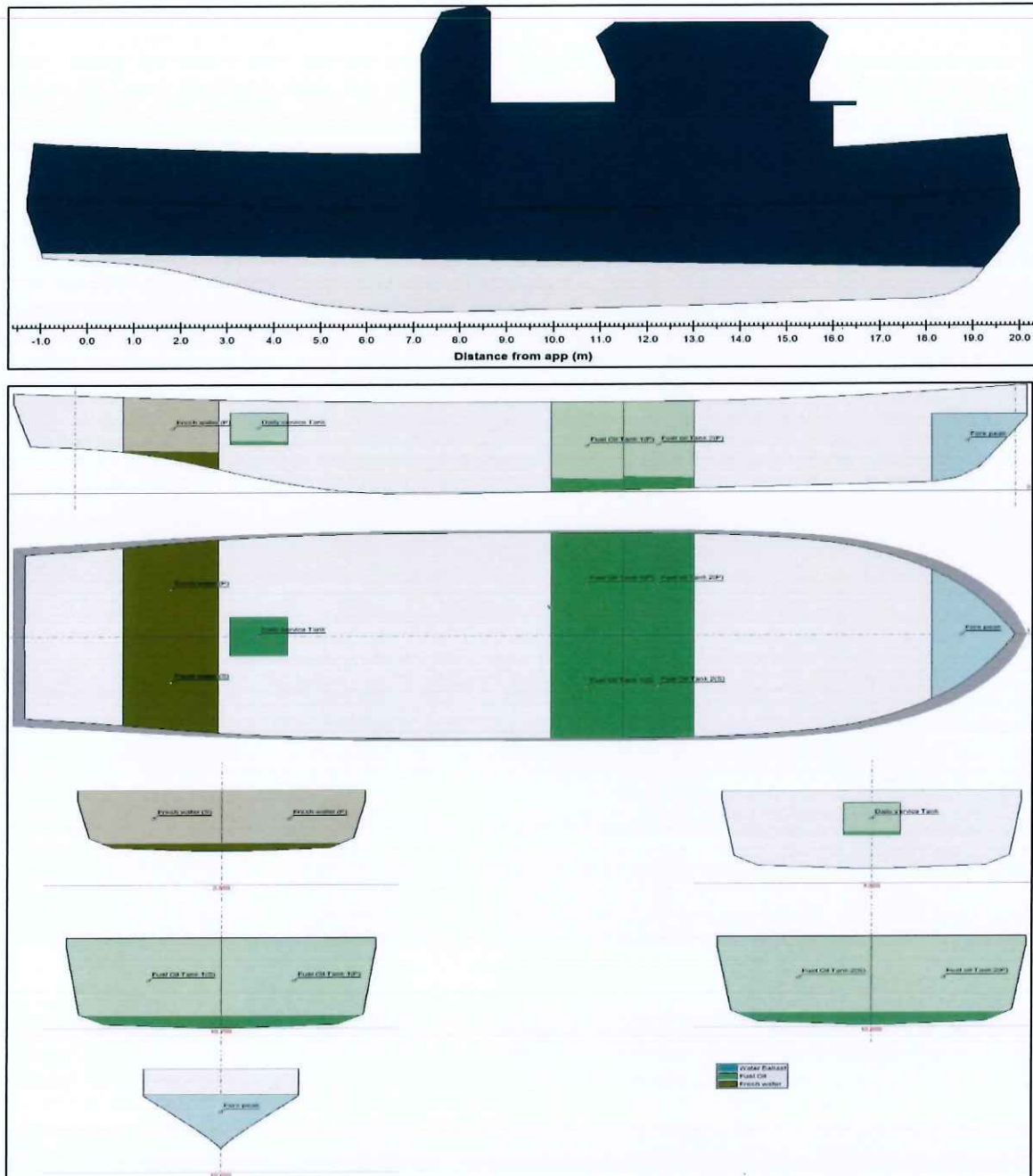
Evaluation of criteria						
Minimum design criteria applicable to Tugs						
Description	Attained value		Criterion	Required value		Complies
Area 0° - 30°	0.1471	(mrad)	>=	0.0550	(mrad)	YES
Area 0° - 40°	0.2027	(mrad)	>=	0.0900	(mrad)	YES
Area 30° - 40°	0.0556	(mrad)	>=	0.0300	(mrad)	YES
Max. GZ at 30° or greater	0.467	(m)	>=	0.200	(m)	YES
Angle of max GZ	29.3	(Degr.)	>=	25.0	(Degr.)	YES
Initial metacentric height	0.978	(m)	>	0.150	(m)	YES
Severe wind and rolling criterion						
Wind pressure	51.4	(kg/m^2)				
Wind area	79.98	(m^2)				
Steady wind lever	0.076	(m)				
Deck immersion angle	15.54	(Degr.)				
Wind gust lever	0.115	(m)				
Ratio of areaA/areaB	0.462		<=	1.000		YES
Maximum allowed static heeling angle	4.1	(Degr.)	<=	16.0	(Degr.)	YES
Max allowed ratio static angle/deck immersion angle	0.264		<=	0.800		YES
The vessel complies with the stability criteria						

Towing criteria						
Description	Attained value		Criterion	Required value		Complies
Heeling moment	27.30	(Tonnes*m)				
Area between heeling lever and righting lever calculated from the 1st intercept upto an angle 40° beyond the first intercept or upto the down flooding angle, whichever is less.	0.113	m-rad	>=	0.09	m-rad	YES
The ratio of area under a righting lever curve to the area under a heeling lever curve upto an angle of 40° or the downflooding angle, whichever is less	1.966	-	>=	1.4		YES

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12.3 Fully Loaded Arrival condition

The below load case gives the Intact Stability Analysis of the vessel loaded with 10% consumables and other miscellaneous items.



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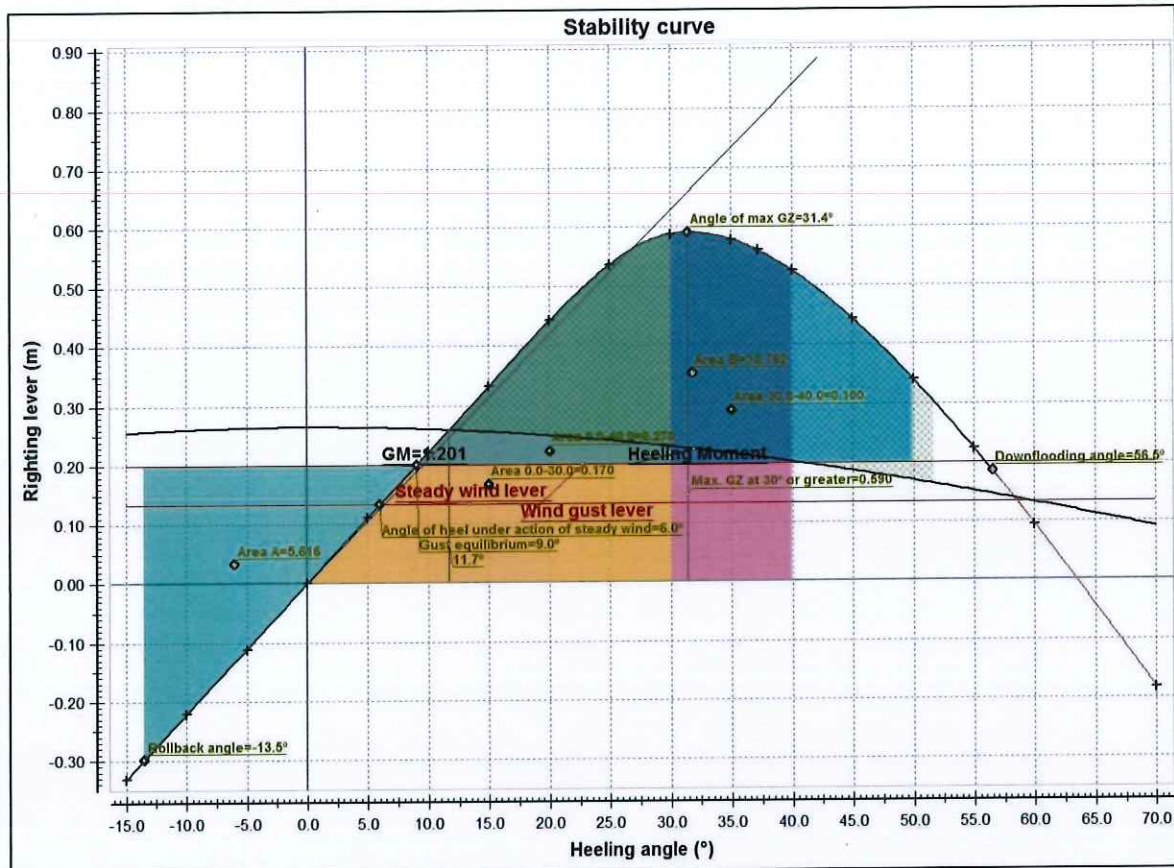
Hydrostatic particulars					
List	0.00 (CL)	(Degr.)	KM	3.797	(m)
Draft aft pp	1.581	(m)	GM solid	1.428	(m)
Mean draft	1.348	(m)	GG'	0.227	(m)
Draft forward pp	1.116	(m)	GM liquid	1.201	(m)
Trim	-0.465	(m)	MCT	1.581	(t*m/cm)
Volume	100.79	(m^3)	Relative water density	1.0250	
LCF	8.642	(m)	Immersion rate	1.120	(t/cm)

GZ values	
Heeling angle (Degr.)	GZ (m)
0.00 (CL)	0.000
5.00 (PS)	0.111
10.00 (PS)	0.221
15.00 (PS)	0.332
20.00 (PS)	0.443
25.00 (PS)	0.537
30.00 (PS)	0.588
35.00 (PS)	0.579
37.20 (PS)	0.560
40.00 (PS)	0.526
45.00 (PS)	0.443
50.00 (PS)	0.340
55.00 (PS)	0.223
60.00 (PS)	0.095
70.00 (PS)	-0.183

Description	Density (t/m^3)	Fill%	Weight (tonnes)	VCG (m)	LCG (m)	TCG (m)	FSM (Tonnes*m)
Fore peak	1.0250	0.0	Water Ballast 0.000	0.000	0.000	0.000 (CL)	0.000
Crew and Misc.			Miscellaneous 3.000	5.200	12.300	0.000 (CL)	
			Fuel Oil				
Fuel Oil Tank 1(P)	0.8500	10.0	1.045	0.242	10.726	1.250 (PS)	3.573
Fuel Oil Tank 1(S)	0.8500	10.0	1.045	0.242	10.726	-1.250 (SB)	3.573
Fuel oil Tank 2(P)	0.8500	10.0	1.021	0.294	12.222	1.216 (PS)	3.569
Fuel Oil Tank 2(S)	0.8500	10.0	1.021	0.294	12.222	-1.216 (SB)	3.569
Daily service Tank	0.8500	10.0	0.120	1.537	3.850	0.000 (CL)	0.144
Totals for Fuel Oil			4.253	0.303	11.250	0.000 (CL)	14.428
			Fresh water				
Fresh water (P)	1.0000	10.0	1.030	1.123	2.440	1.168 (PS)	4.517
Fresh water (S)	1.0000	10.0	1.030	1.123	2.440	-1.168 (SB)	4.518
Totals for Fresh water			2.060	1.123	2.440	0.000 (CL)	9.035
Lightship			94.000	2.400	9.200	0.000 (CL)	
Deadweight			9.313	2.062	9.640	0.000 (CL)	23.463
Displacement			103.313	2.370	9.240	0.000 (CL)	23.463



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Description	Critical points Type	Dist. to wl (m)	Submersion angle (Degr.)
Engine Room ventilation sup/Exhaust	Downflooding	2.172	56.53
Cowl Vent 1	Downflooding	2.628	68.51



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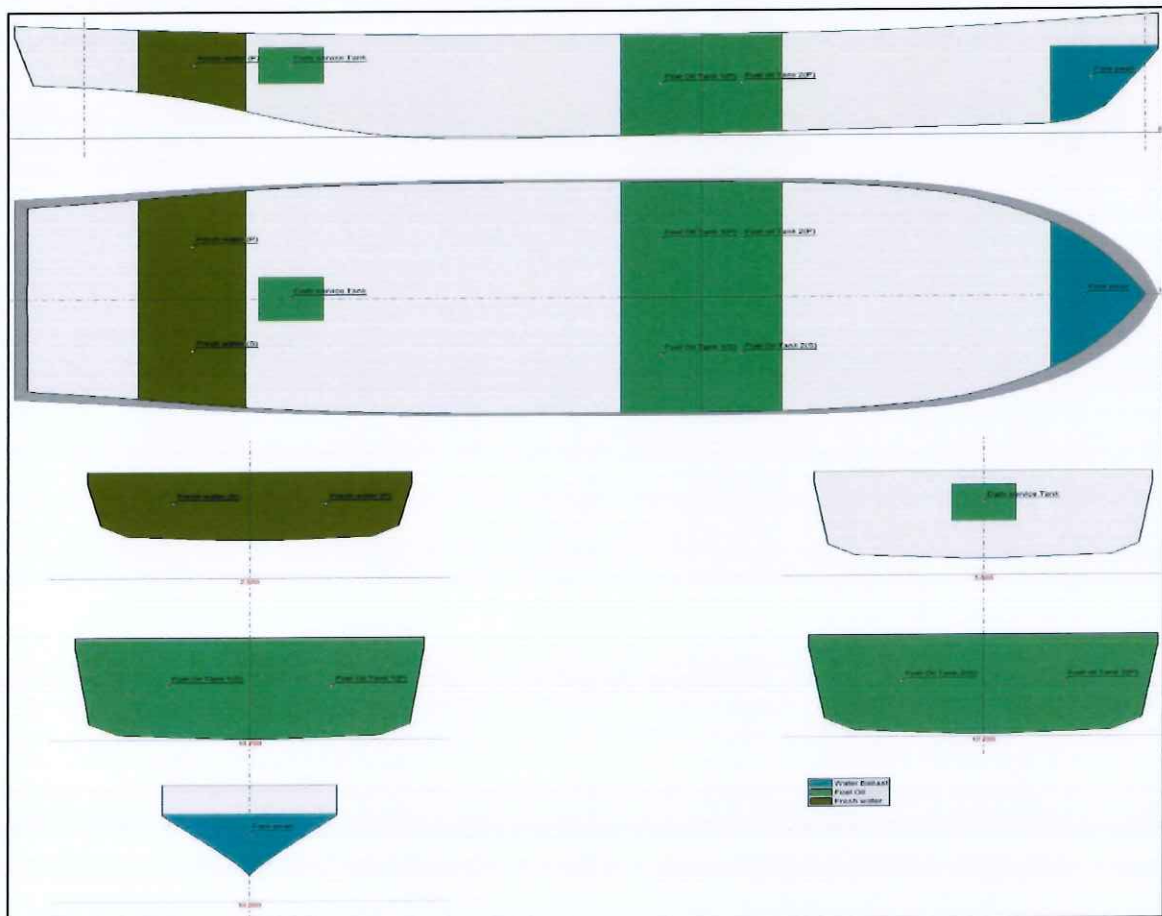
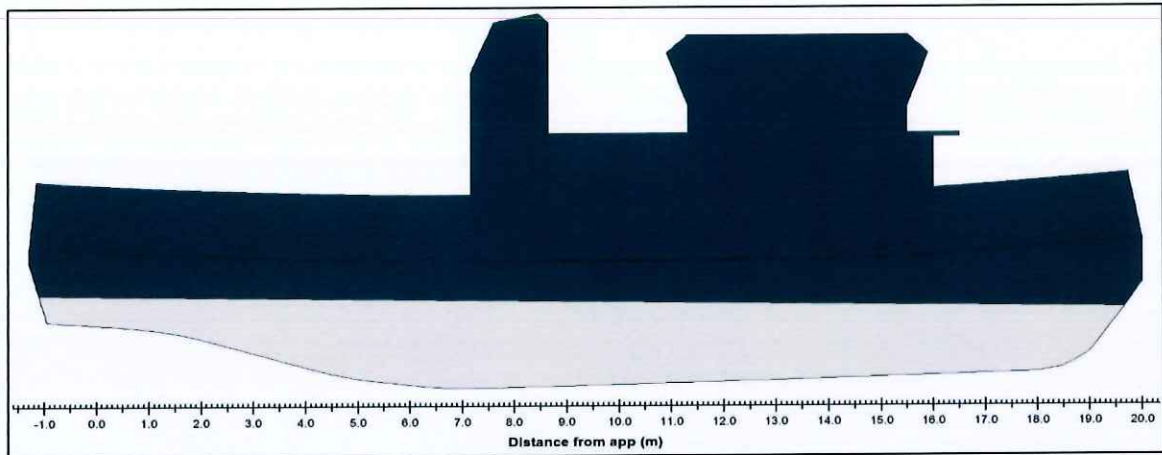
Evaluation of criteria						
Minimum design criteria applicable to Tugs						
Description	Attained value		Criterion	Required value		Complies
Area 0° - 30°	0.1698	(mrad)	>=	0.0550	(mrad)	YES
Area 0° - 40°	0.2696	(mrad)	>=	0.0900	(mrad)	YES
Area 30° - 40°	0.0997	(mrad)	>=	0.0300	(mrad)	YES
Max. GZ at 30° or greater	0.590	(m)	>=	0.200	(m)	YES
Angle of max GZ	31.4	(Degr.)	>=	25.0	(Degr.)	YES
Initial metacentric height	1.201	(m)	>	0.150	(m)	YES
Severe wind and rolling criterion						
Wind pressure	51.4	(kg/m ²)				YES
Wind area	89.76	(m ²)				
Steady wind lever	0.133	(m)				
Deck immersion angle	25.63	(Degr.)				
Wind gust lever	0.200	(m)				
Ratio of areaA/areaB	0.522		<=	1.000		YES
Maximum allowed static heeling angle	6.0	(Degr.)	<=	16.0	(Degr.)	YES
Max allowed ratio static angle/deck immersion angle	0.234		<=	0.800		YES
The vessel complies with the stability criteria						

Towing criteria						
Description	Attained value		Criterion	Required value		Complies
Heeling moment	27.30	(Tonnes*m)				
Area between heeling lever and righting lever calculated from the 1st intercept upto an angle 40° beyond the first intercept or upto the down flooding angle, whichever is less.	0.1756	m-rad	>=	0.09	m-rad	YES
The ratio of area under a righting lever curve to the area under a heeling lever curve upto an angle of 40° or the downflooding angle, whichever is less	1.587	-	>=	1.4		YES

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12.4 Ballast Departure condition

The below load case gives the Intact Stability Analysis of the vessel loaded with 100 %consumables and Ballast and other miscellaneous items



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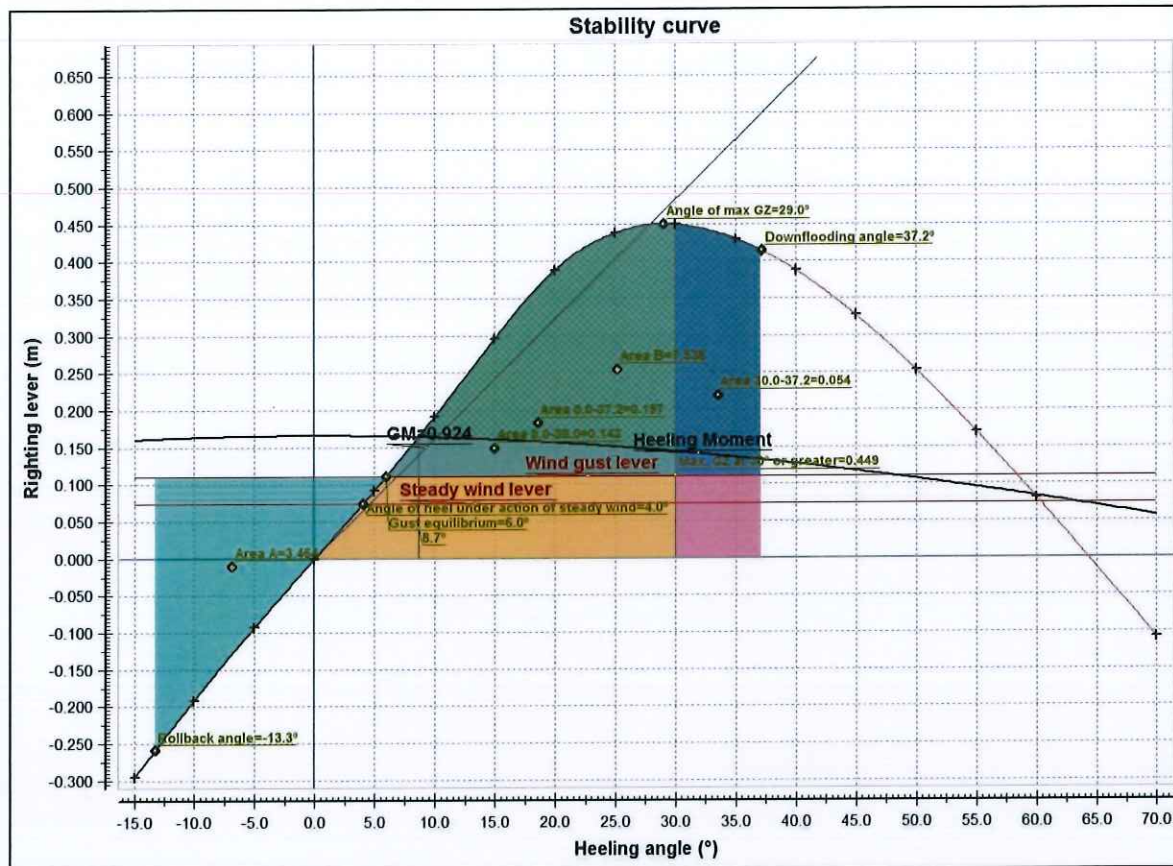
Hydrostatic particulars					
List	0.00 (CL)	(Degr.)	KM	3.259	(m)
Draft aft pp	2.032	(m)	GM solid	1.106	(m)
Mean draft	1.886	(m)	GG'	0.182	(m)
Draft forward pp	1.740	(m)	GM liquid	0.924	(m)
Trim	-0.291	(m)	MCT	1.851	(t*m/cm)
Volume	160.70	(m^3)	Relative water density	1.0250	
LCF	8.720	(m)	Immersion rate	1.202	(t/cm)

GZ values	
Heeling angle (Degr.)	GZ (m)
0.00 (CL)	0.000
5.00 (PS)	0.092
10.00 (PS)	0.191
15.00 (PS)	0.296
20.00 (PS)	0.388
25.00 (PS)	0.438
30.00 (PS)	0.449
35.00 (PS)	0.430
37.20 (PS)	0.414
40.00 (PS)	0.388
45.00 (PS)	0.327
50.00 (PS)	0.253
55.00 (PS)	0.170
60.00 (PS)	0.081
70.00 (PS)	-0.108

Description	Density (t/m^3)	Fill%	Weight (tonnes)	VCG (m)	LCG (m)	TCG (m)	FSM (Tonnes*m)
Water Ballast							
Fore peak	1.0250	100.0	5.432	1.768	18.604	0.000 (CL)	6.582
Miscellaneous							
Crew and Misc.			3.000	5.200	12.300	0.000 (CL)	
Fuel Oil							
Fuel Oil Tank 1(P)	0.8500	98.0	10.245	1.455	10.747	1.534 (PS)	3.573
Fuel Oil Tank 1(S)	0.8500	98.0	10.245	1.455	10.747	-1.534 (SB)	3.573
Fuel oil Tank 2(P)	0.8500	98.0	10.006	1.492	12.246	1.523 (PS)	3.569
Fuel Oil Tank 2(S)	0.8500	98.0	10.006	1.492	12.246	-1.523 (SB)	3.569
Daily service Tank	0.8500	98.0	1.176	1.977	3.850	0.000 (CL)	0.144
Totals for Fuel Oil			41.677	1.487	11.272	0.000 (CL)	14.428
Fresh water							
Fresh water (P)	1.0000	100.0	10.304	2.028	2.051	1.421 (PS)	4.517
Fresh water (S)	1.0000	100.0	10.304	2.028	2.051	-1.421 (SB)	4.518
Totals for Fresh water			20.608	2.028	2.051	0.000 (CL)	9.035
Lightship			94.000	2.400	9.200	0.000 (CL)	
Deadweight			70.717	1.824	9.192	0.000 (CL)	30.045
Displacement			164.717	2.153	9.197	0.000 (CL)	30.045



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Description	Critical points		
	Type	Dist. to wl (m)	Submersion angle (Degr.)
Engine Room ventilation sup/Exhaust	Downflooding	1.669	37.16
Cowl Vent 1	Downflooding	2.033	49.98
Engine Room Funnel	Downflooding	5.772	67.41
Sky light	Downflooding	1.528	56.25



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Evaluation of criteria						
Minimum design criteria applicable to Tugs						
Description	Attained value		Criterion	Required value		Complies
Area 0° - 30°	0.1430	(mrad)	>=	0.0550	(mrad)	YES
Area 0° - 40°	0.1975	(mrad)	>=	0.0900	(mrad)	YES
Area 30° - 40°	0.0545	(mrad)	>=	0.0300	(mrad)	YES
Max. GZ at 30° or greater	0.449	(m)	>=	0.200	(m)	YES
Angle of max GZ	29.0	(Degr.)	>=	25.0	(Degr.)	YES
Initial metacentric height	0.924	(m)	>	0.150	(m)	YES
Severe wind and rolling criterion						
Wind pressure	51.4	(kg/m^2)				YES
Wind area	78.85	(m^2)				
Steady wind lever	0.074	(m)				
Deck immersion angle	15.57	(Degr.)				
Wind gust lever	0.111	(m)				
Ratio of areaA/areaB	0.460		<=	1.000		YES
Maximum allowed static heeling angle	4.0	(Degr.)	<=	16.0	(Degr.)	YES
Max allowed ratio static angle/deck immersion angle	0.258		<=	0.800		YES
The vessel complies with the stability criteria						

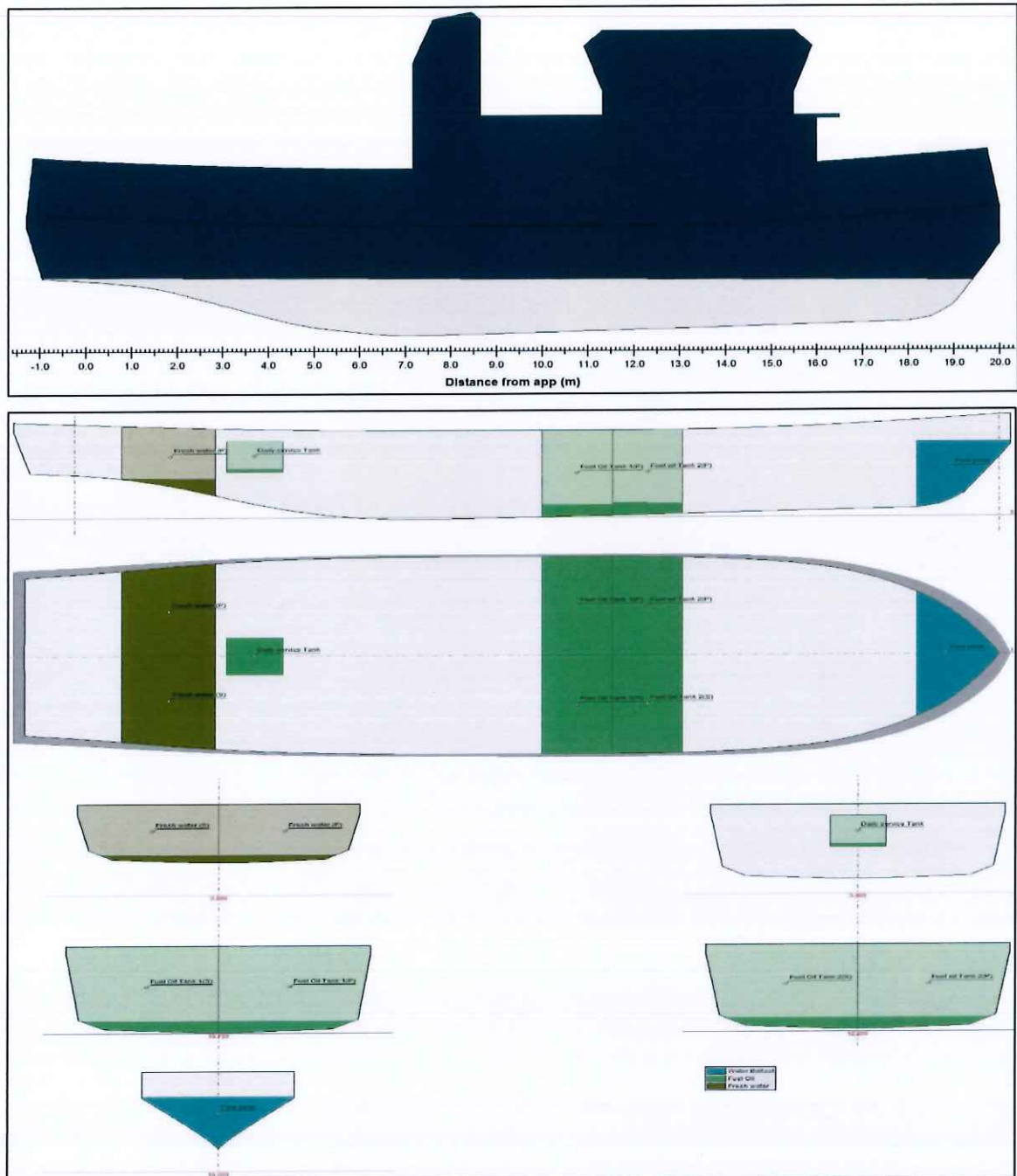
Towing criteria						
Description	Attained value		Criterion	Required value		Complies
Heeling moment	27.30	(Tonnes*m)				
Area between heeling lever and righting lever calculated from the 1st intercept upto an angle 40° beyond the first intercept or upto the down flooding angle, whichever is less.	0.1101	m-rad	>=	0.09	m-rad	YES
The ratio of area under a righting lever curve to the area under a heeling lever curve upto an angle of 40° or the downflooding angle, whichever is less	1.971	-	>=	1.4		YES



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12.5 Ballast Arrival condition

The below load case gives the Intact Stability Analysis of the vessel loaded with 10 %consumables and Ballast and other miscellaneous items.



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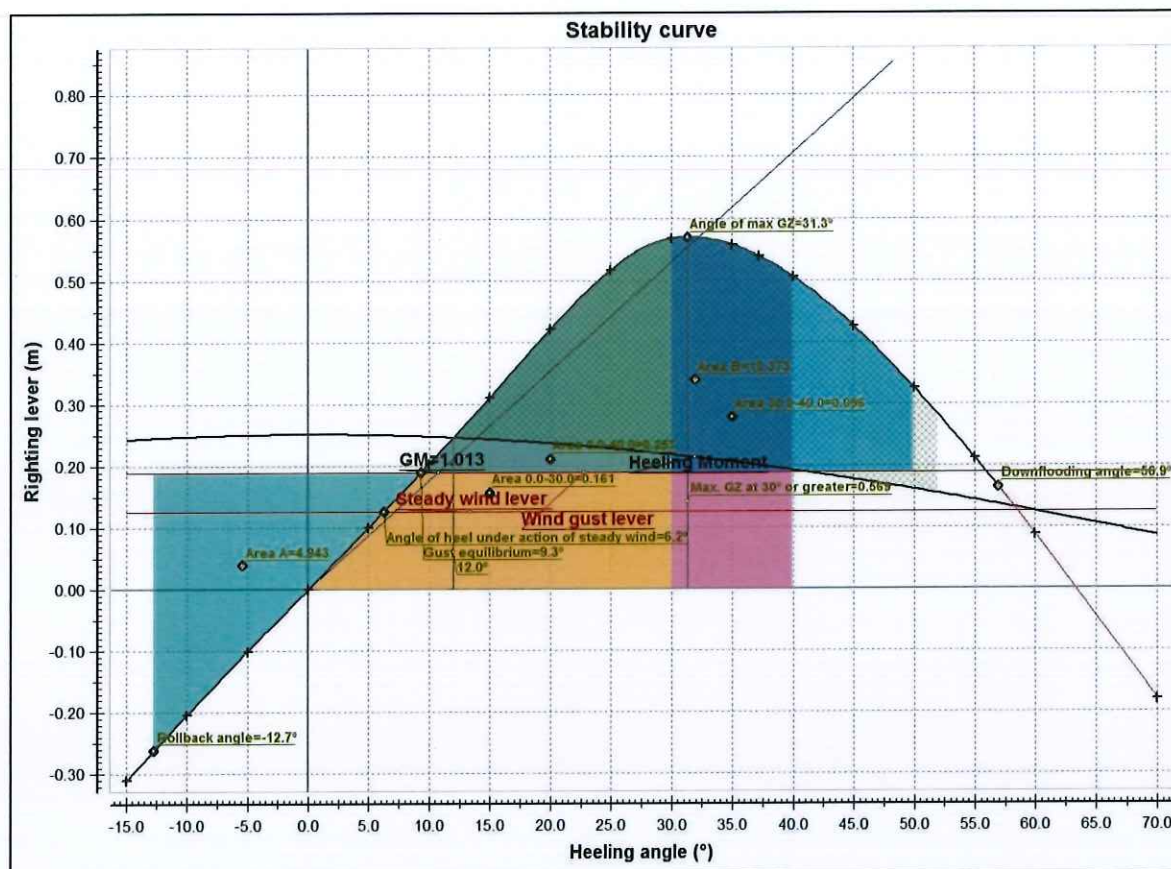
Hydrostatic particulars					
List	0.00 (CL)	(Degr.)	KM	3.629	(m)
Draft aft pp	1.469	(m)	GM solid	1.289	(m)
Mean draft	1.417	(m)	GG'	0.276	(m)
Draft forward pp	1.364	(m)	GM liquid	1.013	(m)
Trim	-0.105	(m)	MCT	1.515	(t*m/cm)
Volume	106.09	(m^3)	Relative water density	1.0250	
LCF	8.996	(m)	Immersion rate	1.106	(t/cm)

GZ values	
Heeling angle (Degr.)	GZ (m)
0.00 (CL)	0.000
5.00 (PS)	0.101
10.00 (PS)	0.204
15.00 (PS)	0.311
20.00 (PS)	0.421
25.00 (PS)	0.517
30.00 (PS)	0.568
35.00 (PS)	0.557
37.20 (PS)	0.539
40.00 (PS)	0.506
45.00 (PS)	0.426
50.00 (PS)	0.326
55.00 (PS)	0.212
60.00 (PS)	0.088
70.00 (PS)	-0.179

Description	Density (t/m^3)	Fill%	Weight (tonnes)	VCG (m)	LCG (m)	TCG (m)	FSM (Tonnes*m)
Water Ballast							
Fore peak	1.0250	100.0	5.432	1.768	18.604	0.000 (CL)	6.582
Miscellaneous							
Crew and Misc.			3.000	5.200	12.300	0.000 (CL)	
Fuel Oil							
Fuel Oil Tank 1(P)	0.8500	10.0	1.045	0.242	10.726	1.250 (PS)	3.573
Fuel Oil Tank 1(S)	0.8500	10.0	1.045	0.242	10.726	-1.250 (SB)	3.573
Fuel oil Tank 2(P)	0.8500	10.0	1.021	0.294	12.222	1.216 (PS)	3.569
Fuel Oil Tank 2(S)	0.8500	10.0	1.021	0.294	12.222	-1.216 (SB)	3.569
Daily service Tank	0.8500	10.0	0.120	1.537	3.850	0.000 (CL)	0.144
Totals for Fuel Oil			4.253	0.303	11.250	0.000 (CL)	14.428
Fresh water							
Fresh water (P)	1.0000	10.0	1.030	1.123	2.440	1.168 (PS)	4.517
Fresh water (S)	1.0000	10.0	1.030	1.123	2.440	-1.168 (SB)	4.518
Totals for Fresh water			2.060	1.123	2.440	0.000 (CL)	9.035
Lightship			94.000	2.400	9.200	0.000 (CL)	
Deadweight			14.745	1.954	12.942	0.000 (CL)	30.045
Displacement			108.745	2.339	9.707	0.000 (CL)	30.045



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Description	Critical points		
	Type	Dist. to wl (m)	Submersion angle (Degr.)
Engine Room ventilation sup/Exhaust	Downflooding	2.175	56.92
Cowl Vent 1	Downflooding	2.440	61.91



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Evaluation of criteria					
Minimum design criteria applicable to Tugs					
Description	Attained value	Criterion	Required value		Complies
Area 0° - 30°	0.1609 (mrad)	>=	0.0550 (mrad)		YES
Area 0° - 40°	0.2569 (mrad)	>=	0.0900 (mrad)		YES
Area 30° - 40°	0.0961 (mrad)	>=	0.0300 (mrad)		YES
Max. GZ at 30° or greater	0.569 (m)	>=	0.200 (m)		YES
Angle of max GZ	31.3 (Degr.)	>=	25.0 (Degr.)		YES
Initial metacentric height	1.013 (m)	>	0.150 (m)		YES
Severe wind and rolling criterion					YES
Wind pressure	51.4 (kg/m ²)				
Wind area	88.63 (m ²)				
Steady wind lever	0.126 (m)				
Deck immersion angle	24.71 (Degr.)				
Wind gust lever	0.189 (m)				
Ratio of areaA/areaB	0.477	<=	1.000		YES
Maximum allowed static heeling angle	6.2 (Degr.)	<=	16.0 (Degr.)		YES
Max allowed ratio static angle/deck immersion angle	0.253	<=	0.800		YES
The vessel complies with the stability criteria					

Towing criteria					
Description	Attained value	Criterion	Required value		Complies
Heeling moment	27.30 (Tonnes*m)				
Area between heeling lever and righting lever calculated from the 1st intercept upto an angle 40° beyond the first intercept or upto the down flooding angle, whichever is less.	0.1701 m-rad	>=	0.09 m-rad		YES
The ratio of area under a righting lever curve to the area under a heeling lever curve upto an angle of 40° or the downflooding angle, whichever is less	1.592 -	>=	1.4		YES

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13 Worked out Example

The trim and stability calculations have been carried out for different loading conditions. The calculation for any other loading condition can be carried out if we know the Wt, LCG, VCG and free surface moments of all the dead-weight items in that loading condition. As for example in loaded departure condition, all the items have been considered full. Therefore 98% of all the tank items, their LCG, VCG and free surface moments have been taken. Similarly 98% wt of stores, its LCG, VCG have been taken. The wt of crew & effect will be same for all the loading conditions.

The below example is given for the Load case "Ballast Departure condition (sec-12.4)".

Loading detail:

Description	Density (t/m ³)	Fill%	Weight (tonnes)	VCG (m)	LCG (m)	TCG (m)	FSM (Tonnes*m)
Water Ballast							
Fore peak	1.0250	100.0	5.432	1.768	18.604	0.000 (CL)	6.582
Miscellaneous							
Crew and Misc.			3.000	5.200	12.300	0.000 (CL)	
Fuel Oil							
Fuel Oil Tank 1(P)	0.8500	98.0	10.245	1.455	10.747	1.534 (PS)	3.573
Fuel Oil Tank 1(S)	0.8500	98.0	10.245	1.455	10.747	-1.534 (SB)	3.573
Fuel oil Tank 2(P)	0.8500	98.0	10.006	1.492	12.246	1.523 (PS)	3.569
Fuel Oil Tank 2(S)	0.8500	98.0	10.006	1.492	12.246	-1.523 (SB)	3.569
Daily service Tank	0.8500	98.0	1.176	1.977	3.850	0.000 (CL)	0.144
Totals for Fuel Oil			41.677	1.487	11.272	0.000 (CL)	14.428
Fresh water							
Fresh water (P)	1.0000	100.0	10.304	2.028	2.051	1.421 (PS)	4.517
Fresh water (S)	1.0000	100.0	10.304	2.028	2.051	-1.421 (SB)	4.518
Totals for Fresh water			20.608	2.028	2.051	0.000 (CL)	9.035
Lightship			94.000	2.400	9.200	0.000 (CL)	
Deadweight			70.717	1.824	9.192	0.000 (CL)	30.045
Displacement			164.717	2.153	9.197	0.000 (CL)	30.045



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The above table is to be evaluated to obtain displacements and centre of gravities for the loading condition as shown below:

DESCRIPTION	WEIGHT	LCG	LMOM	TCG	TMOM	VCG	VMOM	FSM
Weight-1	W1	X1	W1X1	Y1	W1Y1	Z1	W1Z1	M1
Weight-2	W2	X2	W2X2	Y2	W2Y2	Z2	W2Z2	M2
.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.
Weight-n	Wn	Xn	WnXn	Yn	WnYn	Zn	WnZn	Mn
Total Weight	ΣWn	$\Sigma WnXn/\Sigma Wn$	$\Sigma WnXn$	$\Sigma WnYn/\Sigma Wn$	$\Sigma WnYn$	$\Sigma WnZn/\Sigma Wn$	$\Sigma WnZn$	ΣMn

$$\begin{aligned}\text{TOTAL DISPLACEMENT} &= \Sigma Wn \\ &= 164.717 \text{ T}\end{aligned}$$

$$\begin{aligned}\text{LCG} &= (\Sigma Wn \times Xn) / \Sigma Wn \\ &= 9.197 \text{ m}\end{aligned}$$

$$\begin{aligned}\text{VCG} &= (\Sigma Wn \times Zn) / \Sigma Wn \\ &= 2.153 \text{ m}\end{aligned}$$

$$\begin{aligned}\text{TCG} &= (\Sigma Wn \times Yn) / \Sigma Wn \\ &= 0.000 \text{ m}\end{aligned}$$

Once the displacement and Centroids are found, the draft values and other hydrostatic calculations are calculated from hydrostatic table for corresponding displacement at trim=0. The hydrostatics for the required displacement is calculated by interpolation.

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Trim: 0.000 (m)									
Draft (m)	Vol. mould (m ³)	Displ. (tonnes)	LCB (m)	VCB (m)	LCF (m)	S (m ²)	KMI (m)	KMI (m)	TpCm (t/cm)
1.400	103.907	106.504	9.852	0.817	9.337	130.64	3.611	25.95	1.068
1.450	109.192	111.922	9.822	0.846	9.110	135.48	3.569	27.19	1.102
1.500	114.636	117.501	9.784	0.876	8.956	139.62	3.537	27.71	1.128
1.550	120.184	123.189	9.743	0.906	8.868	143.07	3.505	27.63	1.146
1.600	125.800	128.945	9.704	0.936	8.843	145.85	3.462	27.03	1.156
1.650	131.457	134.743	9.667	0.966	8.828	148.54	3.422	26.40	1.166
1.700	137.171	140.600	9.631	0.995	8.816	151.23	3.385	25.80	1.175
1.750	142.923	146.496	9.598	1.025	8.808	153.90	3.352	25.24	1.183
1.800	148.718	152.436	9.567	1.054	8.805	156.55	3.322	24.69	1.192
1.850	154.545	158.408	9.539	1.083	8.810	159.14	3.291	24.14	1.199
1.900	160.407	164.417	9.512	1.112	8.822	161.70	3.261	23.58	1.205
1.950	166.303	170.461	9.488	1.141	8.835	164.27	3.235	23.07	1.212
2.000	172.232	176.538	9.466	1.169	8.847	166.84	3.213	22.59	1.218
2.050	178.188	182.642	9.445	1.198	8.860	169.43	3.194	22.14	1.225
2.100	184.178	188.782	9.426	1.227	8.874	172.02	3.178	21.73	1.232
2.150	190.202	194.957	9.409	1.255	8.888	174.63	3.165	21.35	1.239
2.200	196.265	201.171	9.393	1.283	8.902	177.24	3.154	20.99	1.245

Interpolated Hydrostatics								
Draft (m)	Displ. (tonnes)	LCB (m)	VCB (m)	LCF (m)	S (m ²)	KMI (m)	KMI (m)	TpCm (t/cm)
1.900	164.417	9.512	1.112	8.822	161.7	3.261	23.58	1.205
1.950	170.461	9.488	1.141	8.835	164.27	3.235	23.07	1.212
1.902	164.717	9.511	1.113	8.823	161.828	3.260	23.555	1.205

Trim calculation:

$$\begin{aligned}
 \text{Trim} &= \frac{(LCG - LCB) \times LBP}{(KMI - KG)} \\
 &= \frac{(9.197 - 9.511) \times 19.76}{(23.555 - 2.153)} \\
 &= 0.290 \text{ m by aft}
 \end{aligned}$$

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Hydrostatics for Trim=-0.25m								
Draft (m)	Displ. (tonnes)	LCB (m)	VCB (m)	LCF (m)	S (m^2)	KMt (m)	KMI (m)	TpCm (t/cm)
1.850	160.096	9.243	1.093	8.726	159.8	3.281	23.78	1.197
1.900	166.099	9.224	1.122	8.737	162.35	3.252	23.23	1.203
1.888	164.717	9.228	1.115	8.734	161.763	3.259	23.357	1.202

Hydrostatics for Trim=-0.50m								
Draft (m)	Displ. (tonnes)	LCB (m)	VCB (m)	LCF (m)	S (m^2)	KMt (m)	KMI (m)	TpCm (t/cm)
1.850	161.901	8.954	1.107	8.646	160.530	3.275	23.450	1.196
1.900	167.900	8.943	1.136	8.655	163.070	3.248	22.920	1.203
1.873	164.717	8.949	1.121	8.650	161.722	3.262	23.201	1.199

Hydrostatics for Trim =-0.29 m									
Trim (m)	Draft (m)	Displ. (tonnes)	LCB (m)	VCB (m)	LCF (m)	S (m^2)	KMt (m)	KMI (m)	TpCm (t/cm)
-0.250	1.888	164.717	9.228	1.115	8.734	161.763	3.259	23.357	1.202
-0.500	1.873	164.717	8.949	1.121	8.650	161.722	3.262	23.201	1.199
-0.290	1.886	164.717	9.184	1.116	8.721	161.756	3.259	23.332	1.201

So,

$$\text{Mean Draft } T_m = 1.886 \text{ m}$$

$$\text{Draft at LCF } T_{LCF} = T_m - \left(\frac{LBP - LCF}{LBP} \times TRIM \right)$$

$$= 1.903 \text{ m}$$

$$\text{Draft at Aft perpendicular } T_{AP} = T_{LCF} - \frac{LCF \times TRIM}{LBP}$$

$$= 2.031 \text{ m}$$

$$\text{Draft at Fwd perpendiculars } T_{FP} = T_{AP} + TRIM$$

$$= 1.741 \text{ m}$$



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Free Surface Correction: Provided a tank is completely filled with a liquid no movement of the liquid is possible and the effect on the ship's stability is precisely the same as if the tank contained solid material. When a quantity of liquid is withdrawn from the tank, the situation changes completely and stability of ship is adversely affected by what is known as „free surface effects“. The adverse effect on the stability is referred to as „loss in GM“ or as a virtual rise in VCG and is calculated as follows.

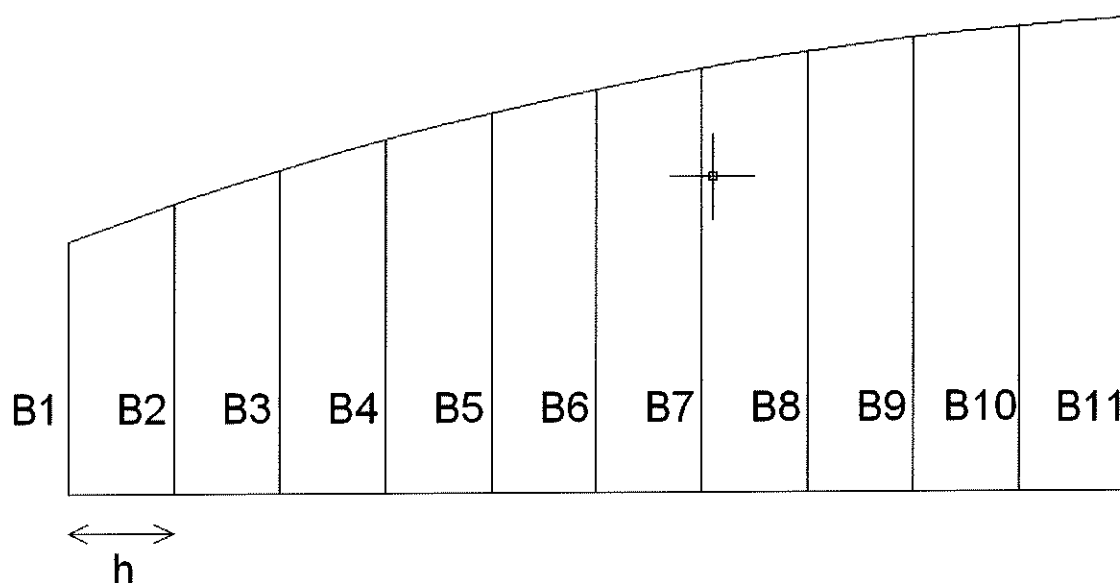
Loss in GM due to free surface effects (in meter) = Free surface moment (Tonnes Meters)/Displacement of Vessel (Tonnes)

Notes:

1. Slack must not be kept in all tanks. It must be collected in one tank.
2. The Master has to ensure that Actual KG < Maximum Permissible KG for corresponding displacement.

Example for computation of FSM

Tank FSM computation example for a random shape:





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Let the half breadths of a cross section be as shown above

The second moment of area is calculated as shown below using Simpsons rule:

S.no	B	B ³	SM	PRODUCT
01	B ₁	(B ₁) ³	1	1x(B ₁) ³
02	B ₂	(B ₂) ³	2	2x(B ₂) ³
03	B ₃	(B ₃) ³	4	4x(B ₃) ³
04	B ₄	(B ₄) ³	2	2x(B ₄) ³
05	B ₅	(B ₅) ³	4	4x(B ₅) ³
06	B ₆	(B ₆) ³	2	2x(B ₆) ³
07	B ₇	(B ₇) ³	4	4x(B ₇) ³
08	B ₈	(B ₈) ³	2	2x(B ₈) ³
09	B ₉	(B ₉) ³	4	4x(B ₉) ³
10	B ₁₀	(B ₁₀) ³	2	2x(B ₁₀) ³
11	B ₁₁	(B ₁₁) ³	1	1x(B ₁₁) ³

$$I = 2/9(h \times \Sigma \text{PRODUCT})$$

Free surface moment = $I \times \rho$ (density of the contents in the tank)

Here we have considered the maximum FSM for the tanks which are filled with their liquids for conservative approach.

Σ Free surface moment for all tanks = 30.045 t-m

$$\begin{aligned} \text{Free surface effect} &= \text{FSM} / \Delta \\ &= 30.045 / 164.717 \\ &= 0.182\text{m} \end{aligned}$$

$$\begin{aligned} \text{KG corrected} &= 2.153 + 0.182 \\ &= 2.335 \text{ m} \end{aligned}$$

$$\begin{aligned} \text{GM corrected} &= \text{KM} - \text{KG corrected} \\ &= 3.259 - 2.335 \\ &= 0.924\text{m} \end{aligned}$$

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Heel Calculation:

$$\begin{aligned}
 \text{HEEL} &= \text{TAN}^{-1}(\text{TCG} / \text{GM corrected}) \\
 &= \text{TAN}^{-1}(0.00/0.924) \\
 &= 0.00 \text{ m}
 \end{aligned}$$

Stability Criteria:

The purpose of cross curve is to enable stability curve to be drawn for the ship in any sailing condition.

$$\text{GZ} = \text{KN} - \text{KG} \times \sin \theta - \text{TCG} \times \cos \theta$$

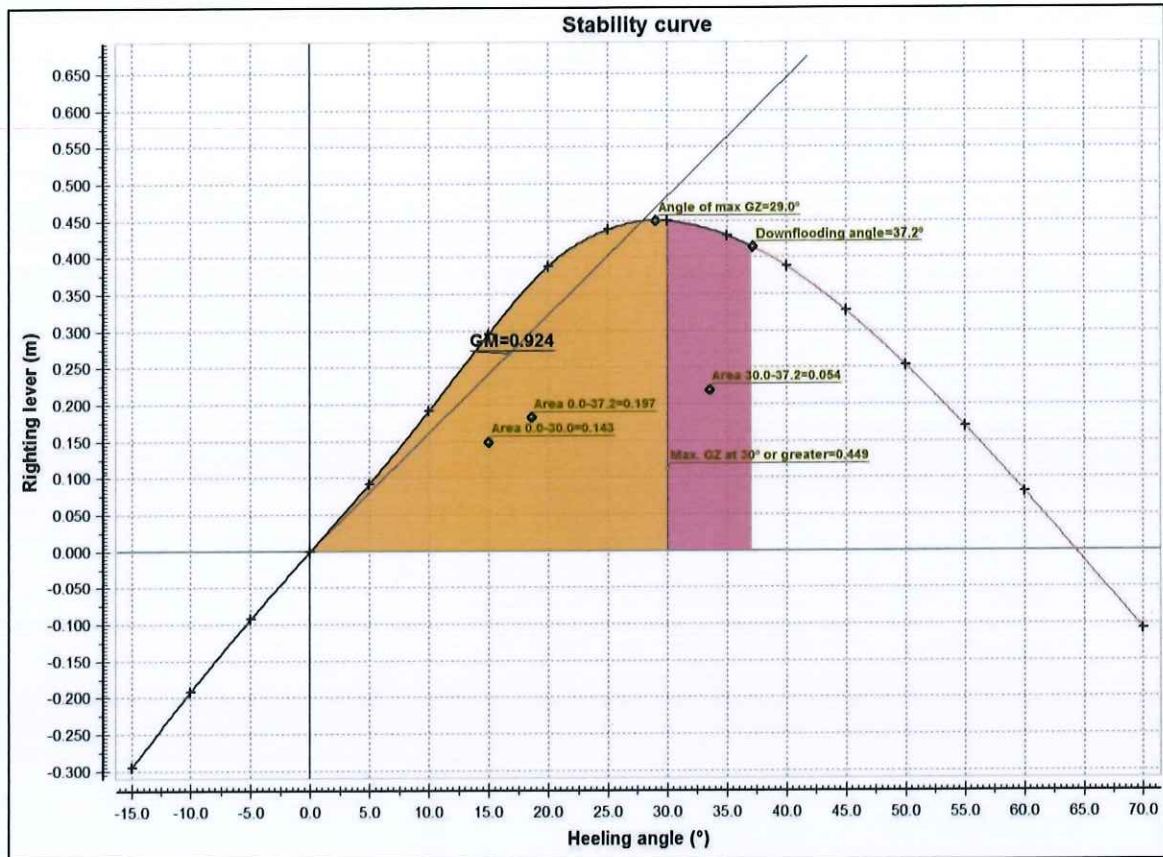
The values for Trim = -0.29m are evaluated as below

Heeling angle (Degr.)	GZ values				
	KN (m)	VCG sin(θ) (m)	GG' sin(θ) (m)	TCG cos(θ) (m)	GZ (m)
0.00 (CL)	0.000	0.000	0.000	0.000	0.000
5.00 (PS)	0.330	0.207	0.013	0.000	0.111
10.00 (PS)	0.657	0.411	0.025	0.000	0.221
15.00 (PS)	0.981	0.613	0.036	0.000	0.332
20.00 (PS)	1.297	0.810	0.044	0.000	0.443
25.00 (PS)	1.587	1.001	0.049	0.000	0.537
30.00 (PS)	1.826	1.185	0.053	0.000	0.588
35.00 (PS)	1.994	1.359	0.056	0.000	0.579
37.20 (PS)	2.049	1.433	0.057	0.000	0.560
40.00 (PS)	2.107	1.523	0.058	0.000	0.526
45.00 (PS)	2.178	1.676	0.059	0.000	0.443
50.00 (PS)	2.216	1.815	0.060	0.000	0.340
55.00 (PS)	2.225	1.941	0.061	0.000	0.223
60.00 (PS)	2.208	2.052	0.062	0.000	0.095
70.00 (PS)	2.109	2.227	0.065	0.000	-0.183



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A graph is plotted between angle of heel and GZ value:



The area under the curve is determined for the following regions in order to satisfy the stability criteria.

1. Area under the curve between 0 and 30 is 0.143 m-rad (minimum required: 0.055m-rad)
2. Area under the curve between 0 and 40 is 0.1975 m-rad (minimum required: 0.09 m-rad)
3. Area under the curve between 30 and 40 is 0.0545 m-rad (minimum required: 0.03 m-rad)
4. Maximum GZ at an angle 25 or greater is 0.449 and the angle is 29.60 deg.
5. Initial metacentric height is 0.924 m



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Severe Wind and Rolling Criteria:

Wind Pressure = 51.4 kg/m²

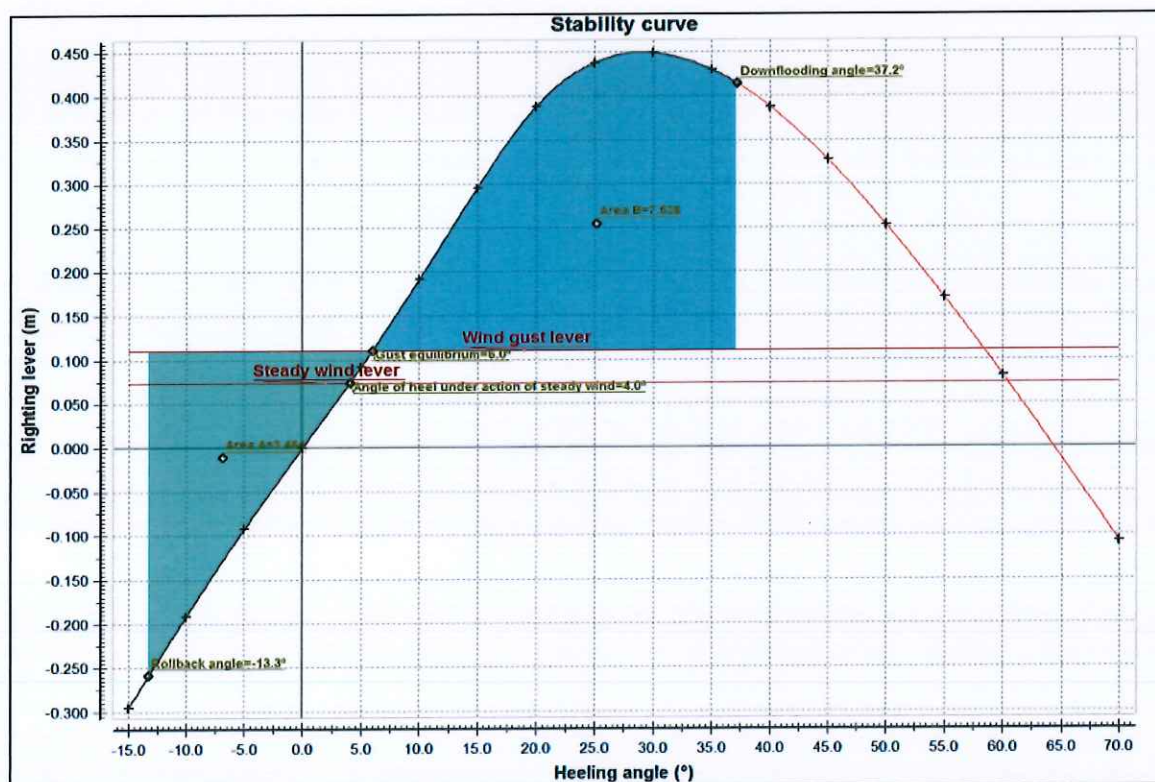
Wind Area = 78.85 m² (Interpolated Windage area table)

Wind Lever = 3.01 m (Interpolated Windage area table)

$$\begin{aligned}
 LW_1 &= (P \times A \times Z) / (1000 \times g \times \Delta) \text{ m} \\
 &= (504 \times 78.85 \times 3.01) / (1000 \times 9.81 \times 164.717) \\
 &= 0.074 \text{ m}
 \end{aligned}$$

$$\begin{aligned}
 LW_2 &= 1.5 \times LW_1 \\
 &= 1.5 \times 0.074 \\
 &= 0.111 \text{ m}
 \end{aligned}$$

Evaluation of Criteria:





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Deck immersion angle = 15.57 deg

Ratio of areaA/areaB = 0.460 (Maximum = 1)

Maximum allowed static heeling angle = 4.00 deg (Maximum 16.00 deg)

Max allowed ratio static angle/deck immersion angle = 0.258 (maximum 0.8)

Towing Criteria:

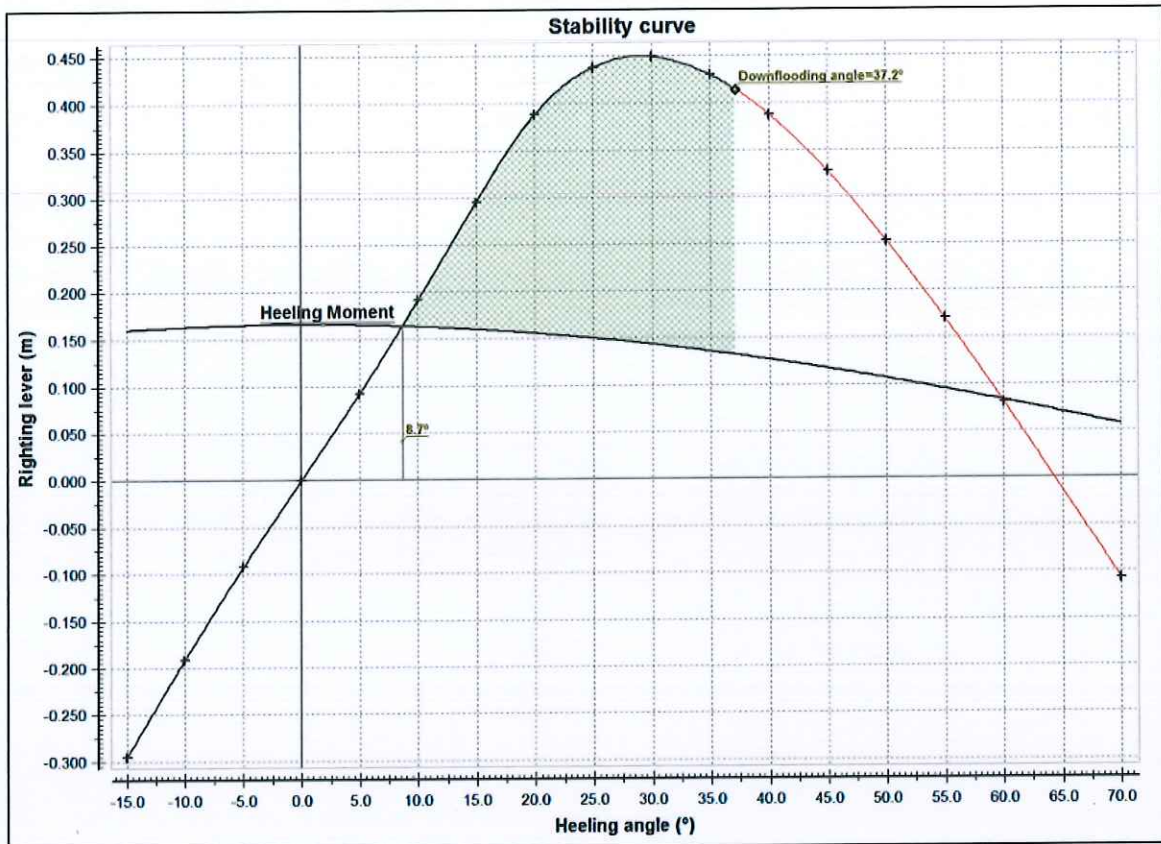
We calculate the heeling lever for every angle of heel as shown below

Heeling Lever = $0.7 \times T \times H \times \cos\theta / 9.81 \Delta$

Ballast Departure						
Calculation of Heeling Lever due to tow						
θ Degr.	$\cos\theta$ -	Bollard pull T	H m	disp T	Heeling Lever m	Area m2
0	1.000	10.000	3.900	164.717	0.166	0.000
5	0.996	10.000	3.900	164.717	0.165	0.014
10	0.985	10.000	3.900	164.717	0.163	0.029
15	0.966	10.000	3.900	164.717	0.160	0.043
20	0.940	10.000	3.900	164.717	0.156	0.057
25	0.906	10.000	3.900	164.717	0.150	0.070
30	0.866	10.000	3.900	164.717	0.144	0.083
35	0.819	10.000	3.900	164.717	0.136	0.095
40	0.766	10.000	3.900	164.717	0.127	0.107
45	0.707	10.000	3.900	164.717	0.117	0.117
50	0.643	10.000	3.900	164.717	0.107	0.127
55	0.574	10.000	3.900	164.717	0.095	0.136
60	0.500	10.000	3.900	164.717	0.083	0.144
65	0.423	10.000	3.900	164.717	0.070	0.150
70	0.342	10.000	3.900	164.717	0.057	0.156



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Area between heeling lever and righting lever calculated from the 1st intercept upto an angle 40° beyond the first intercept or up to the down flooding angle, whichever is less is 0.1101 m-rad (minimum 0.09 m-rad)

or

The ratio of area under a righting lever curve to the area under a heeling lever curve upto an angle of 40° or the Down flooding angle, whichever is less is 1.971 (minimum 1.4)

THE VESSEL COMPLIES WITH THE CRITERIA



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14 Dead weight table

Displ. (SW)	Draft	DWT(SW)	Draft	Displ.(FW)	Draft	DWT(FW)	Draft
(tonnes)	(m)	(tonnes)	(m)	(tonnes)	(m)	(tonnes)	(m)
30.00	0.57	0.00	1.28	30.00	0.58	0	1.3
35.00	0.63	5.00	1.33	35.00	0.64	5.00	1.35
40.00	0.69	10.00	1.38	40.00	0.70	10.00	1.4
45.00	0.75	15.00	1.42	45.00	0.76	15.00	1.45
50.00	0.81	20.00	1.47	50.00	0.82	20.00	1.49
55.00	0.87	25.00	1.51	55.00	0.88	25.00	1.54
60.00	0.92	30.00	1.56	60.00	0.94	30.00	1.58
65.00	0.98	35.00	1.6	65.00	1.00	35.00	1.63
70.00	1.03	40.00	1.64	70.00	1.05	40.00	1.67
75.00	1.09	45.00	1.69	75.00	1.11	45.00	1.72
80.00	1.14	50.00	1.73	80.00	1.16	50.00	1.76
85.00	1.19	55.00	1.77	85.00	1.21	55.00	1.8
90.00	1.24	60.00	1.81	90.00	1.26	60.00	1.85
95.00	1.29	65.00	1.85	95.00	1.31	65.00	1.89
100.00	1.34	70.00	1.9	100.00	1.36	70.00	1.93
105.00	1.39	75.00	1.94	105.00	1.41	75.00	1.97
110.00	1.43			110.00	1.46		
115.00	1.48			115.00	1.50		
120.00	1.52			120.00	1.55		
125.00	1.57			125.00	1.59		
130.00	1.61			130.00	1.64		
135.00	1.65			135.00	1.68		
140.00	1.69			140.00	1.72		
145.00	1.74			145.00	1.77		
150.00	1.78			150.00	1.81		
155.00	1.82			155.00	1.85		
160.00	1.86			160.00	1.90		
165.00	1.9			165.00	1.94		



K.B. SHIPPING & CO.	21.30M TUG BOAT	16-1011-701
7/21/2016	PRELIMINARY INTACT STABILITY BOOKLET	Rev No: 0

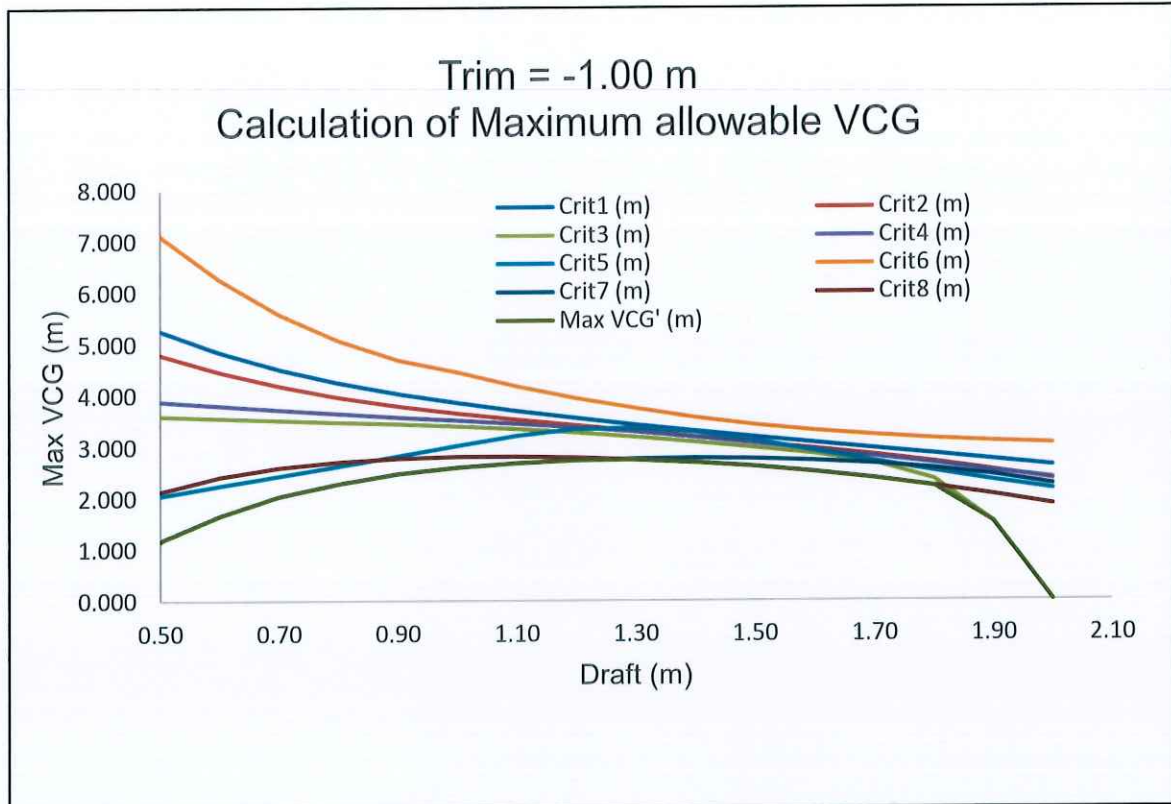
15 Max VCG

Minimum design criteria applicable to Tugs	
Criterion	description
Crit1	Area 0° - 30°
Crit2	Area 0° - 40°
Crit3	Area 30° - 40°
Crit4	Max. GZ at 30° or greater
Crit5	Angle of max GZ
Crit6	Initial metacentric height
Crit7	Severe wind and rolling criterion
Crit9	Area Between GZ curve and heeling arm curve(Tug criteria)

Initial trim: -1.000 (by stern)										
Draft (m)	Crit1 (m)	Crit2 (m)	Crit3 (m)	Crit4 (m)	Crit5 (m)	Crit6 (m)	Crit7 (m)	Crit8 (m)	Max VCG' (m)	Min GM' (m)
0.50	5.271	4.808	3.612	3.897	2.061	7.119	1.183	2.140	1.183	6.109
0.60	4.855	4.470	3.571	3.815	2.257	6.272	1.669	2.425	1.669	4.753
0.70	4.523	4.197	3.531	3.735	2.446	5.599	2.046	2.604	2.046	3.703
0.80	4.257	3.979	3.493	3.659	2.635	5.083	2.290	2.715	2.290	2.943
0.90	4.043	3.804	3.454	3.588	2.828	4.698	2.483	2.780	2.483	2.366
1.00	3.866	3.662	3.411	3.522	3.030	4.464	2.608	2.813	2.608	2.006
1.10	3.707	3.539	3.355	3.457	3.221	4.184	2.690	2.815	2.690	1.644
1.20	3.564	3.423	3.282	3.384	3.347	3.949	2.743	2.793	2.743	1.356
1.30	3.432	3.308	3.193	3.294	3.368	3.755	2.768	2.751	2.751	1.154
1.40	3.309	3.195	3.093	3.190	3.290	3.573	2.779	2.694	2.694	1.030
1.50	3.191	3.081	2.983	3.071	3.126	3.430	2.767	2.619	2.619	0.962
1.60	3.078	2.965	2.864	2.944	2.917	3.316	2.731	2.514	2.514	0.953
1.70	2.965	2.840	2.688	2.809	2.712	3.226	2.665	2.383	2.383	0.994
1.80	2.855	2.698	2.362	2.671	2.521	3.156	2.573	2.231	2.231	1.075
1.90	2.742	2.539	1.531	2.529	2.337	3.100	2.443	2.060	1.531	1.720
2.00	2.625	2.362	0.000	2.386	2.169	3.059	2.265	1.868	0.000	3.210



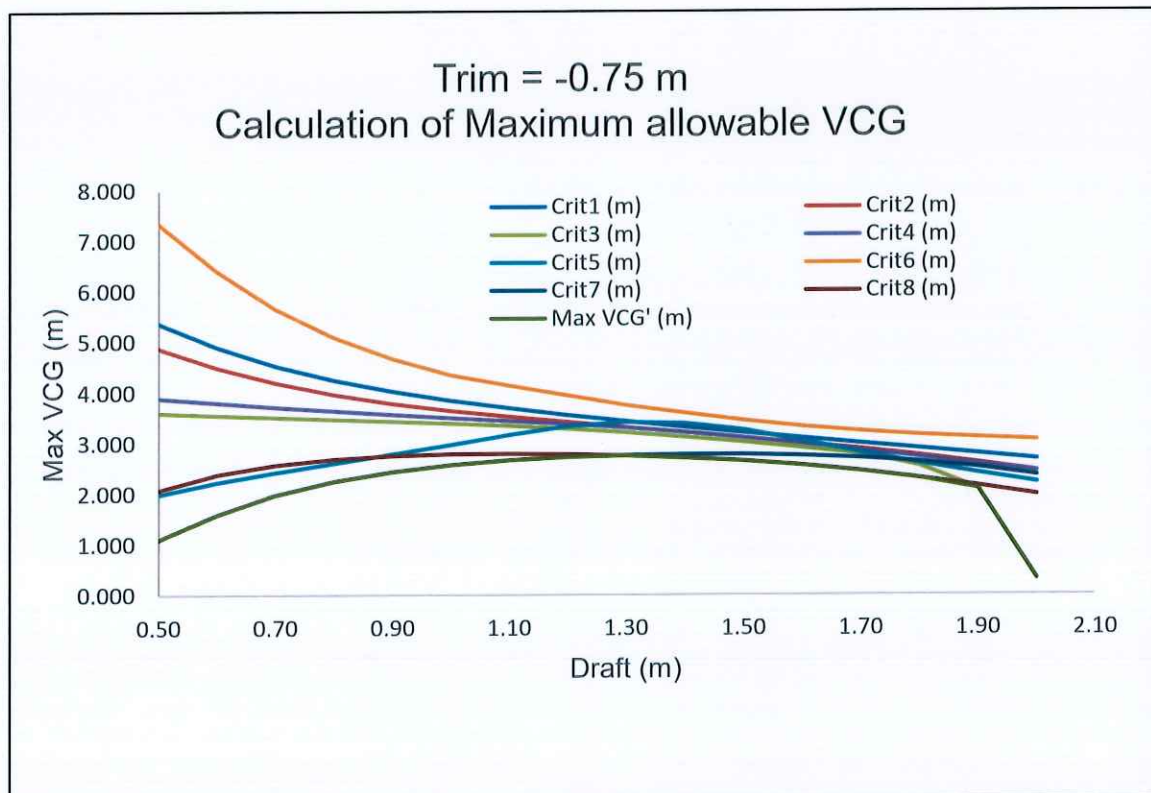
K.B. SHIPPING & CO.	21.30M TUG BOAT	16-1011-701
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K.B. SHIPPING & CO.	21.30M TUG BOAT	16-1011-701
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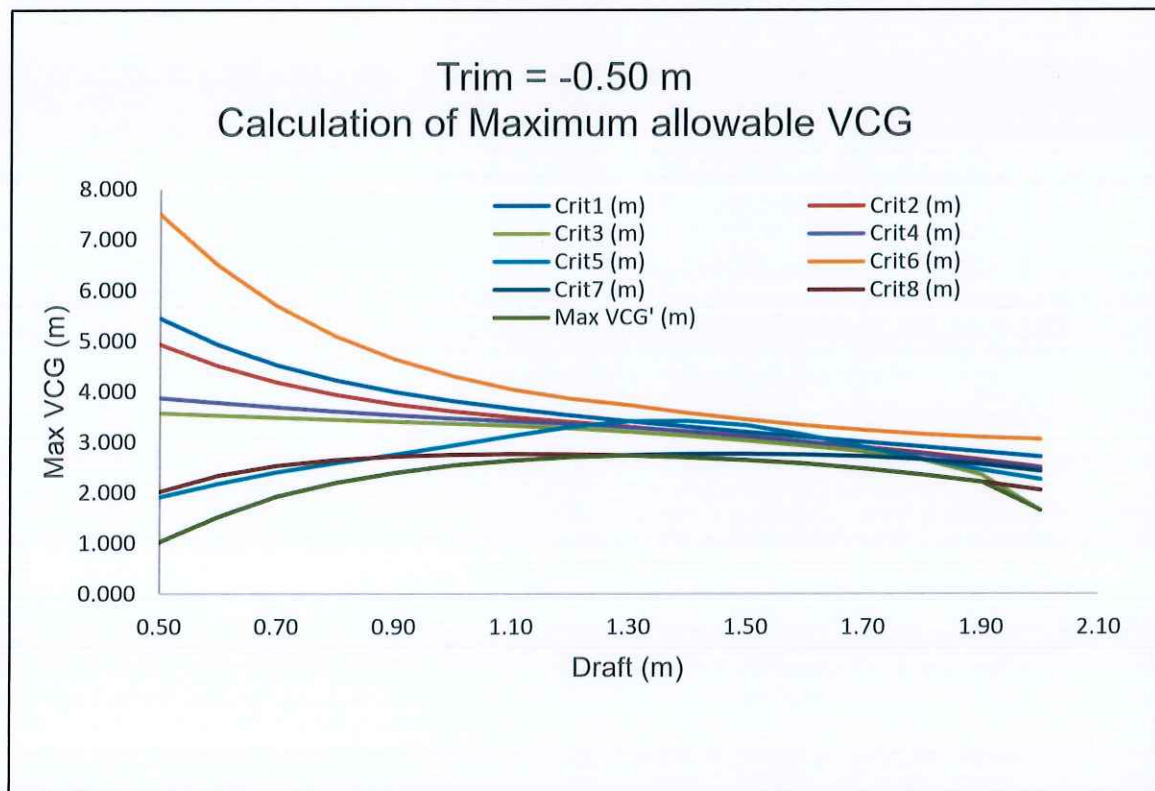
Initial trim: -0.750 (by stern)										
Draft (m)	Crit1 (m)	Crit2 (m)	Crit3 (m)	Crit4 (m)	Crit5 (m)	Crit6 (m)	Crit7 (m)	Crit8 (m)	Max VCG' (m)	Min GM' (m)
0.50	5.368	4.878	3.595	3.885	1.987	7.342	1.094	2.072	1.094	6.442
0.60	4.893	4.493	3.551	3.798	2.223	6.400	1.588	2.378	1.588	4.962
0.70	4.527	4.194	3.509	3.715	2.419	5.657	1.979	2.566	1.979	3.828
0.80	4.242	3.961	3.471	3.636	2.603	5.094	2.243	2.681	2.243	3.001
0.90	4.020	3.780	3.432	3.564	2.784	4.666	2.433	2.748	2.433	2.383
1.00	3.843	3.639	3.391	3.497	2.966	4.345	2.571	2.782	2.571	1.924
1.10	3.690	3.523	3.343	3.436	3.158	4.136	2.666	2.792	2.666	1.621
1.20	3.552	3.416	3.283	3.375	3.327	3.940	2.725	2.780	2.725	1.365
1.30	3.427	3.311	3.206	3.301	3.407	3.747	2.759	2.749	2.749	1.148
1.40	3.311	3.206	3.115	3.209	3.382	3.589	2.772	2.702	2.702	1.038
1.50	3.200	3.099	3.013	3.102	3.254	3.442	2.771	2.640	2.640	0.953
1.60	3.094	2.990	2.901	2.981	3.040	3.323	2.746	2.553	2.553	0.920
1.70	2.989	2.879	2.776	2.853	2.805	3.228	2.699	2.438	2.438	0.940
1.80	2.884	2.749	2.542	2.719	2.586	3.154	2.623	2.300	2.300	1.004
1.90	2.779	2.604	2.093	2.581	2.391	3.097	2.510	2.144	2.093	1.154
2.00	2.669	2.442	0.314	2.440	2.215	3.054	2.351	1.969	0.314	2.890





K.B. SHIPPING & CO.	21.30M TUG BOAT	16-1011-701
7/21/2016	PRELIMINARY INTACT STABILITY BOOKLET	Rev No: 0

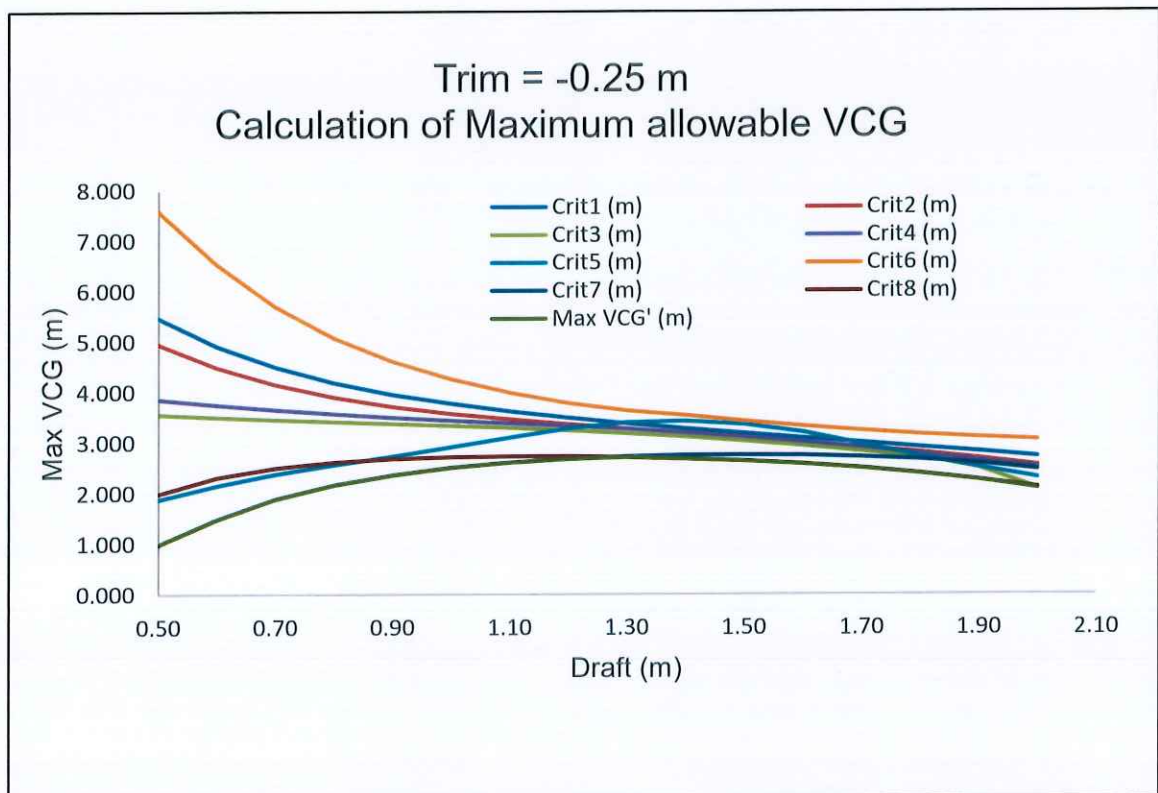
Initial trim: -0.500 (by stern)										
Draft (m)	Crit1 (m)	Crit2 (m)	Crit3 (m)	Crit4 (m)	Crit5 (m)	Crit6 (m)	Crit7 (m)	Crit8 (m)	Max VCG' (m)	Min GM' (m)
0.50	5.442	4.931	3.577	3.872	1.914	7.504	1.027	2.022	1.027	6.696
0.60	4.916	4.504	3.530	3.777	2.176	6.490	1.521	2.342	1.521	5.120
0.70	4.519	4.182	3.486	3.690	2.405	5.692	1.926	2.534	1.926	3.916
0.80	4.219	3.937	3.446	3.609	2.588	5.095	2.200	2.650	2.200	3.046
0.90	3.990	3.751	3.407	3.535	2.751	4.644	2.396	2.717	2.396	2.399
1.00	3.812	3.610	3.367	3.469	2.933	4.301	2.538	2.753	2.538	1.913
1.10	3.660	3.496	3.323	3.411	3.120	4.041	2.638	2.764	2.638	1.553
1.20	3.528	3.396	3.269	3.355	3.297	3.861	2.704	2.758	2.704	1.308
1.30	3.411	3.301	3.203	3.292	3.412	3.730	2.744	2.735	2.735	1.145
1.40	3.302	3.204	3.123	3.212	3.420	3.573	2.763	2.697	2.697	1.027
1.50	3.199	3.105	3.029	3.117	3.334	3.447	2.766	2.644	2.644	0.953
1.60	3.100	3.003	2.924	3.005	3.145	3.332	2.749	2.575	2.575	0.908
1.70	3.001	2.899	2.811	2.883	2.899	3.234	2.714	2.477	2.477	0.907
1.80	2.903	2.785	2.651	2.754	2.660	3.157	2.655	2.354	2.354	0.953
1.90	2.805	2.652	2.354	2.621	2.444	3.097	2.558	2.212	2.212	1.036
2.00	2.702	2.504	1.653	2.484	2.260	3.053	2.419	2.050	1.653	1.550





K.B. SHIPPING & CO.	21.30M TUG BOAT	16-1011-701
7/21/2016	PRELIMINARY INTACT STABILITY BOOKLET	Rev No: 0

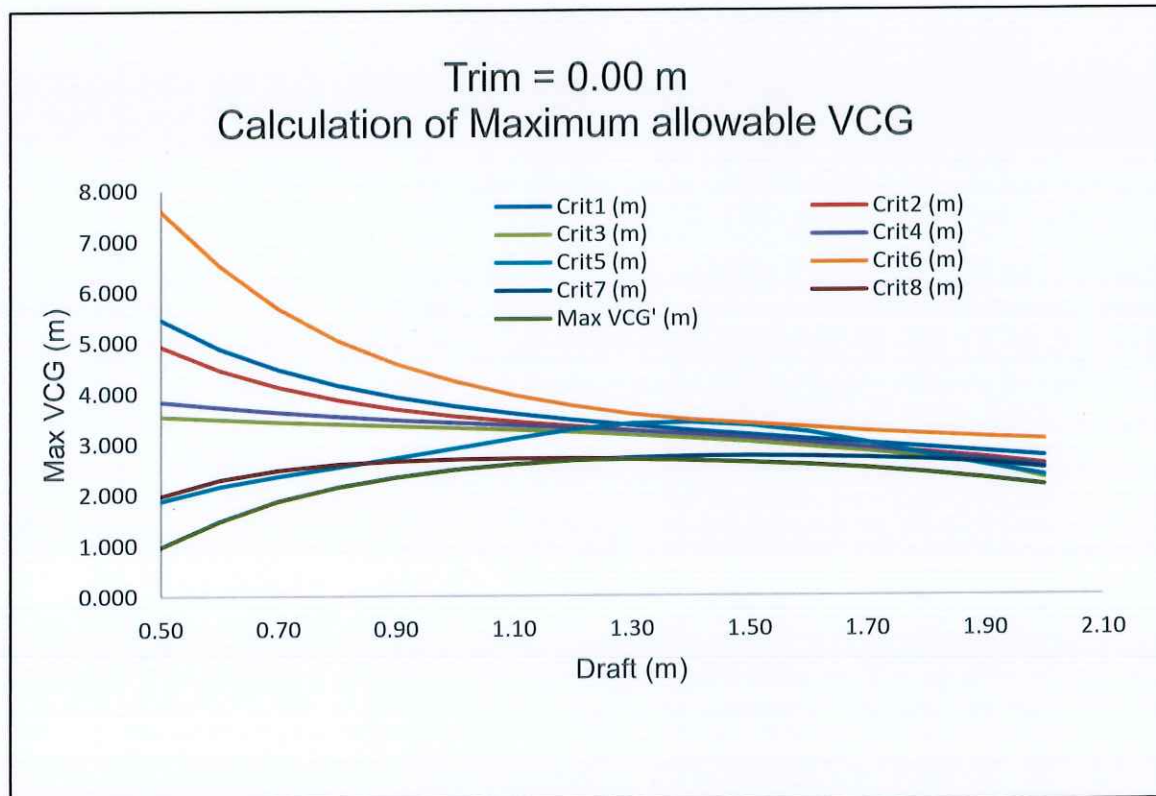
Initial trim: -0.250 (by stern)										
Draft (m)	Crit1 (m)	Crit2 (m)	Crit3 (m)	Crit4 (m)	Crit5 (m)	Crit6 (m)	Crit7 (m)	Crit8 (m)	Max VCG' (m)	Min GM' (m)
0.50	5.471	4.949	3.561	3.857	1.878	7.590	0.987	1.992	0.987	6.849
0.60	4.913	4.495	3.508	3.753	2.155	6.534	1.488	2.316	1.488	5.196
0.70	4.500	4.158	3.461	3.659	2.382	5.698	1.888	2.507	1.888	3.961
0.80	4.191	3.907	3.419	3.576	2.564	5.082	2.169	2.622	2.169	3.063
0.90	3.957	3.719	3.381	3.504	2.736	4.618	2.364	2.688	2.364	2.405
1.00	3.777	3.578	3.342	3.442	2.915	4.265	2.510	2.723	2.510	1.905
1.10	3.625	3.466	3.299	3.385	3.102	3.994	2.613	2.736	2.613	1.531
1.20	3.496	3.369	3.249	3.331	3.281	3.787	2.682	2.731	2.682	1.256
1.30	3.383	3.279	3.188	3.273	3.401	3.638	2.728	2.713	2.713	1.076
1.40	3.282	3.190	3.117	3.201	3.427	3.544	2.751	2.682	2.682	1.013
1.50	3.187	3.099	3.031	3.114	3.358	3.429	2.755	2.637	2.637	0.942
1.60	3.094	3.005	2.934	3.013	3.204	3.328	2.745	2.580	2.580	0.899
1.70	3.004	2.907	2.828	2.899	2.971	3.242	2.716	2.499	2.499	0.894
1.80	2.912	2.806	2.713	2.776	2.724	3.163	2.668	2.391	2.391	0.922
1.90	2.820	2.686	2.496	2.649	2.502	3.102	2.586	2.262	2.262	0.990
2.00	2.723	2.551	2.091	2.516	2.304	3.056	2.465	2.114	2.091	1.115





K.B. SHIPPING & CO.	21.30M TUG BOAT	16-1011-701
7/21/2016	PRELIMINARY INTACT STABILITY BOOKLET	Rev No: 0

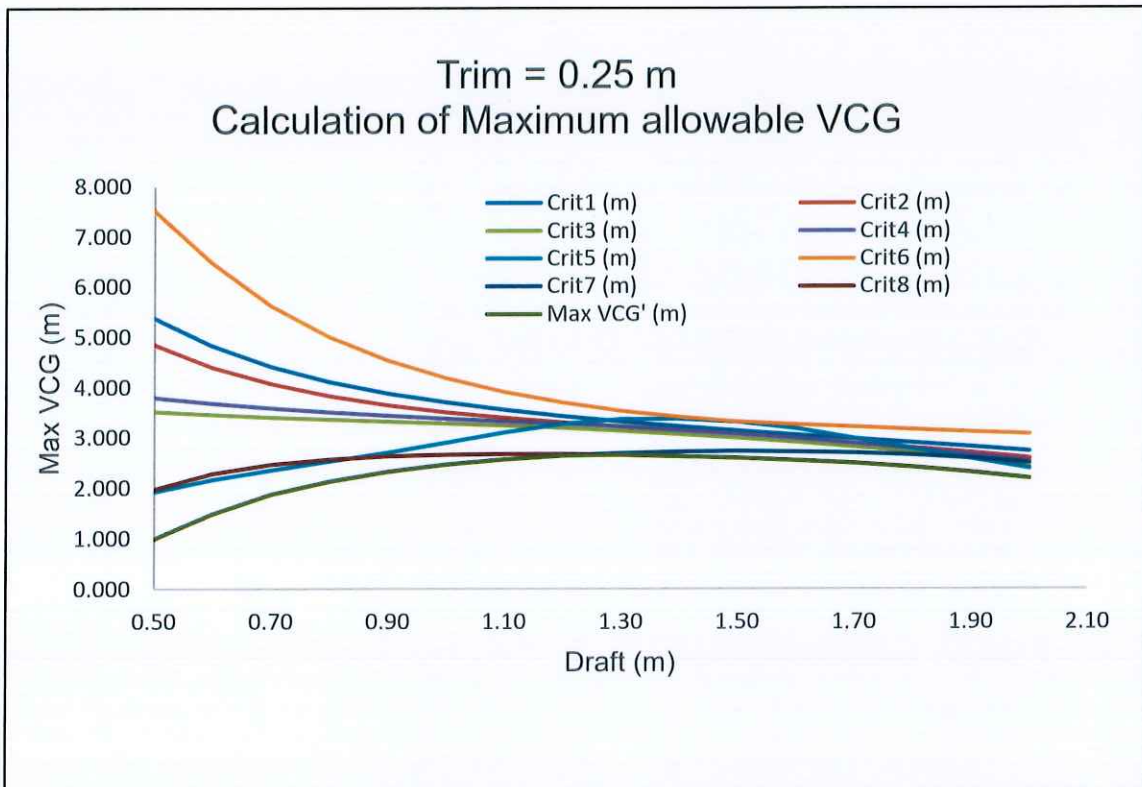
Initial trim: 0.00 m										
Draft (m)	Crit1 (m)	Crit2 (m)	Crit3 (m)	Crit4 (m)	Crit5 (m)	Crit6 (m)	Crit7 (m)	Crit8 (m)	Max VCG' (m)	Min GM' (m)
0.50	5.448	4.922	3.542	3.835	1.880	7.593	0.977	1.980	0.977	6.890
0.60	4.881	4.460	3.486	3.724	2.160	6.526	1.477	2.298	1.477	5.200
0.70	4.465	4.122	3.435	3.626	2.365	5.674	1.876	2.486	1.876	3.949
0.80	4.154	3.871	3.392	3.542	2.539	5.050	2.150	2.596	2.150	3.051
0.90	3.920	3.685	3.353	3.472	2.719	4.584	2.341	2.661	2.341	2.394
1.00	3.741	3.545	3.316	3.412	2.907	4.229	2.488	2.696	2.488	1.891
1.10	3.590	3.435	3.274	3.358	3.098	3.954	2.591	2.708	2.591	1.513
1.20	3.462	3.339	3.224	3.307	3.276	3.742	2.663	2.705	2.663	1.229
1.30	3.351	3.251	3.166	3.249	3.386	3.579	2.712	2.687	2.687	1.043
1.40	3.253	3.166	3.099	3.181	3.407	3.460	2.739	2.659	2.659	0.952
1.50	3.165	3.082	3.021	3.100	3.347	3.387	2.744	2.620	2.620	0.917
1.60	3.080	2.995	2.932	3.006	3.214	3.312	2.735	2.571	2.571	0.891
1.70	2.996	2.904	2.832	2.899	3.003	3.235	2.710	2.505	2.505	0.880
1.80	2.911	2.810	2.723	2.784	2.769	3.171	2.667	2.412	2.412	0.910
1.90	2.824	2.705	2.576	2.663	2.553	3.111	2.597	2.296	2.296	0.965
2.00	2.733	2.581	2.300	2.535	2.351	3.063	2.492	2.162	2.162	1.051

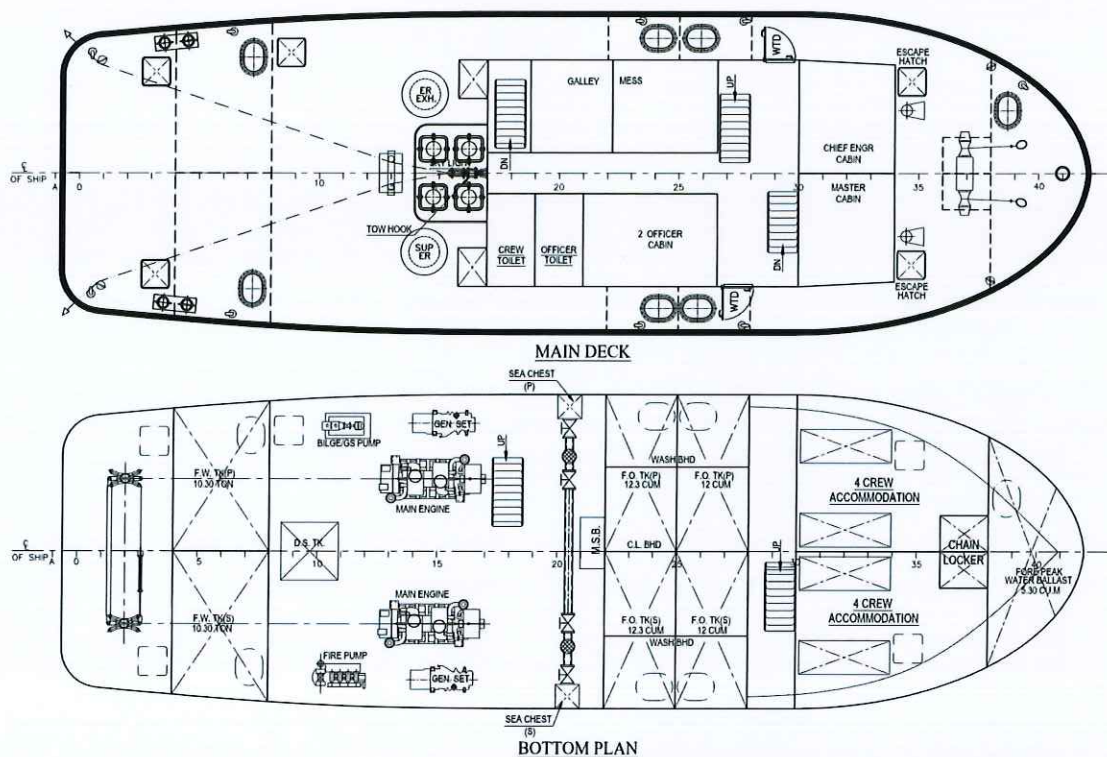
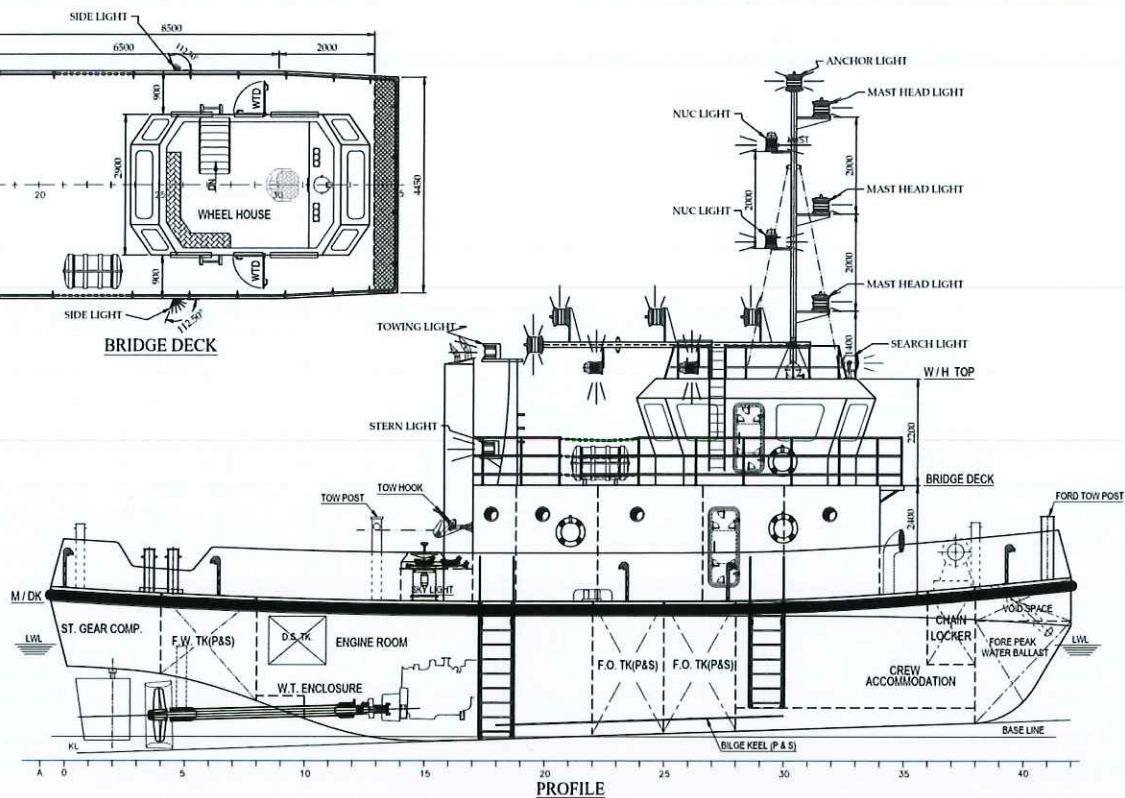




K.B. SHIPPING & CO.	21.30M TUG BOAT	16-1011-701
7/21/2016	PRELIMINARY INTACT STABILITY BOOKLET	Rev No: 0

Initial trim: 0.250 (by stem)										
Draft (m)	Crit1 (m)	Crit2 (m)	Crit3 (m)	Crit4 (m)	Crit5 (m)	Crit6 (m)	Crit7 (m)	Crit8 (m)	Max VCG' (m)	Min GM' (m)
0.50	5.374	4.852	3.520	3.801	1.922	7.515	0.996	1.979	0.996	6.819
0.60	4.821	4.402	3.460	3.687	2.167	6.468	1.486	2.288	1.486	5.132
0.70	4.413	4.073	3.408	3.589	2.357	5.620	1.877	2.468	1.877	3.893
0.80	4.110	3.830	3.365	3.508	2.530	5.004	2.135	2.575	2.135	3.019
0.90	3.881	3.648	3.326	3.441	2.710	4.542	2.326	2.636	2.326	2.366
1.00	3.705	3.512	3.289	3.383	2.900	4.188	2.469	2.669	2.469	1.870
1.10	3.556	3.403	3.247	3.330	3.102	3.915	2.572	2.680	2.572	1.494
1.20	3.428	3.309	3.198	3.280	3.276	3.703	2.643	2.677	2.643	1.211
1.30	3.319	3.221	3.140	3.223	3.368	3.539	2.693	2.661	2.661	1.028
1.40	3.223	3.138	3.074	3.155	3.377	3.413	2.723	2.634	2.634	0.930
1.50	3.137	3.057	2.999	3.076	3.314	3.321	2.731	2.598	2.598	0.874
1.60	3.058	2.975	2.915	2.987	3.186	3.265	2.722	2.552	2.552	0.863
1.70	2.980	2.891	2.821	2.887	2.993	3.221	2.697	2.496	2.496	0.876
1.80	2.901	2.802	2.720	2.778	2.786	3.164	2.657	2.417	2.417	0.898
1.90	2.820	2.709	2.610	2.663	2.583	3.117	2.593	2.315	2.315	0.953
2.00	2.733	2.597	2.409	2.541	2.387	3.075	2.500	2.191	2.191	1.034



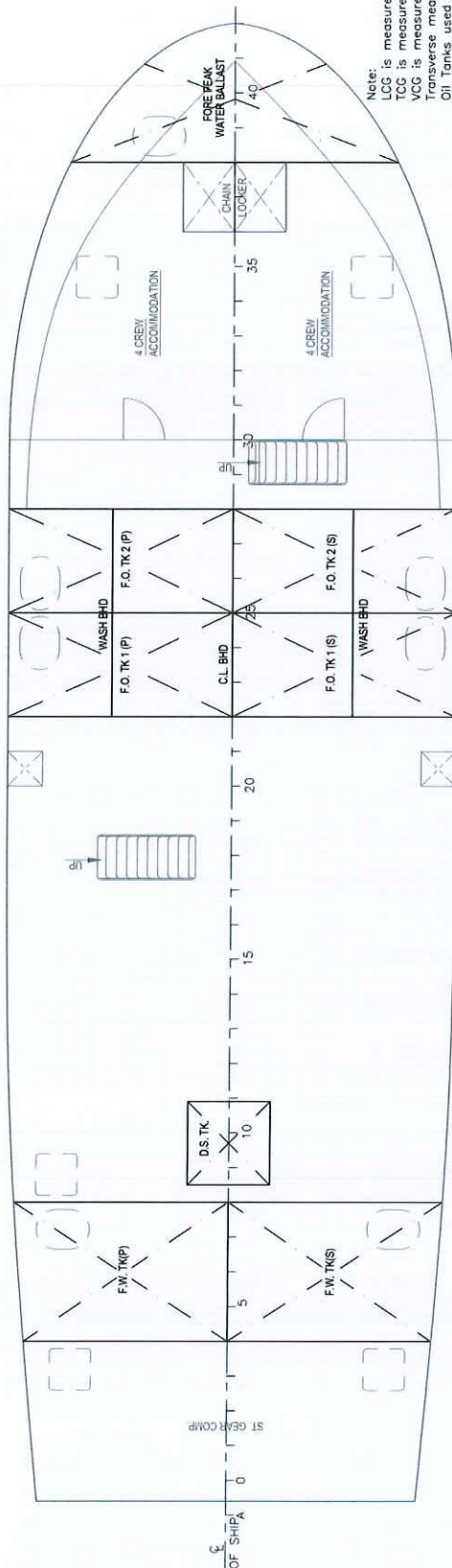


PRINCIPAL PARTICULARS

LENGTH OVERALL	:-	21.30M
LENGTH B.P.	:-	19.76 M
BREADTH	:-	06.50 M
DEPTH	:-	02.80 M
DRAFT(DESIGN)	:-	01.90 M
DRAFT(SCANTLING)	:-	02.00M
FRAME SPG.	:-	500 MM
BOLLARD PULL	:-	10 T
ENGINE POWER	:-	2 x 400 BHP
DESIGN SPEED	:-	10 KNOTS
COMPLEMENT	:-	12 PERSONS
CLASS NOTATION	:-	[45] SUL TUG [45] 1 Y INDIAN RIVER-SEA VESSEL TYPE - IV
CLASS	:-	INDIAN REGISTER OF SHIPPING
FUEL OIL	:-	48.60 CU. MTR
FRESH WATER	:-	20.60 T

Rev.	Date	Description						Drawn	Review
		Issued for :						Client Approval	
Client		KB SHIPPING & CO.							
Project		21.30 M TUG BOAT							
Drawing Title		GENERAL ARRANGEMENT PLAN							
Drawn	Checked	Review	Date	Scale	Paper	Drawing No :-			Sheet
HARSHAD	VG	SC	20.07.16	1:75	A2	S	B	M	- 1 6 - 1 0 1 1 - 0 0 1 1 of 1
 S.B. MARINE CONSULTANT Tel : 91 22 40244296 , Fax : 25862318 E-mail : info@sb-marine.com Web : www.sbmarineconsultant.com						SHIPYARD :			
						CHISTI MARINE ENGINEERING WORKS, SIKKA			
						YARD NO :			
						002			
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Note:
 LCG is measured from Fr-2.
 TCG is measured from CL.
 WCG is measured from BL.
 Transverse measurements toward
 Oil Tanks used are to be filled
 Maximum FSM observed for a

TANK NO.	ITEMS	SP. GR.	VOLUME (M ³)	WEIGHT (T)	VCG (M)	LCG (M)	TGC (M)	FSMT (M ⁴)
1	Fuel Oil Tank 1(P)	0.85	12,299	10,454	1.481	10,747	1,536	3,573
2	Fuel Oil Tank 1(S)	0.85	12,299	10,454	1.481	10,747	1,536	3,573
3	Fuel Oil Tank 2(P)	0.85	12,011	10,210	1.518	12,247	1,525	3,569
4	Fuel Oil Tank 2(S)	0.85	12,012	10,210	1.518	12,247	1,525	3,569
5	Daily service Tank	0.85	1,411	1,200	1.987	1,850	0.000	0.144

TANK NO.		ITEMS	SP. GR.	VOLUME (M) ³	WEIGHT (T)	VCG (M)	LCG (M)	TCG (M)	FSMT (M) ⁴
1	FRESH WATER (P)	1.00	10.304	10.304	10.304	2.028	2.051	1.421	4.517
2	FRESH WATER (S)	1.00	10.304	10.304	10.304	2.028	2.051	1.421	4.517

FRESH WATER								
TANK NO.	ITEMS	SP. GR.	VOLUME (M) ³	WEIGHT (T)	VCG (M)	LCG (M)	TCG (M)	FSMT (M) ⁴
1	FORE PEAK	1.025	5.300	5.432	1.768	18.604	0.00	6.592

PRINCIPAL PARTICULARS	
LENGTH OVERALL :-	21.30M
LENGTH B.P.	19.76 M
BREADTH	06.50 M
DEPTH	02.80 M
DRAUGHT(DSIGN)	01.90 M
DRAUGHT(SCANTLING)	02.00M
FRAME SPC.	500 MM
ROLL ROLL PULL	10 T
ENGINE POWER	2 x 400 BHP
CRUISING SPEED	10 KNOTS
COMPLEMENT	12 PERSONS
CLASS NOTATION	140 SUL TUG
CLASS	RIVER-SEA VES
FUEL OIL	INDIAN REGIS
FRESH WATER	48.60 CU. MTR
	26.60 T

[illegible]

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